QED on S^3: Spectral Geometry of Perturbative Transcendentals

GeoVac Project

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Abstract

We compute perturbative QED spectral sums on the three-sphere S^3 using the Camporesi–Higuchi Dirac spectrum and the Hodge-1 (photon) spectrum with SO(4) vertex selection rules. Eight structural results emerge. (1) **T9 theorem:** the spectral zeta function of the squared Dirac operator D^2 is a two-term polynomial in π^2 with rational coefficients at every integer s, with no odd-zeta content there (half-integer $s \geq 5/2$ carries odd zeta values; Theorem 1). (2) **Even/odd parity discriminant:** operator order (2nd vs 1st) is the primary transcendental discriminant (conditional on the period conjecture, $\zeta(2k{+}1) \notin \mathbb{Q}(\pi)$)—even s produces π^{even}, odd $s \geq 5$ produces $\zeta(3)$, $\zeta(5)$, etc. (Theorem 2). (3) χ_{-4} **identity:** $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s{-}2))$ at every integer $s \geq 2$, a concrete spectral-zeta to Dirichlet-L bridge (Theorem 3). (4) **Self-energy structural zero:** the one-loop electron self-energy on S^3 vanishes identically at the ground state $n_{\text{ext}} = 0$ as a consequence of vertex parity (Theorem 4). (5) **Anomalous magnetic moment:** the one-loop vertex correction with a $q = 1$ probe photon yields $F_2/[\alpha/(2\pi)] = 1.084$ at the lowest external mode ($n_{\text{ext}} = 1$), approximately 8.4% above the flat-space Schwinger value, matching the Parker–Toms leading curvature correction within 0.4%. The next-order coefficient $c_3 = -5.946(3) \times 10^{-7}$ is nonzero at $200.9\,\sigma$, so the curvature expansion does not terminate at second order (Sec. 7.7). (6) **Depth-k tower:** loop order bounds the motivic depth of the level-2 (alternating-MZV) content; it does not automatically mint new irreducibles. The two-loop $S_{\min}$ decomposes exactly into Hoffman double t-values (classical periods of $\mathcal{MT}(\mathbb{Z}[1/2])$; the prior "PSLQ-irreducible" claim was a basis-coverage artifact); the three-loop $S^{(3)} = 31.5726\ldots$ (rigorous bracket $[31.57063, 31.57300]$) has realized depth ≤ 2: the catalogued depth-2 generators $t(5,1)$, $t(5,3)$, $t(7,1)$ enter the components at coefficients $\pm\frac{1}{2}$ and cancel in the total, where the sole surviving depth-2 constant is $\zeta(5,3) \in \mathcal{MT}(\mathbb{Z})$ (Proposition 4, Sec. 10.4). (7) $\zeta(3)$ **complementarity:** the degeneracy-weighted fine-structure shell sum $\mathcal{S}(n) = n(5n-4)/2$ and the Dirac mode degeneracy $g_n^{\text{Dirac}} = 2n(n+1)$ satisfy $\mathcal{S}(n) + g_n^{\text{Dirac}} = 9n^2/2$ exactly, so the Dirichlet series identity $D_{\mathcal{S}}(s) + D_g(s) = \frac{9}{2}\,\zeta(s{-}2)$ holds for all s: the $\zeta(s{-}1)$ terms—all odd-zeta content—cancel identically (Theorem 7). (8) **Fine-structure Euler sum cancellation and product survival rule:** the degeneracy-weighted Dirichlet sums D_p at order $(Z\alpha)^{2p}$ in the exact Dirac hydrogen spectrum, computed through D_6 ($(Z\alpha)^{12}$, weight 11), satisfy a product survival rule: $\zeta(2) \times \zeta(\text{odd})$ cancels algebraically at every order, while exactly $\max(0, \lfloor(2p{-}5)/2\rfloor)$ products survive (Observations 1, 2). The cancellation mechanism is the n^2 Fock degeneracy — the same weighting that produces $F = \zeta(2) = D_{n^2}(4)$ in Paper 2. However, PSLQ finds no small-coefficient $\mathbb{Q}$-linear relation between Paper 2's $K = \pi(B + F - \Delta)$ and $D_2, \ldots, D_6$: K lies in $\mathbb{Q}[\pi]$ (B a rational Casimir trace, $F = \zeta(2)$, Δ rational), while the Sommerfeld sums carry odd-zeta content from the Euler–Zagier MZV algebra. A complementary *graph-native* formulation (Sec. 15) computes all one-loop diagrams as finite matrix traces on the GeoVac Fock graph, requiring no infinite sums, regularization, or renormalization. All graph-native QED quantities are π-free (algebraic); the projection exchange constant relating graph and continuum VP is *rational* at every finite truncation, confirming Paper 18's calibration-tier classification. The continuum self-energy structural zero (Theorem 4) is broken on the finite graph: the pendant-edge theorem (Proposition 5) proves $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max} \to 2$ exactly, never healing in the graph limit. The root cause

is that the graph photon is a scalar 1-cochain, lacking the spin-1 vector structure whose SO(4) selection rules enforce the continuum zero. The graph-native anomalous moment $F_2(t)$ is an even rational function of the hopping coupling t over $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$, with $F_2(\kappa)$ decreasing as a power law $\sim 3.5\, n_{\max}^{-0.57}$ ($F_2 = 2.353, 1.873, 1.589, 1.396, 1.253$ at $n_{\max} = 2, \ldots, 6$). A systematic selection rule census shows that only 1 of 8 continuum QED selection rules survives on the scalar Fock graph (Gaunt/CG sparsity). Building QED on the native Dirac graph with (n, κ, m_j) node labels recovers 4 of 8 rules, establishing a three-tier partition: 1 always survives, 3 are spinor-recoverable (intrinsic to spinor quantum numbers), and 4 require vector photon structure (calibration tier). A five-track vector QED sprint (VQ-1–5) confirms this partition from a complementary direction: neither Pauli spin matrices at the vertex, nor direction-resolved edge Hodge decomposition, nor their combination recovers any of the 4 vector-required rules. However, the transverse photon propagator $G_T = (d_1 d_1^\top)^+$ built from Paper 30's plaquette boundary operator *does* recover the ground-state structural zero ($\Sigma_T(\mathrm{GS}) = 0$ exactly), refining the partition to 1+3+1+3: the plaquette-recoverable tier is topological (pendant-edge exclusion from closed walks), requiring no photon quantum numbers. Adding explicit vector photon quantum numbers (q, m_q) with Clebsch–Gordan vertex coupling recovers 7 of 8 continuum selection rules for scalar electrons and **8 of 8 for Dirac electrons** (Sec. 15.12). The calibration cost is exactly one transcendental per loop: π, entering as the S^2 Weyl exchange constant $1/(4\pi)$ from Paper 18's taxonomy. For scalar electrons, the sole failure is Furry's theorem; for Dirac electrons, the diagonal vertex coupling $V(a, a, q, m_q) = 0$ vanishes identically by the *Dirac spinor phase constraint*: the α-matrix is off-diagonal in large/small component space, and for $a = b$ the cross terms combine, via Hermiticity and the i-factor on the small component, into an imaginary part that m_l conservation forces to vanish. This is a kinematic identity of the single-particle vertex, not the field-theoretic charge conjugation mechanism. For Dirac electrons, the 8 rules partition as 1–7–8: graph topology → angular momentum conservation → Dirac spinor phase constraint (zero additional cost). The VP projection constant C does *not* serve as a universal graph-to-continuum projector: $C \times F_2$ diverges from $\alpha/(2\pi)$, confirming that different diagram topologies require independent projection factors. Comprehensive transcendental tables for $D(s)$, $D_{\mathrm{even}}(s)$, $D_{\mathrm{odd}}(s)$, and the one-loop self-energy are presented, all backed by the test suite.

1 Introduction

The GeoVac framework [1, 2] computes quantum-mechanical observables from the spectral data of operators on compact manifolds, on a discrete lattice of their quantum-number labels (on S^3 the scalar spectrum is the diagonal $SO(4)$ Casimir on those labels; the lattice's graph Laplacian does not converge to the S^3 Laplacian, Paper 7). On the three-sphere S^3, the Dirac operator has a discrete spectrum with known eigenvalues and degeneracies [12], and the framework produces structurally sparse qubit Hamiltonians [2] whose Pauli count is exactly linear in Q across molecules at fixed basis ($N_{\mathrm{Pauli}} = 27.90\, Q$, exact global-$M_L$ rule).

Paper 18 [3] introduced an exchange-constant taxonomy that classifies where transcendental numbers enter quantum-mechanical computations: intrinsic (from angular structure), calibration (from the S^3 projection), and embedding (from the electron–electron cusp). The present paper extends this taxonomy to perturbative QED by computing spectral sums that probe the interaction structure of quantum electrodynamics on S^3.

All calculations in the paper use four modules in the `geovac` codebase: `qed_vacuum_polarization` (one-loop heat kernel), `qed_vertex` (two-loop sunset with vertex parity), `qed_self_energy` (one-loop self-energy and vertex correction), and `qed_three_loop` (three-loop chain topology with $O(N^3)$ factorization). Every stated numerical value is backed by a test in `tests/test_qed_*.py` (the four named modules' suites collect 150 tests; the full glob collects 645 as of 2026-07-04 — all passing). Two scope notes (2026-07-04): the Sommerfeld product-survival closed forms D_4–D_6 are pinned at value level in `tests/test_paper28_gap_closure_pins.py` (their Euler-sum derivation pins are a

named follow-up), and the flat-space consistency checks are frozen in `tests/test_qed_flat_limit.py`.

The central finding is that the S^3 spectral geometry produces the odd zeta values that appear in flat-space perturbative QED—$\zeta(3)$, $\zeta(5)$—together with vertex-parity content specific to S^3 (Catalan's G, Dirichlet $\beta(4)$, which do not appear in the flat-space a_e through three loops), through a purely geometric mechanism (operator order plus vertex topology), together with multi-loop constants that climb the depth filtration of the classical level-2 (alternating) MZV ring: the two-loop $S_{\min}$ is identified in closed form in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ (Sec. 9).

2 Spectral Data on S^3

2.1 Dirac spectrum

On the unit three-sphere, the Dirac operator has eigenvalues and degeneracies [12]

$$|\lambda_n| = n + \tfrac{3}{2}, \qquad g_n = 2(n+1)(n+2), \qquad n = 0, 1, 2, \ldots \tag{1}$$

The Dirac Dirichlet series is $D(s) = \sum_{n=0}^{\infty} g_n / |\lambda_n|^s$. Via Hurwitz zeta,

$$D(s) = 2\,\zeta(s-2, \tfrac{3}{2}) - \tfrac{1}{2}\,\zeta(s, \tfrac{3}{2}). \tag{2}$$

Using $\zeta(s, 3/2) = (2^s - 1)\zeta_R(s) - 2^s$, this becomes

$$D(s) = 2(2^{s-2} - 1)\,\zeta_R(s-2) - \tfrac{1}{2}(2^s - 1)\,\zeta_R(s). \tag{3}$$

We define the even-n and odd-n sub-sums:

$$D_{\text{even}}(s) = \sum_{\substack{n=0 \\ n\,\text{even}}}^{\infty} \frac{g_n}{|\lambda_n|^s}, \qquad D_{\text{odd}}(s) = \sum_{\substack{n=1 \\ n\,\text{odd}}}^{\infty} \frac{g_n}{|\lambda_n|^s}, \tag{4}$$

so that $D(s) = D_{\text{even}}(s) + D_{\text{odd}}(s)$.

2.2 Hodge-1 (photon) spectrum

The transverse vector Laplacian (Hodge-1 operator) on S^3 has eigenvalues and degeneracies

$$\mu_q = q(q+2), \qquad d_q^T = q(q+2), \qquad q = 1, 2, 3, \ldots \tag{5}$$

(Labeling-convention note, 2026-07-04: this is the framework's q-labeling as implemented and documented in `geovac/hodge1_s3.py`; the literature's coclosed/transverse family — Ikeda–Taniguchi 1978, Camporesi 1990 — is indexed so that the eigenvalue reads $(k+1)^2$ with multiplicity $2k(k+2)$. Reconciling the two labelings at the operator level is the named hodge1/P28 follow-up, deferred as its own item; the module's anchor test documents the divergence.) The Ricci shift $+2$ relative to the scalar Laplacian eigenvalues $\ell(\ell+2)$ follows from the Bochner–Weitzenböck identity $\Delta_1 = \nabla^*\nabla + \mathrm{Ric}$ on S^3 (Ricci scalar $R = 6$ on unit S^3, giving $\mathrm{Ric} = 2g$). There are no zero modes ($\beta_1 = 0$ for S^3).

2.3 Vertex selection rule

The QED vertex on S^3 couples spinor harmonics at levels n_1, n_2 to a vector harmonic at level q with the SO(4) selection rule:

$$|n_1 - n_2| \le q \le n_1 + n_2, \qquad n_1 + n_2 + q \text{ is odd.} \tag{6}$$

The parity constraint arises from the γ^μ coupling, which flips the chirality representation of $\mathrm{SO}(4) = \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$.

The SO(4) Clebsch–Gordan channel count $W(n_1, n_2, q) \in \{0, 1, 2\}$ weights each allowed triple by the number of independent coupling channels (the two components of the transverse vector harmonic under $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$).

3 Theorem 1: Squared Dirac Spectral Zeta (T9)

Theorem 1 (T9: D^2 spectral zeta is π^{even})**.** *The spectral zeta function of the squared Dirac operator on unit S^3 satisfies*

$$\zeta_{D^2}(s) = 2^{2s-1}\big[\lambda(2s{-}2) - \lambda(2s)\big], \tag{7}$$

where $\lambda(2k) = (1 - 2^{-2k})\,\zeta_R(2k) = \text{rational} \times \pi^{2k}$. At every integer $s \ge 1$, $\zeta_{D^2}(s)$ is a two-term polynomial in π^2 with rational coefficients; no odd-zeta content appears at any integer $s \ge 1$. The statement is an integer-s one: at half-integer $s \ge 5/2$ the same formula (7) carries odd zeta values (e.g. $\zeta_{D^2}(5/2) = 14\,\zeta(3) - \frac{31}{2}\,\zeta(5)$), and $s = 1/2, 3/2$ are simple poles with rational residues.

Proof. The squared Dirac eigenvalues are $\lambda_n^2 = (n + 3/2)^2 = (2n + 3)^2/4$, so $\zeta_{D^2}(s) = \sum_{n\ge 0} g_n \cdot (4/(2n + 3)^2)^s = 4^s \sum_{n\ge 0} 2(n + 1)(n + 2)/(2n + 3)^{2s}$. Substituting $m = 2n + 3$ (which runs over odd integers ≥ 3), the sum becomes $\sum_{m\ge 3, m\,\text{odd}} (m^2 - 1)/2 \cdot 4^s/m^{2s}$. The $(m^2 - 1)/2$ factor splits as $(m^{2-2s} - m^{-2s})/2$, giving two Dirichlet-type sums over odd integers. Each odd-integer sum is $\lambda(2k) = (1 - 2^{-2k})\zeta_R(2k)$ with the first three odd values ($m = 1$) added or subtracted to correct the lower limit. By the Euler–Bernoulli formula, $\zeta_R(2k) = (-1)^{k+1}(2\pi)^{2k} B_{2k}/(2 \cdot (2k)!)$, so $\lambda(2k) = \text{rational} \times \pi^{2k}$. At integer s both arguments $2s - 2$ and $2s$ are even, so odd-zeta values $\zeta_R(2k + 1)$ cannot appear; at half-integer $s \ge 5/2$ both arguments are odd and ≥ 3, so $\lambda(2s - 2)$, $\lambda(2s)$ are rational multiples of odd zeta values (at $s = 1/2, 3/2$ an argument equals 1, a pole; at negative half-integer s the values $\zeta(1 - 2k) = -B_{2k}/(2k)$ are rational). □

Verification: Symbolic identity confirmed in `sympy` for $s = 1, \ldots, 10$. Numerical agreement to 10^{-60} at $s = 2, \ldots, 20$. The integer-s closed forms at $s = 2, 3$ and the half-integer value $\zeta_{D^2}(5/2) = 14\,\zeta(3) - \frac{31}{2}\,\zeta(5)$ are checked against the direct spectral sum in `tests/test_paper28_t9_half_integer.py`.

Table 1: Squared Dirac spectral zeta $\zeta_{D^2}(s)$.

s	$\zeta_{D^2}(s)$
1	$-\pi^2/4$
2	$\pi^2 - \pi^4/12$
3	$\pi^4/3 - \pi^6/30$
4	$2\pi^6/15 - 17\pi^8/1260$

Remark 1 (Where $\sqrt{\pi}$ lives: heat-trace coefficients, not ζ_{D^2} (Sprint Mixed-Tate Test, June 2026; corrected 2026-09-30))**.** *The $\sqrt{\pi}$ met in heat-kernel-side analyses of ζ_{D^2} does not live in the spectral zeta function at any s. It lives only in the heat-trace coefficients: the short-time expansion $\operatorname{Tr} e^{-tD^2} \sim \frac{\sqrt{\pi}}{2}\, t^{-3/2} - \frac{\sqrt{\pi}}{4}\, t^{-1/2}$ carries the $(4\pi t)^{-3/2}$-type prefactor of an odd-dimensional heat kernel, and the Mellin transform $\zeta_{D^2}(s) = \Gamma(s)^{-1} \int_0^\infty t^{s-1} \operatorname{Tr} e^{-tD^2}\, dt$ divides each coefficient by $\Gamma(s)$ at its pole. At $s = 3/2$ this gives the residue $(\sqrt{\pi}/2)/\Gamma(3/2) = 1$, and at $s = 1/2$ the residue $(-\sqrt{\pi}/4)/\Gamma(1/2) = -1/4$: both rational. Away from the poles, the values at half-integer $s \geq 5/2$ are odd-zeta combinations (Theorem 1), not multiples of $\sqrt{\pi}$. At integer $s \geq 1$ the values are pure $\pi^{2k} \cdot \mathbb{Q}$ (Table 1 above). After the standard Connes–Chamseddine volume normalization (multiplication of the heat-trace coefficients by $(4\pi)^{d/2}$ with $d = 3$), the coefficients also cleanly enter $\mathbb{Q}[\pi^2]$: Paper 51 [8] Cor. 2.1 gives $a_0^{D^2} = 4\pi^2$, $a_1^{D^2} = -2\pi^2$, $a_k = 0$ for $k \geq 2$. At integer s the Geo-Vac S^3 M2 sub-mechanism is therefore mixed-Tate over $\mathbb{Q}$ in the F–M sense (Paper 32 [7] § VIII Corollary), in the strictly smaller pure-Tate sub-ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$.*

Remark 2 (No functional equation for ζ_{D^2}: the quadratic multiplicity is the obstruction (Sprint GB-2, 2026-05-29))**.** *Sprint RH-O (Apr. 2026) reported, by a 13,080-template numerical search, that the Dirac spectral zeta $D(s)$ admits no spectral-triple functional equation (best residual* 0.594 *vs. classical anchors at 10^{-51}). The closed form* (7) *upgrades this to an* analytic *statement and pinpoints the mechanism. Since $\zeta_{D^2}(s) = D(2s)$, both $\zeta_{D^2}(s)$ and its reflection $\zeta_{D^2}(c-s)$ lie in the two-dimensional space* $\operatorname{span}\{\zeta_R(2s),\, \zeta_R(2s-2)\}$ *over the field of elementary (gamma/exponential/trigonometric) functions; a functional equation $\zeta_{D^2}(s) = \chi(s)\, \zeta_{D^2}(c-s)$ with χ elementary exists iff the two coefficient vectors are parallel, i.e.* $\det = AD - BC = 0$.

The Riemann reflection $z \mapsto 1 - z$ matches the argument set $\{2s-2,\, 2s\}$ to its reflected partner only at the natural axis $c = 3/2$ (equivalently $c = 1/2$), the Dirac ground-state value $|\lambda_0| = 3/2$ — so the axis is not *the problem. The obstruction is the multiplier: the two terms acquire the gamma ratio $\Gamma(2s)/\Gamma(2s-2) = (2s-1)(2s-2)$, which forces* $\det \neq 0$ *(verified* $|\det| = 2.8,\, 8.4,\, 60.3,\, 8.2$ *at $s = 1+i,\, \frac{3}{2}+3i,\, 2+5i,\, \frac{1}{2}+10i$; the candidate-multiplier ratio χ_1/χ_2 tracks $(2s-1)(2s-2)$). The exponent gap of* 2 *between the two λ-terms is produced* precisely *by the quadratic Weyl multiplicity $g_n = 2(n+1)(n+2)$: a constant-multiplicity (single-power) spectrum lies in a one-dimensional span, is trivially parallel, and* does *inherit a functional equation. The reflection algebra is cross-checked against direct evaluation of* (7) *to 10^{-41}.*

Consequence (unification of two RH walls). *The "wrong Weyl law" wall (RH-N: $\gamma_n \sim \sqrt{n}$, harmonic-oscillator density) and the "no functional equation" wall (RH-O) are the* same *structural fact: both are consequences of the S^3 quadratic degeneracy. The very multiplicity n^2 that places the framework's even-s values on the Bernoulli ladder (Paper 32's Bernoulli-ladder remark: $F = \zeta(2) = \pi^2 B_2$, etc.) is the multiplicity that denies ζ_{D^2} a critical line. This is consistent with — and sharpens — WH6: the framework's zeta carries ζ_R's special values but not its zero-controlling functional equation. Source:* `debug/sprint_gb2_zeta_dsq_functional_eq.py`, `debug/sprint_gb2_zeta_dsq_memo.md`.

4 Theorem 2: Even/Odd Parity Discriminant

Theorem 2 (Parity discriminant)**.** *At even integer s, the Dirac Dirichlet series $D(s)$ is purely π^{even}. At odd integer s, $D(s)$ involves $\zeta_R(s-2)$ and $\zeta_R(s)$ (odd-zeta values; $s \geq 5$, since $\zeta_R(1)$ is a pole). Operator order (2nd vs 1st) is the primary transcendental discriminant, conditional on the period conjecture $\zeta(2k+1) \notin \mathbb{Q}(\pi)$: the second-order operators Δ_{LB} and D^2 produce π^{even} exclusively at integer s (this holds for Δ_{LB} and D^2, with spectra $n^2 - 1$ and $(n + 3/2)^2$, whose partial fractions have rational poles; it is not a property of every second-order operator — the spinor connection*

Laplacian $\nabla^*\nabla = D^2 - 3/2$ *has* $\zeta_{\nabla^*\nabla}(2) = \frac{1}{12}\big(14\,\pi\tan(\frac{\pi\sqrt{6}}{2})/\sqrt{6} + 5\pi^2\sec^2(\frac{\pi\sqrt{6}}{2})\big)$*, outside* $\mathbb{Q}[\pi^2]$*) (their spectral determinants are not covered:* $\zeta'_{D^2}(0) = 3\zeta_R(3)/(4\pi^2) + \frac{1}{2}\ln 2$*); first-order operators produce odd-zeta values at odd* s*.*

Proof. From Eq. (3), $D(s) = 2(2^{s-2} - 1)\,\zeta_R(s{-}2) - \frac{1}{2}(2^s - 1)\,\zeta_R(s)$. At even s, both $\zeta_R(s{-}2)$ and $\zeta_R(s)$ are Bernoulli-type: $\zeta_R(2k) = (-1)^{k+1}(2\pi)^{2k}B_{2k}/(2\cdot(2k)!) = \text{rational} \times \pi^{2k}$. At odd s, both $\zeta_R(s{-}2)$ and $\zeta_R(s)$ are genuine odd-zeta values ($\zeta(3)$, $\zeta(5)$, ...), conjecturally $\mathbb{Q}$-linearly independent of all π^{2k} (open: Apéry's theorem gives the irrationality of $\zeta(3)$ and Rivoal's infinitely many irrational odd zeta values [27, 28]; neither gives independence from π). □

Table 2: Dirac Dirichlet series $D(s)$: PSLQ identifications at 80 dps.

s	$D(s)$	Class
4	$\pi^2 - \pi^4/12$	π^{even}
5	$14\,\zeta(3) - \frac{31}{2}\,\zeta(5)$	odd-zeta
6	$\pi^4/3 - \pi^6/30$	π^{even}
7	$62\,\zeta(5) - \frac{127}{2}\,\zeta(7)$	odd-zeta
8	$2\pi^6/15 - 17\pi^8/1260$	π^{even}
9	$254\,\zeta(7) - \frac{511}{2}\,\zeta(9)$	odd-zeta
10	$17\pi^8/315 - 31\pi^{10}/5670$	π^{even}

The appearance of $\zeta(3)$ at $s = 5$ is the S^3 analog of the $\zeta(3)$ that enters flat-space two-loop QED through the Schwinger–Rosner vertex correction [13].

5 Theorem 3: The χ_{-4} Identity

Theorem 3 (χ_{-4} identity)**.** *For every integer* $s \geq 2$ *(the even/odd spectral sums converge for* $s \geq 4$*; at* $s = 2, 3$ *by Hurwitz continuation),*

$$D_{\text{even}}(s) - D_{\text{odd}}(s) \;=\; 2^{s-1}\big(\beta(s) - \beta(s{-}2)\big), \tag{8}$$

where $\beta(s) = L(s, \chi_{-4})$ *is the Dirichlet beta function.*

Proof. Express $D_{\text{even}}(s)$ and $D_{\text{odd}}(s)$ via Hurwitz zeta at quarter-integer shifts. With $m = n + 3/2$ and full degeneracy $g_n = 2(n+1)(n+2) = 2(m^2 - 1/4)$, the even-$n$ eigenvalues are $m = 2(k + 3/4)$ and the odd-n ones $m = 2(k + 5/4)$, $k \geq 0$, so $D_{\text{even}}(s) = 2^{-(s-3)}\zeta(s{-}2, 3/4) - 2^{-(s+1)}\zeta(s, 3/4)$ and $D_{\text{odd}}(s) = 2^{-(s-3)}\zeta(s{-}2, 5/4) - 2^{-(s+1)}\zeta(s, 5/4)$. Two elementary Hurwitz identities: $\zeta(s, 5/4) = \zeta(s, 1/4) - 4^s$ (shift recurrence) and $\zeta(s, 3/4) - \zeta(s, 1/4) = -4^s\beta(s)$ (the Dirichlet beta definition). Together, $\zeta(s, 3/4) - \zeta(s, 5/4) = 4^s(1 - \beta(s))$; substituting into the difference, $D_{\text{even}} - D_{\text{odd}} = 2^{s-1}(1{-}\beta(s{-}2)) - 2^{s-1}(1{-}\beta(s)) = 2^{s-1}(\beta(s){-}\beta(s{-}2))$, which is (8). The Hurwitz decompositions and the closed form are verified bit-exactly against direct spectral sums at $s \in \{4, 5, 6, 7\}$ to 40 digits. □

Verification: Eq. (8) verified symbolically (`sympy` residual zero) at $s \in \{2, 3, \ldots, 10, 15\}$ and by PSLQ at 100-digit precision at every integer s in the tested range. At $s = 4$, $D_{\text{even}}(4) - D_{\text{odd}}(4) = 8(\beta(4) - G)$ (since $G = \beta(2)$), consistent with the explicit decompositions (9)–(10) below.

5.1 Vertex-parity decomposition at $s = 4$

The two-loop sunset diagram couples spinor pairs (n_1, n_2) subject to the vertex parity constraint. The even-n and odd-n sub-sums of $D(4)$ decompose as:

$$D_{\text{even}}(4) = \tfrac{\pi^2}{2} - \tfrac{\pi^4}{24} - 4G + 4\beta(4), \tag{9}$$

$$D_{\text{odd}}(4) = \tfrac{\pi^2}{2} - \tfrac{\pi^4}{24} + 4G - 4\beta(4), \tag{10}$$

$$D(4) = \pi^2 - \tfrac{\pi^4}{12} \qquad \text{(Catalan and } \beta(4) \text{ cancel)}. \tag{11}$$

Here $G = \beta(2) = \sum_{n=0}^{\infty}(-1)^n/(2n+1)^2 \approx 0.9159656$ is Catalan's constant and $\beta(4) \approx 0.9889446$ is the Dirichlet beta function at $s = 4$. The mechanism: vertex parity acts as the Dirichlet character χ_{-4} on the mode sum, and the quarter-integer Hurwitz shifts expose $L(s, \chi_{-4})$ in the even/odd difference. Verified by PSLQ at 80-digit precision with basis $\{1, \pi^2, \pi^4, G, \beta(4)\}$.

Corollary 1. *The χ_{-4} identity is a structural realization of the vertex-parity-as-χ_{-4} mechanism, but does not constrain the zeros of $\beta(s)$: a zero of β at $s = \rho$ propagates to zeros of D_{diff} at both ρ and $\rho + 2$, consistent with but not enforcing $\Re\rho = 1/2$.*

Remark 3 (Motivic home of Theorem 3)**.** *Each of $D_{\text{even}}(s)$, $D_{\text{odd}}(s)$, and their difference is a period of mixed Tate motives at level 4 over $\mathcal{O}_4[1/4] = \mathbb{Z}[i, 1/2]$, by direct application of Deligne's classification at $N = 4$ and the explicit basis of Glanois (Sprint M3 Cyclotomic Mixed-Tate, June 2026, memo `debug/sprint_m3_cyclotomic_mixed_tate_memo.md`; Paper 32 § VIII Corollary `cor:m3_cyclotomic_mixed_tate`; Paper 18 § III.7). The vertex-parity-as-χ_{-4} correspondence IS the Deligne–Glanois Galois descent from level 4 to level 2 on the framework's natural QED observables: the un-restricted $D(s) = D_{\text{even}} + D_{\text{odd}}$ lives in the smaller level-≤ 2 ring (Theorem 1 and Theorem 2); the $\zeta(s, 1/4)$ pieces in $D_{\text{even}}, D_{\text{odd}}$ individually cancel in the sum but survive in the difference, exposing the level-4 Dirichlet L-value content. Catalan $G = \beta(2)$ and $\beta(4)$ are periods in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at depth 1 but conjecturally not in $\mathcal{MT}(\mathbb{Z}[1/2])$ at any depth — the level-4 lift is structural to the vertex-parity mechanism.*

5.2 Three-axis taxonomy

The transcendental content of any QED spectral sum on S^3 is classified by three independent axes:

1. **Operator order.** Second-order $(D^2) \to \pi^{\text{even}}$ at integer s (Theorem 1); first-order $(|D|) \to$ odd-zeta at odd $s \geq 5$ (Theorem 2).
2. **Bundle type.** Scalar vs spinor modulates coefficients but not the transcendental class (Paper 18 Sec IV).
3. **Diagram topology.** Vertex parity introduces Dirichlet L-function characters (G, $\beta(4)$) that are absent from unrestricted sums (Theorem 3).

6 Theorem 4: Self-Energy Structural Zero

The one-loop electron self-energy on S^3 (Feynman gauge) is:

$$\Sigma(n_{\text{ext}}) = \sum_{n_{\text{int}}, q} \frac{W(n_{\text{ext}}, n_{\text{int}}, q)\; g_{n_{\text{int}}}\; d_q^T}{|\lambda_{n_{\text{int}}}|^{2s_e}\; \mu_q^{s_\gamma}}, \tag{12}$$

where the sum is over internal electron mode n_{int} and photon mode q subject to the vertex selection rule (6), $s_e = 2$ (so $|\lambda|^4$ in the denominator), $s_\gamma = 1$, and W is the SO(4) channel count.

Theorem 4 (Self-energy structural zero)**.** $\Sigma(0) = 0$ *exactly. The one-loop self-energy vanishes identically at the Dirac ground state* $n_{\text{ext}} = 0$.

Proof. At $n_{\text{ext}} = 0$, the vertex selection rule requires $|0 - n_{\text{int}}| \leq q \leq 0 + n_{\text{int}}$ and $0 + n_{\text{int}} + q$ odd. The first condition forces $q = n_{\text{int}}$. The second then requires $2\,n_{\text{int}}$ to be odd, which is impossible for any integer n_{int}. Therefore $W(0, n_{\text{int}}, q) = 0$ for all (n_{int}, q), and the sum is empty. □

This is the S^3 analog of the fact that the ground state of a compact system cannot emit into a vacuum mode: the vertex parity constraint is a spectral selection rule, not a dynamical cancellation.

6.1 Self-energy for excited states

For $n_{\text{ext}} \geq 1$, the self-energy is nonzero and convergent. We compute $\Sigma(n_{\text{ext}})$ at physical propagator powers ($s_e = 2$, $s_\gamma = 1$) for $n_{\text{ext}} = 0, 1, \ldots, 5$ at cutoff $n_{\max} = 30$.

Table 3: One-loop self-energy $\Sigma(n_{\text{ext}})$ on S^3, $n_{\max} = 30$.

n_{ext}	$\Sigma(n_{\text{ext}})$
0	0 (exact, Theorem 4)
1	1.2854
2	1.7862
3	2.0843
4	2.2884
5	2.4398

The self-energy increases monotonically with n_{ext}, consistent with stronger coupling at higher momentum shells. At $n_{\text{ext}} = 1$, PSLQ at 50 dps identifies $\Sigma(1)$ as containing π^{even} and Catalan G components (vertex parity active on the asymmetric coupling to the $n_{\text{ext}} = 1$ shell).

6.2 Vertex correction

The one-loop vertex correction involves two vertex insertions:

$$\Lambda(n_{\text{ext}}) = \sum_{n_{\text{int}}, q_1, q_2} \frac{W(n_{\text{ext}}, n_{\text{int}}, q_1)\; W(n_{\text{int}}, n_{\text{ext}}, q_2)\; g_{n_{\text{int}}}\; d^T_{q_1}\; d^T_{q_2}}{|\lambda_{n_{\text{int}}}|^{2s_e}\; \mu^{s_\gamma}_{q_1}\; \mu^{s_\gamma}_{q_2}}. \tag{13}$$

This is $O(N^3)$ in the cutoff. Both $\Lambda(n_{\text{ext}})$ and $\Sigma(n_{\text{ext}})$ are backed by 35 tests in `tests/test_qed_self_energy.py`.
Note: The shell-summed vertex correction $\Lambda(n_{\text{ext}})$ defined above uses the CG-weight channel count W and is m_j-independent (it gives $B = 0$ by the Wigner–Eckart theorem). The anomalous magnetic moment extraction requires the m_j-resolved three-point function developed in the next section.

7 Anomalous Magnetic Moment on S^3

7.1 Why the two-point vertex correction gives $B = 0$

The two-point vertex correction $\Lambda(n_{\text{ext}})$ defined in Eq. (13) sums over all internal modes and photon polarizations at *fixed* external quantum numbers $(n_{\text{ext}}, j_{\text{ext}}, m_j)$. On S^3, the vertex amplitude at each junction involves a product of four $\text{SU}(2)_L \times \text{SU}(2)_R$ Clebsch–Gordan coefficients. When the

photon polarizations $(m_{g,L}, m_{g,R})$ are summed over at both vertices, the SO(4) Wigner–Eckart theorem forces the result to be independent of m_j:

$$\Lambda(n_{\text{ext}}, j, m_j = +\tfrac{1}{2}) = \Lambda(n_{\text{ext}}, j, m_j = -\tfrac{1}{2}). \tag{14}$$

Therefore the spin-dependent piece $B_{2\text{pt}} = \Lambda(+\frac{1}{2}) - \Lambda(-\frac{1}{2}) = 0$ identically. The two-point vertex correction is a scalar under SO(4) rotations and cannot distinguish spin orientations. To extract the anomalous magnetic moment (F_2 form factor), an external probe must break the rotational symmetry.

7.2 Three-point vertex correction with probe photon

The anomalous magnetic moment is extracted from a three-point function in which an external probe photon of mode $q_{\text{probe}} = 1$ (the lowest transverse vector harmonic, the S^3 analog of a uniform magnetic field) is inserted between the two loop-photon vertices. The physical topology is:

$$\langle \text{ext}, m_j |\, \Gamma^{1-\text{loop}}_{\text{probe}} \,| \text{ext}, m_j \rangle = \sum_{\substack{n_{\text{int}},\, j_{\text{int}} \\ m_{j,\text{int}},\, m'_{j,\text{int}} \\ q_{\text{loop}}}} \frac{V_1 \;\cdot\; V_{\text{probe}} \;\cdot\; V_3}{|\lambda_{n_{\text{int}}}|^4 \; \mu_{q_{\text{loop}}}}, \tag{15}$$

where:

- V_1: amplitude for the external electron (m_j) to emit a loop photon and transition to the internal state $(m_{j,\text{int}})$;
- V_{probe}: amplitude for the probe photon at $q = 1$ to scatter the internal electron from $m_{j,\text{int}}$ to $m'_{j,\text{int}}$, summed over probe polarizations;
- V_3: amplitude for the internal state $(m'_{j,\text{int}})$ to reabsorb the loop photon and return to the external state (m_j).

Each vertex amplitude is a product of four Clebsch–Gordan coefficients coupling the $\text{SU}(2)_L \times \text{SU}(2)_R$ representations of source, target, and photon:

$$V = \sum_{m_{L,1}} C^{j_s}_{j_{s,L}\, j_{s,R}}(m_{L,1},\, m_j - m_{L,1};\, m_j) \cdot C^{j_t}_{j_{t,L}\, j_{t,R}}(m_{L,2},\, m_{R,2};\, m'_j) \cdot C^{j_{t,L}}_{j_{s,L}\, j_{g,L}}(m_{L,1},\, m_{g,L};\, m_{L,2}) \cdot C^{j_{t,R}}_{j_{s,R}\, j_{g,R}}(m_{R,1},\, m$$

(16)

where $m_{L,2} = m_{L,1} + m_{g,L}$ and $m_{R,2} = m_{R,1} + m_{g,R}$, and the sum runs over all $m_{L,1} \in \{-j_{s,L}, \ldots, +j_{s,L}\}$.

The probe photon at $q = 1$ has channels $(j_{g,L}, j_{g,R}) = (1, 0)$ or $(0, 1)$, corresponding to the two components of the transverse vector harmonic. The probe couples internal states $m_{j,\text{int}} \to m'_{j,\text{int}}$, enabling spin-dependent amplitudes that the polarization-summed two-point function cannot access.

The spin-dependent piece of the vertex correction is:

$$B = \Gamma^{1-\text{loop}}_{\text{probe}}(m_j = +\tfrac{1}{2}) - \Gamma^{1-\text{loop}}_{\text{probe}}(m_j = -\tfrac{1}{2}). \tag{17}$$

7.3 Tree-level normalization

The tree-level (zeroth-order) coupling of the external electron to the $q = 1$ probe photon provides the normalization for F_2 extraction. At the external state $n_{\text{ext}} = 1$, $j = 1/2$ (Camporesi–Higuchi

level $n_{\mathrm{CH}} = 1$, with $j_L = 1$, $j_R = 1/2$):

$$V_{\mathrm{tree}}(m_j = +\tfrac{1}{2}) = -V_{\mathrm{tree}}(m_j = -\tfrac{1}{2}) = 0.2790, \tag{18}$$

$$V_{\mathrm{magnetic}} = V_{\mathrm{tree}}(+\tfrac{1}{2}) - V_{\mathrm{tree}}(-\tfrac{1}{2}) = 0.5579. \tag{19}$$

The charge piece $[V_{\mathrm{tree}}(+\frac{1}{2}) + V_{\mathrm{tree}}(-\frac{1}{2})]/2 = 0$ vanishes—the $q = 1$ probe photon couples purely to the magnetic moment at tree level, as expected for the S^3 analog of a uniform magnetic field.

Analytically, $V_{\mathrm{magnetic}} = 2(\sqrt{2}/3 - \sqrt{3}/9)$, a pure algebraic number from the Clebsch–Gordan coefficients.

The Pauli form factor is then:

$$F_2 = \frac{B}{V_{\mathrm{magnetic}}}, \tag{20}$$

and the anomalous magnetic moment on S^3 is $a_e^{S^3} = F_2$.

7.4 Convergence and results

The three-point vertex correction converges rapidly. Table 4 shows the per-level contributions for the external state at $n_{\mathrm{ext}} = 1$.

Table 4: Per-level contributions to the anomalous magnetic moment on S^3. External state $n_{\mathrm{ext}} = 1$, $j = 1/2$. The $n_{\mathrm{int}} = 0$ contribution vanishes by the self-energy structural zero (Theorem 4). Contributions drop by approximately 100× per successive allowed level pair.

n_{int}	ΔB	B_{cumul}	$F_2/[\alpha/(2\pi)]$	Status
0	0	0	—	$\Sigma(0) = 0$ structural zero
1	6.96×10^{-4}	6.96×10^{-4}	1.074	99% of final
2	3.76×10^{-6}	7.00×10^{-4}	1.080	
3	2.85×10^{-6}	7.02×10^{-4}	1.084	
4	1.34×10^{-7}	7.025×10^{-4}	1.084	
5	1.25×10^{-7}	7.027×10^{-4}	1.084	
6	1.25×10^{-8}	7.027×10^{-4}	1.084	
7	1.36×10^{-8}	7.027×10^{-4}	1.084	
8	2.00×10^{-9}	7.027×10^{-4}	1.084	

Three structural features are visible:

1. **Rapid convergence.** The $n_{\mathrm{int}} = 1$ level contributes 99% of the final value. Each subsequent allowed level pair contributes roughly 100× less than the previous one.

2. **Vertex parity filtering.** The $n_{\mathrm{int}} = 0$ contribution vanishes by Theorem 4 (the loop-photon vertex at $n_{\mathrm{ext}} = 1$, $n_{\mathrm{int}} = 0$ requires $q_{\mathrm{loop}} = 1$ and $1 + 0 + 1 = 2$ is even, forbidden). The dominant contribution comes from $n_{\mathrm{int}} = 1$ (the nearest allowed level to the external state). Higher even-n_{int} levels contribute but are suppressed relative to odd-n_{int} neighbors.

3. **Curvature correction.** The ratio $F_2/[\alpha/(2\pi)]$ converges to 1.084, not to 1.

7.5 Headline result

At $n_{\max} = 12$ (converged to 5 significant figures; the value is already stable at $n_{\max} = 8$):

$$\frac{F_2}{\alpha/(2\pi)} = 1.084, \tag{21}$$

with $B = 7.027 \times 10^{-4}$ and $V_{\text{magnetic}} = 0.5579$.

The anomalous magnetic moment on unit S^3 exceeds the flat-space Schwinger value by approximately 8.4%. This excess is a curvature correction: on a three-sphere of radius R, the electron propagator samples a compact geometry where the shortest-wavelength virtual photon modes are modified relative to flat space. In the flat-space limit $R \to \infty$, the curvature correction vanishes and $F_2/[\alpha/(2\pi)] \to 1$, recovering the Schwinger result [10] exactly.

7.6 Mode dependence

The anomalous magnetic moment depends strongly on the external electron mode. Table 5 shows $F_2/[\alpha/(2\pi)]$ computed at $n_{\text{ext}} = 1, \ldots, 5$.

Table 5: Mode-dependent anomalous magnetic moment on unit S^3. Each row reports the tree-level magnetic coupling V_{mag}, the one-loop correction B, and the normalized form factor ratio. The $n_{\text{ext}} = 1$ result uses $n_{\max} = 12$ (converged to 5 digits); $n_{\text{ext}} = 2$–5 use $n_{\max} = 5$ (converged to 3 digits via n^{-7} power-law tail). Exact algebraic forms for V_{mag} and the dominant $B(n_{\text{int}} = 1)$ are available at each n_{ext}; all expressions are sums of square roots with rational coefficients (no transcendentals).

n_{ext}	λ_{ext}	V_{mag}	B	$F_2/[\alpha/(2\pi)]$	vs. Schwinger
1	5/2	0.5579	7.027×10^{-4}	1.084	+8.4%
2	7/2	0.3893	2.772×10^{-4}	0.613	−39%
3	9/2	0.3001	1.296×10^{-4}	0.370	−63%
4	11/2	0.2445	6.940×10^{-5}	0.244	−76%
5	13/2	0.2064	4.139×10^{-5}	0.173	−83%

The physical mechanism is that the loop correction B drops faster with n_{ext} than the tree-level coupling V_{mag}. Across the five modes, B spans a factor of 17× while V_{mag} spans only 2.7×. The tree-level amplitude follows $V_{\text{mag}} \to (4/3)/|\lambda|$ asymptotically (Richardson-extrapolated to 4-digit agreement with 4/3), while the loop correction decays as a steeper power of $|\lambda|$. Higher-mode external electrons are "heavier" relative to the curvature scale ($\lambda = n + 3/2$ sets the effective mass), and the loop integral is progressively suppressed.

Comparison with Parker–Toms. The leading curvature correction to the anomalous magnetic moment in a general curved spacetime is [11]:

$$\frac{F_2}{\alpha/(2\pi)} = 1 + \frac{R_{\text{scalar}}}{12\,|\lambda|^2} + O(|\lambda|^{-4}), \tag{22}$$

where $R_{\text{scalar}} = 6$ on unit S^3 and $|\lambda|$ is the Dirac eigenvalue of the external state. At $n_{\text{ext}} = 1$ ($|\lambda| = 5/2$), this predicts $1 + 6/(12 \cdot 25/4) = 1 + 2/25 = 1.080$, which matches the computed 1.084 within 0.4%. The expansion parameter is $\xi = R_{\text{scalar}}/(12\,|\lambda|^2) \approx 0.08$ at $n_{\text{ext}} = 1$, small enough for the leading term to dominate.

However, the perturbative expansion in $1/|\lambda|^2$ diverges badly for the full mode-dependent curve: a two-term fit gives $c_1 \approx -20$, far from the Parker–Toms $c_1 = 1/2$. The discrepancy arises because the expansion (22) applies to a *single external mode at fixed curvature*, treating R/λ^2 as the small parameter. When we vary n_{ext} at fixed $R = 1$, we change not only $|\lambda|$ but also the angular structure of *every* CG coefficient in the vertex amplitude and the internal sum. The resulting mode dependence is non-perturbative in $1/|\lambda|^2$.

The five computed values are well described by a power law:

$$\frac{F_2}{\alpha/(2\pi)} = (5.91 \pm 0.36)\ |\lambda|^{-1.84\pm0.06}. \tag{23}$$

The fitted exponent is 2.8σ from -2; the forced $p = -2$ fit has twice the RMS residual. Analysis of the dominant internal-mode contribution ($n_{\text{int}} = 1$) shows its local log-log slope converging toward -2 ($-2.57 \to -2.26 \to -2.16 \to -2.11$ across $n_{\text{ext}} = 1, \ldots, 5$), but higher internal modes contribute an increasing fraction at larger n_{ext} (99% at $n_{\text{ext}} = 1$ vs. 69% at $n_{\text{ext}} = 3$), pulling the effective exponent above -2. Whether the true asymptotic exponent is exactly -2 (with finite-curvature corrections) or a distinct value remains an open question.

The power law implies $F_2 \to 0$ as $|\lambda| \to \infty$ on unit S^3. This is not inconsistent with the flat-space Schwinger limit: the flat-space limit is $R \to \infty$ at fixed physical mass (requiring $n \sim mR \to \infty$ simultaneously), not $|\lambda| \to \infty$ on a fixed-size sphere. Taking $|\lambda|$ large at fixed $R = 1$ corresponds to a heavy particle on a small sphere, where form-factor suppression is expected.

Verification: The computation uses exact `sympy` Clebsch–Gordan coefficients at every vertex, with no numerical approximation in the angular algebra. The propagator denominators are exact rationals ($|\lambda_n|^4$ and μ_q from Eqs. (1) and (5)). Computation scripts: `debug/alpha_g_minus_2_vertex.py`, `debug/alpha_g_minus_2_curvature_fit.py`.

7.7 Curvature expansion coefficients

The Parker–Toms expansion (22) at $n_{\text{ext}} = 1$ provides a high-precision extraction of the curvature coefficients. Writing

$$\frac{F_2}{\alpha/(2\pi)} = 1 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots, \qquad x \equiv \frac{1}{|\lambda|^2} = \frac{4}{25} \tag{24}$$

at $n_{\text{ext}} = 1$ ($|\lambda| = 5/2$), the leading coefficient $c_1 = R_{\text{scalar}}/12 = 1/2$ follows from the Parker–Toms theorem [11] for a Dirac field on unit S^3.

The converged $n_{\text{ext}} = 1$ spectral sum ($n_{\text{int}} = 0 \ldots 25$ with power-law tail correction, 13 significant figures) gives a corrected ratio of 1.0844529226552, hence $\delta \equiv F_2/[\alpha/(2\pi)] - 1 = 0.0844529226552$. Subtracting the leading term $c_1 x = 0.08$ and dividing by x^2:

$$c_2^{\text{apparent}} = \frac{\delta - c_1 x}{x^2} = 0.173942291220598 \pm 10^{-12}. \tag{25}$$

We take $c_2 = a + b\pi^2$ for rationals a, b as a PSLQ ansatz. (T9, Theorem 1, constrains only the integer-s values of ζ_{D^2}; it does not constrain every one-loop quantity on S^3 — the one-loop determinant $\zeta'_{D^2}(0) = 3\zeta(3)/(4\pi^2) + \frac{1}{2}\ln 2$ is a counterexample, and so (by a PSLQ identification, Sec. 6; no regression test pins it) is the one-loop self-energy $\Sigma(1)$, reported to carry Catalan G; beyond one loop, so are the vertex-parity-restricted sums of Theorem 3 (two-loop sunset), which carry G and $\beta(4)$ — so the ansatz is not a T9 consequence.)

PSLQ identification against the Paper 2 spectral invariants $B = 42$, $F = \pi^2/6$, $\Delta = 1/40$ [9] yields

$$\boxed{c_2 = \frac{2 - B\Delta - F\Delta - F/B}{5} = \frac{19}{100} - \frac{41\pi^2}{25200}}, \tag{26}$$

numerically 0.173942310300. The match to (25) is 8 digits (residual 1.9×10^{-8}), compared with 5.5 digits for the next-best candidate $(6\pi^2 - 58)/7$ and 3 digits for the purely rational 25/144.

Structural decomposition. The three Paper 2 invariants enter (26) through three bilinear combinations:

$$B\Delta = \frac{42}{40} = \frac{21}{20}, \qquad F\Delta = \frac{\pi^2}{240}, \qquad \frac{F}{B} = \frac{\pi^2}{252}. \tag{27}$$

The numerator $2 - B\Delta - F\Delta - F/B = \frac{19}{20} - \frac{41\pi^2}{5040}$ is rational plus a single π^2 term, as the ansatz assumes. The denominator 5 is a simple integer.

High-precision c_3 extraction. Extending the $n_{\text{ext}} = 1$ spectral sum to $n_{\text{int}} = 50$ with interleaved even/odd power-law tail correction (Hurwitz zeta analytic tail; $p_{\text{even}} \approx 7.62$, $p_{\text{odd}} \approx 7.71$) gives a converged ratio of 1.0844529207082, hence $\delta = 0.0844529207082$. The residual after subtracting $c_1 x + c_2 x^2$ is -2.44×10^{-9}, yielding

$$c_3 = \frac{\delta - c_1 x - c_2 x^2}{x^3} = -5.946(3) \times 10^{-7}, \tag{28}$$

nonzero at $200.9\,\sigma$ (uncertainty from the tail-correction power-law fit). The convergence is slow: the estimate drifts by $\sim 10^{-8}$ per five internal levels at $n_{\text{int}} = 50$, so the true asymptote lies near -5.95 to -5.97×10^{-7}. The ratio $|c_3 x^3|/|c_2 x^2| \approx 5.4 \times 10^{-6}$ confirms the expansion is rapidly converging but *does not* terminate at second order. PSLQ identification against the Paper-2 invariant basis $\{1, \pi^2, \pi^4, B\Delta, F\Delta, F/B, c_2, c_2^2, c_1 c_2, \Delta, \Delta^2, B\Delta^2, F\Delta^2, F^2/B\}$ at $n_{\text{int}} = 50$ finds no relation with integer coefficients bounded by 10^4; c_3 is not expressible in these invariants at current precision.

Multi-n_{ext} obstruction. The curvature coefficients in (24) are defined for the $j = 1/2$ sector of the external Dirac mode. At $n_{\text{ext}} \geq 2$, the Dirac level carries multiple total angular momentum sectors ($j = 1/2, 3/2, \ldots, n_{\text{ext}} + 1/2$) with distinct form factors. The $j = 1/2$ component is a $2/[(n_{\text{ext}} + 1)(n_{\text{ext}} + 2)]$ fraction of the total degeneracy, shrinking as $\sim 1/n^2$; extracting universal curvature coefficients from a full trace requires Wigner–Eckart decomposition into multipole form factors, which is not yet implemented. Consequently, the high-precision extraction is currently restricted to $n_{\text{ext}} = 1$, where $j = 1/2$ is the unique sector.

Paper 2 invariants in c_2: a numerical coincidence, not a derivation. Since α cancels in the ratio $F_2/[\alpha/(2\pi)]$, c_2 is a pure geometric invariant of the Dirac spectral sum on S^3, fitted in the ansatz form $a + b\pi^2$ with rationals a, b (an ansatz, not a T9 consequence; see above). The PSLQ identification (Eq. 26) writes a and b as a small-integer combination of B, F, Δ, but this should be read with caution.

First, the form $(c_0 - c_1 B\Delta - c_2 F\Delta - c_3 F/B)/c_4$ has more free small-integer coefficients than the two real numbers (a, b) it explains. Finding *some* small-integer combination matching $a + b\pi^2$ to high precision is expected by chance at this search size; the cross-invariant beats the next-best PSLQ candidate $(6\pi^2 - 58)/7$ by roughly 2.4 digits, which is suggestive but not conclusive.

Second, and decisively: the *same* form does not predict c_3. Eq. (28) measures $c_3 = -5.946(3) \times 10^{-7}$ at 200σ, and the extended Paper-2 invariant basis $\{1, \pi^2, \pi^4, B\Delta, F\Delta, F/B, c_2, c_2^2, c_1 c_2, \Delta, \Delta^2, B\Delta^2, F\Delta^2, F^2/B$ admits no PSLQ relation at integer coefficients bounded by 10^4. A genuine structural identity

should extend. This single-term failure is consistent with c_2's identification being a numerical coincidence between two natural Tier-A spectral invariants of the same manifold, both built from $|\lambda_n|$ and g_n, rather than evidence of shared structural origin.

A genuine bridge between the Paper 2 invariants and the curvature expansion would require deriving c_2 (and ideally c_3) from the Schwinger–DeWitt heat-kernel expansion of the spinor on S^3, with B, F, Δ appearing as identifiable terms in the expansion rather than via post-hoc PSLQ matching. This derivation is not done in this work and is flagged as an open question.

Computation scripts: `debug/g2_c2_verification.py`, `debug/g2_c2_diagnostic.py`.

8 One-Loop Vacuum Structure

The one-loop QED effective action involves traces of D^2. The Seeley–DeWitt heat kernel expansion on unit S^3 (four-component spinors; the printed values include the $(4\pi)^{-3/2}$ prefactor) gives

$$a_0 = \sqrt{\pi}, \qquad a_1 = -\sqrt{\pi}/2, \qquad a_k = 0 \ (k \geq 2) \tag{29}$$

(exact in `sympy`, from the Vassilevich curvature polynomials with the Lichnerowicz endomorphism $E = -R/4$; twice the two-component closed form (31) below). An earlier version printed $a_0 = a_1 = \sqrt{\pi}$, $a_2 = \sqrt{\pi}/8$, produced by a sign error in the production coefficient function (corrected 2026-10-02, with the function; `tests/test_qed_vacuum_polarization.py`). The standard flat-space one-loop vacuum polarization coefficient and QED beta function are

$$\Pi = \frac{1}{48\pi^2}, \qquad \beta(\alpha) = \frac{2\alpha^2}{3\pi}. \tag{30}$$

These are the standard flat-space values, entered here rather than computed from the S^3 data (an earlier version said they were reproduced from S^3 spectral data alone; the code returns the standard constants). They are consistent with the flat-space limit, since the gauge-field term of the four-dimensional a_2 is local.

8.1 SD two-term exactness for the Dirac heat kernel

The Dirac heat kernel on unit S^3, $K(t) = \mathrm{Tr}\, e^{-tD^2} = \sum_{n=0}^{\infty} 2(n+1)(n+2)\, e^{-(n+3/2)^2 t}$, has a Seeley–DeWitt expansion that terminates at exactly two terms:

$$K(t) = \frac{\sqrt{\pi}}{2}\, t^{-3/2} - \frac{\sqrt{\pi}}{4}\, t^{-1/2} + O(e^{-\pi^2/t}). \tag{31}$$

All higher SD coefficients $a_k = 0$ identically for $k \geq 2$. The proof uses the Jacobi theta modular identity $\theta_2(0, e^{-t}) = \sqrt{\pi/t}\ \theta_4(0, e^{-\pi^2/t})$, which maps the small-$t$ expansion to a large-π^2/t regime where $\theta_4 \to 1 + O(e^{-\pi^2/t})$. The vanishing of the higher terms up to an exponentially small remainder is Chamseddine and Connes' [32], stated there for $S^3 \times S^1$. Numerical verification at 80 dps (50,000 terms) confirms $a_2 = -3.88 \times 10^{-74}$ (numerical noise, not a physical coefficient), and the residual does not scale as t^2.

Setting $\mathrm{Tr}\, e^{-D^2/\Lambda^2} = K/\pi$ gives the depressed cubic

$$2\Lambda^3 - \Lambda = \frac{4(K/\pi)}{\sqrt{\pi}}, \tag{32}$$

with Cardano closed form $\Lambda = \sqrt[3]{A + \sqrt{A^2 - 1/216}} + \sqrt[3]{A - \sqrt{A^2 - 1/216}}$, where $A = (B + F - \Delta)/\sqrt{\pi} \approx 24.610$ (an earlier version printed a spurious factor 2; the value Λ_∞ below used the correct

A). This yields $\Lambda_\infty = 3.7102454679060528505$ (50 dps). At this cutoff, a sharp spectral truncation includes exactly $40 = \Delta^{-1}$ modes ($n = 0, 1, 2$), since $|\lambda_2| = 3.5 < 3.71 < 4.5 = |\lambda_3|$.

The SD ratio $a_0(D^2)/a_0(\Delta_{\mathrm{LB}}) = 2 = \mathrm{rank}(\text{spinor bundle})$ clarifies the supertrace sprint F1 result (ratio = 4): the discrepancy is the Dirac vs. Weyl counting convention ($4 = 2 \times 2$).

Structural conclusion: the Connes–Chamseddine framework provides an exact 2-term functional form for the spectral action but does *not* provide a mechanism to fix Λ. The value $\Lambda = 3.710\ldots$ is determined by inverting the spectral action at K/π, not selected autonomously.

8.2 Parametric spectral action on S^3_R and $\zeta_{\mathrm{unit}}(-k) = 0$

The two-term exactness theorem (31) has an equivalent formulation on the spectral-zeta side which makes the spectral action on the family $\{S^3_R : R > 0\}$ *exactly* two-term for any reasonable cutoff function f.

The $\zeta_{\mathrm{unit}}(-k) = 0$ identity. The spectral zeta of D^2 on unit S^3 admits a Hurwitz decomposition:

$$\zeta_{\mathrm{unit}}(s) = \sum_n 2(n{+}1)(n{+}2)\,(n+\tfrac{3}{2})^{-2s} = 2\,\zeta_H(2s{-}2,\,\tfrac{3}{2}) - \tfrac{1}{2}\,\zeta_H(2s,\,\tfrac{3}{2}). \tag{33}$$

At non-positive integers $s = -k$ ($k = 0, 1, 2, \ldots$), using $\zeta_H(-n, a) = -B_{n+1}(a)/(n+1)$:

$$\zeta_{\mathrm{unit}}(-k) = -\frac{2\,B_{2k+3}(3/2)}{2k+3} + \frac{B_{2k+1}(3/2)}{2(2k+1)}. \tag{34}$$

Theorem 5 ($\zeta_{\mathrm{unit}}(-k) = 0$ for all $k \geq 0$)**.** *The right-hand side of* (34) *vanishes identically for every non-negative integer k. Equivalently,*

$$4(2k+1)\,B_{2k+3}(3/2) = (2k+3)\,B_{2k+1}(3/2) \qquad \forall\, k \geq 0. \tag{35}$$

Proof sketch. Via the Mellin transform $\mathrm{Tr}\, e^{-tD^2} = (1/2\pi i)\int \Gamma(s)\,\zeta_{\mathrm{unit}}(s)\,t^{-s}\,ds$, shifting the contour through the right half-plane picks up contributions from poles of $\Gamma(s)\zeta_{\mathrm{unit}}(s)$. The two-term exactness theorem (31) states that the asymptotic expansion at small t has only the leading $t^{-3/2}$ and $t^{-1/2}$ terms, with no Taylor terms t^k ($k \geq 0$). The Taylor coefficients would otherwise be sourced by poles of $\Gamma(s)$ at $s = -k$ with residue $(-1)^k/k!$, giving contributions proportional to $\zeta_{\mathrm{unit}}(-k)$. Their absence forces $\zeta_{\mathrm{unit}}(-k) = 0$. Verified symbolically in `sympy` for $k = 0, 1, 2, 3, 4, 5$. □

Exact two-term spectral action on S^3_R. For any cutoff f with Mellin transform $\phi(s) = \int_0^\infty f(x)\,x^{s-1}\,dx$ that is meromorphic with poles only at non-positive integers (e.g. Gaussian e^{-x}, polynomial e^{-x^2}, sharp $\Theta(1-x)$), the spectral action on S^3_R is

$$S(R, \Lambda) := \mathrm{Tr}\, f(D^2/\Lambda^2) = \phi(3/2)\,(\Lambda R)^3 - \tfrac{1}{4}\,\phi(1/2)\,(\Lambda R) + O(\text{exp small}) \tag{36}$$

modulo exponentially small corrections in $(\Lambda R)^2$. The two-term form is *exact*: there are no power-law corrections at any order. Corrections from poles of $\phi(s)$ at non-positive integers vanish identically because each one is proportional to $\zeta_{\mathrm{unit}}(-k) = 0$.

This is a clean structural consequence of two-term exactness: on the family of round S^3_R at fixed cutoff function, the spectral action is EXACTLY of Einstein–Hilbert plus cosmological-constant form ($\phi(3/2)(\Lambda R)^3$ is the cosmological-constant slot via $\mathrm{Vol}(S^3_R) \propto R^3$; $-\frac{1}{4}\phi(1/2)\Lambda R$ is the Einstein–Hilbert slot via $R_{\mathrm{scalar}} \cdot V \propto R$).

Formal extremum and the CC spectral-action reading. Three natural cutoffs give:

Cutoff	$\phi(3/2)$	$\phi(1/2)$	$u_{\rm crit} = \sqrt{\phi(1/2)/(12\,\phi(3/2))}$
Gaussian e^{-x}	$\sqrt{\pi}/2$	$\sqrt{\pi}$	$1/\sqrt{6} \approx 0.408$
Polynomial e^{-x^2}	$\frac{1}{2}\Gamma(3/4)$	$\frac{1}{2}\Gamma(1/4)$	≈ 0.497
Sharp $\Theta(1-x)$	$2/3$	2	$1/2$

The asymptotic spectral action $S_{\rm asymp}(u) = Au^3 - Bu$ with $A = \phi(3/2) > 0$, $B = \frac{1}{4}\phi(1/2) > 0$ has a unique minimum at $u_{\rm crit} = \sqrt{B/(3A)} = O(1)$. Under the Connes–Chamseddine spectral action principle (Chamseddine–Connes 1997), $S_{\rm asymp}$ IS the effective classical action for the geometry; this gives a formal de-Sitter-like preferred radius $R_{\rm crit} \sim 1/\Lambda$, the GeoVac analog of CC's spectral-action-selects-metric mechanism.

Honest scope. The literal QM action $S_{\rm QM}(u) = \sum_n g_n f(\lambda_n^2/\Lambda^2)$ is monotonically increasing in u for all three cutoffs and has no literal extremum: at $u \to 0$ all modes are Boltzmann-suppressed by the cutoff and $S_{\rm QM} \to 0$, while at $u \to \infty$ many modes pass the cutoff and $S_{\rm QM} \to Au^3$. The asymptotic $S_{\rm asymp} = Au^3 - Bu$ agrees with $S_{\rm QM}$ to machine precision for $u \gtrsim 2$ (Gaussian); the formal extremum at $u_{\rm crit} \sim 0.4$–0.5 sits in the IR regime where the asymptotic does NOT faithfully represent the QM exact sum.

Under the CC spectral-action reading (asymptotic = classical action), GeoVac on S^3_R reproduces CC-style Einstein–Hilbert + cosmological constant with a stronger structural result: *the asymptotic terminates at two terms with no higher-curvature corrections at any order*, by Theorem 5. CC's continuum spectral action on a generic 4-manifold has higher-curvature corrections at every order; the GeoVac substrate eliminates them structurally. Sprint G1 driver: `debug/g1_spectral_action_S3_radius.py` and memo `debug/g1_spectral_action_S3_radius_memo.md`.

8.3 Extension to $S^3_R \times S^1_\beta$: factorization and 4D CC structure

The two-term exactness theorem (31) and the $\zeta_{\rm unit}(-k) = 0$ identity (34) PROPAGATE from S^3 to the thermal product $S^3_R \times S^1_\beta$ (Euclidean compactified time with antiperiodic boundary conditions for fermions). On the product, the spectral action acquires the exact 4D Connes–Chamseddine form: Einstein–Hilbert + cosmological constant with no higher-curvature corrections at any order.

Factorization. The Dirac operator on $S^3_R \times S^1_\beta$ is $D = \gamma^0 \otimes \partial_\tau + I \otimes D_{S^3}$ with $\{\gamma^0, D_{S^3}\} = 0$, so $D^2 = D^2_{S^3} \otimes I + I \otimes (-\partial_\tau^2)$. The heat kernel factorizes:

$$\operatorname{Tr} e^{-tD^2}\big|_{S^3_R \times S^1_\beta} = K_{S^3}(t, R) \cdot K_{S^1}(t, \beta). \tag{37}$$

$K_{S^3}(t,R) = K_{\rm unit}(t/R^2)$ has the two-term form (31). $K_{S^1}(t,\beta) = \sum_k e^{-t\omega_k^2}$ with Matsubara frequencies $\omega_k = 2\pi(k+1/2)/\beta$ admits Poisson resummation:

$$K_{S^1}(t,\beta) = \frac{\beta}{2\sqrt{\pi t}}\,\theta_4\Big(0, e^{-\beta^2/(4t)}\Big) = \frac{\beta}{2\sqrt{\pi t}} + O\Big(e^{-\beta^2/(4t)}\Big)\,. \tag{38}$$

Two-term exactness on the product. Both factors have clean asymptotic expansions modulo exp-small. Multiplying:

$$\operatorname{Tr} e^{-tD^2}\big|_{S^3_R \times S^1_\beta} = \frac{\beta R^3}{4}\,t^{-2} - \frac{\beta R}{8}\,t^{-1} + O\big(\text{exp small in } \min(R^2/t, \beta^2/t)\big)\,. \tag{39}$$

EXACTLY two power-law terms, no Taylor terms in t, no higher curvature corrections. The cross-product of "exp small" with "power-law" remains exp small.

Corollary 2 ($\zeta_{S^3\times S^1}(-k)=0$)**.** *For the product space $S^3_R\times S^1_\beta$ with antiperiodic boundary conditions for fermions, $\zeta_{S^3\times S^1}(-k)=0$ for all non-negative integers k.*

Proof. $\zeta_{S^3\times S^1}(-k)$ corresponds to the Taylor coefficient of $\operatorname{Tr} e^{-tD^2}$ at $t=0$ via Mellin inversion (cf. proof of Theorem 5). By (39), $\operatorname{Tr} e^{-tD^2}$ has no Taylor terms in t, only power-law t^{-2}, t^{-1} and exp-small corrections. □

4D Connes–Chamseddine form. For Gaussian cutoff $f(x)=e^{-x}$, the spectral action factorizes and matches the asymptotic to bit-exact precision at large Λ:

$$S(R,\beta,\Lambda)=\operatorname{Tr} e^{-D^2/\Lambda^2}=\frac{\beta R^3}{4}\,\Lambda^4-\frac{\beta R}{8}\,\Lambda^2+O(\text{exp small}). \tag{40}$$

This is the standard 4D CC form: Λ^4 cosmological constant + Λ^2 Einstein–Hilbert.

- Λ^4 coefficient: $\beta R^3/4=\operatorname{Vol}(S^3_R\times S^1_\beta)/(8\pi^2)$ where $\operatorname{Vol}=2\pi^2R^3\beta$.
- Λ^2 coefficient: $-\beta R/8=-\frac{1}{96\pi^2}\int R_{\text{scalar}}\sqrt{g}\,d^4x$ where $\int R_{\text{scalar}}\sqrt{g}\,d^4x=12\pi^2R\beta$ (only S^3 contributes; S^1 is flat).
- Higher Λ powers: exactly zero by Corollary 2.

Formal extremum and de Sitter vacuum. The asymptotic $S(R,\beta,\Lambda)=(\beta R^3/4)\Lambda^4-(\beta R/8)\Lambda^2$ has $\partial S/\partial R=0$ at $R_{\text{crit}}\Lambda=1/\sqrt{6}$ (identical to the one-parameter G1 extremum). At $R=R_{\text{crit}}$, $\partial S/\partial\beta<0$, so the joint minimum is at $\beta\to\infty$. This corresponds to the decompactified $S^1=\mathbb{R}_\tau$ limit, i.e., the **zero-temperature de Sitter vacuum** with radius $R_{\text{crit}}\sim 1/\Lambda$. Same standard de Sitter prediction (and same cosmological-constant-problem gap) as CC's continuum spectral action.

Stefan–Boltzmann lives in the exp-small. The thermal free energy $F_{\text{SB}}(T)=-7\pi^2/180\cdot VT^4$ for free fermions does NOT appear in the UV asymptotic (40). It lives in the exp-small corrections from higher Matsubara modes ($m\geq 1$ in Poisson resum). The CC spectral-action expansion (UV / asymptotic) and Stefan–Boltzmann thermal physics (IR / convergent) are structurally distinct contributions, as already known in CC's continuum literature.

Sprint G2 driver: `debug/g2_spectral_action_S3_x_S1.py` and memo `debug/g2_spectral_action_S3_x_S1_mem`

8.4 Spinor-bundle specificity: scalar and TT-tensor heat traces

The two-term exactness theorem (31), which underwrites both the parametric spectral action on S^3_R (§8.2) and the 4D thermal extension (§8.3), is *specific to the spinor bundle on S^3*. The scalar Laplacian and the Lichnerowicz Laplacian on transverse-traceless symmetric rank-2 tensors (the graviton sector) have heat traces with infinitely many Seeley–DeWitt coefficients, the standard CC continuum expansion.

This sharpens the structural reading of two-term exactness: it reflects the half-integer-shifted Dirac spectrum, NOT the geometry of S^3 per se. We derive closed-form heat traces for the three operators of interest.

Three closed forms. Dirac (Theorem (31), half-integer spectrum $(n+3/2)^2$ with multiplicity $2(n+1)(n+2)$, Jacobi θ_2 inversion):

$$K_{D^2}(t)\big|_{S^3} = \frac{\sqrt{\pi}}{2}\, t^{-3/2} - \frac{\sqrt{\pi}}{4}\, t^{-1/2} + O(e^{-\pi^2/t}). \tag{41}$$

Scalar Laplacian (integer spectrum $n(n+2)$ with multiplicity $(n+1)^2$, $n \geq 0$, Jacobi θ_3 inversion + exponential prefactor):

$$K_{\Delta}(t)\big|_{S^3} = \frac{\sqrt{\pi}}{4}\, \frac{e^t}{t^{3/2}} + O(e^{-\pi^2/t}). \tag{42}$$

TT-tensor Lichnerowicz (integer spectrum $n(n+2)-2$ for $n \geq 2$ with multiplicity $2(n-1)(n+3)$, Jacobi θ_3 inversion + discrete corrections from missing $n = 0, 1$ levels):

$$K_{\Delta_L^{TT}}(t)\big|_{S^3} = e^{3t}\left[\frac{\sqrt{\pi}}{2}\, t^{-3/2} - 4\sqrt{\pi}\, t^{-1/2} + 4\right] + 6e^{2t} + O(e^{-\pi^2/t}). \tag{43}$$

All three closed forms are verified bit-exactly (relative error $< 10^{-50}$ at $t \leq 0.1$) numerically in `debug/g3_graviton_lichnerowicz_S3.py`.

Seeley–DeWitt coefficient comparison. Expanding into standard form $K(t) = (4\pi t)^{-3/2} \sum_k a_k t^k$:

Theorem 6 (Scalar SD closed form on S^3)**.** *For the scalar Laplacian on unit S^3, all Seeley–DeWitt coefficients are nonzero and given by*

$$a_k^{\Delta} = \frac{2\pi^2}{k!} \qquad \forall k = 0, 1, 2, \ldots \tag{44}$$

Proof. From (42), expanding $e^t = \sum_k t^k/k!$ gives $K_\Delta(t) = \sum_k (\sqrt{\pi}/(4k!))\, t^{k-3/2}$. Matching to $K_\Delta = (4\pi t)^{-3/2} \sum_k a_k t^k$ yields $a_k = (4\pi)^{3/2}\sqrt{\pi}/(4k!) = 2\pi^2/k!$. □

For the TT-tensor, the heat trace (43) contains BOTH half-integer-power terms (from the smooth Weyl asymptotic with e^{3t} prefactor) AND integer-power terms (from the constant 4 and the discrete correction $6e^{2t}$). The standard SD form captures only the half-integer-power content; the integer-power content is "non-SD," reflecting the truncation of the spectrum to $n \geq 2$.

Half-integer-power coefficients from $e^{3t}[(\sqrt{\pi}/2)\, t^{-3/2} - 4\sqrt{\pi}\, t^{-1/2}]$:

$$\begin{aligned} a_0^{TT} &= 4\pi^2, \\ a_1^{TT} &= -20\pi^2 = (\sqrt{\pi}/2)(3) - 4\sqrt{\pi} \text{ scaled by } (4\pi)^{3/2}, \\ a_2^{TT} &= -78\pi^2, \quad \text{etc.} \end{aligned}$$

All nonzero, with $a_k^{TT} \propto \pi^2 \cdot p_k(3)$ for explicit polynomials p_k from the e^{3t} expansion.

Structural distinction.

Operator	Spectrum shift	Two-term exact?	Higher SD coefficients
D^2 Dirac	half-integer	YES (Paper 28)	$a_k = 0$ for $k \geq 2$
Δ scalar	integer	NO	$a_k = 2\pi^2/k!$ (Thm 6)
Δ_L^{TT} Lichnerowicz	integer (truncated)	NO	all nonzero + non-SD pieces

Implication. The clean CC Einstein–Hilbert + cosmological constant structure on S^3 that emerges from the spectral action (36) (Sprint G1) and (40) (Sprint G2) is a feature of the *spinor bundle*, not of S^3 as a manifold. Gravitons (spin-2 TT modes) on S^3 have a heat trace with infinite higher-curvature SD coefficients plus non-SD discrete corrections from the spectrum's $n \geq 2$ cutoff. The graviton sector on the GeoVac substrate, were it constructed via CC-style metric variation, would inherit the standard CC continuum expansion with corrections at every order, NOT the clean two-term form of the Dirac sector.

This is consistent with the framework's structural-skeleton scope: clean structure lives in the Dirac sector sourced by the Fock projection (Paper 23 rigidity), with gravity-side coupling constants remaining external calibration data.

Sprint G3 driver: `debug/g3_graviton_lichnerowicz_S3.py` and memo `debug/g3_graviton_lichnerowicz_S3`

8.5 Graviton diagnostic: physical $(1,1)$ modes on the truncated substrate

The structural-test phase of G1–G3 establishes that the spectral action on the GeoVac substrate has EXACT Einstein–Hilbert + cosmological constant form on the Dirac sector but inherits standard CC continuum behavior on the spin-2 sector. The natural follow-up question: does the discrete substrate host graviton-like perturbations of D at the operator level, or are gravitons structurally blocked?

Setup. Perturb $D = D_0 + \epsilon V$ where $V \in \mathrm{Herm}(\mathcal{H}_{n_{\max}})$. Expand the spectral action: $S[D_0+\epsilon V] = S[D_0] + \epsilon S^{(1)}[V] + \frac{1}{2}\epsilon^2 S^{(2)}[V,V] + O(\epsilon^3)$. The quadratic form $S^{(2)}$ is diagonal in the D_0 eigenbasis.

Gauge–physical classification. Modes $V = i[X, D_0]$ for Hermitian X are tangent to the gauge orbit $D_0 \to U D_0 U^*$, and the spectral action is invariant under unitary conjugation. Therefore $S^{(2)}$ eigenvalues on gauge modes measure the curvature of the gauge orbit along the linear direction, not physical kinetic content. For the truncated CH Dirac, the gauge subspace equals the cross-sector off-block Hermitian matrices, and within-sector modes are physical.

Spin-2 content. The $SU(2)_L \times SU(2)_R$ irreps at level n are $((n+1)/2, n/2)$ for +chirality and $(n/2, (n+1)/2)$ for −chirality. Within-sector tensor products $\rho \otimes \rho$ contain the graviton irrep $(j_L, j_R) = (1,1)$ for $n \geq 1$.

Proposition 1 (Physical $(1,1)$ modes on the substrate)**.** *At every $n_{\max} \geq 1$, the second variation $S^{(2)}$ has PHYSICAL (within-sector) eigenmodes carrying $(j_L, j_R) = (1,1)$ irrep content with POSITIVE eigenvalue*

$$A_\lambda = a_\lambda \left(\frac{4\lambda^2}{\Lambda^4} - \frac{2}{\Lambda^2} \right), \quad a_\lambda = e^{-\lambda^2/\Lambda^2}, \tag{45}$$

for each within-sector block at CH eigenvalue $|\lambda|$. Numerically verified by 5-point finite-difference stencil to rel diff $\sim 10^{-9}$ at $\Lambda^2 = 6$, $n_{\max} \in \{1,2,3\}$:

$\lvert\lambda\rvert$	*blocks containing* $(1,1)$	A_λ
5/2	$S_3 \otimes S_3$, $S_4 \otimes S_4$	+0.127
7/2	$S_5 \otimes S_5$, $S_6 \otimes S_6$	+0.133
9/2	$S_7 \otimes S_7$, $S_8 \otimes S_8$	+0.066

Eigenvalues are stable across $n_{\max}$ (no sign changes), and the $|\lambda|$-dependence reflects the standard Gaussian-regulated kinetic behavior of the CC spectral action.

Honest scope. Proposition 1 establishes the NECESSARY condition for graviton dynamics on the discrete substrate: physical (1, 1)-irrep eigenmodes exist and propagate with positive kinetic eigenvalue. SUFFICIENT conditions — Fierz–Pauli kinetic structure separating TT graviton (2 polarizations) from longitudinal/scalar modes (7 of the 9 within $(1,1)$), gauge invariance of the TT projector, propagator structure, and continuum graviton emergence via Paper 38-style state-space GH convergence — are not verified here; the J-blindness leg has since been completed (Paper 51, §Fierz–Pauli + the J-blindness theorem), and the remaining sufficiency legs constitute the extended G6 program. The S^3 Fock projection rigidity (Paper 23) fixes the background metric; graviton candidates here are operator-level perturbations of D, not direct metric perturbations.

Structural reading. Sprint G3 (§8.4) established that the spin-2 sector inherits standard CC continuum expansion. Sprint G6-Diag-Full establishes that this expansion's graviton modes are accessible on the discrete substrate at the necessary-condition level. The clean structure of G1/G2 (Einstein–Hilbert + cosmological constant exactly on Dirac sector) does NOT extend to gravitons; gravitons inherit CC continuum behavior with infinite higher-curvature corrections. Substrate-level existence of $(1,1)$ modes confirms gravitons are not structurally blocked.

DOF count is a continuum-limit statement (Sprint GD-1). The reduction of the 9-dimensional $(1,1)$ irrep to the 2 transverse-traceless physical polarizations does *not* occur at finite truncation. The continuum count is standard: under the diagonal $SU(2)$, $(1,1) = \text{spin-}2 \oplus \text{spin-}1 \oplus \text{spin-}0 = 5 \oplus 3 \oplus 1$, and diffeomorphism gauge (4) plus constraints (3) leave 2 TT. On the discrete substrate, however, the inner-automorphism (diffeomorphism) gauge tangent $i[X, D_0]$ has entries $(\lambda_b - \lambda_a)X_{ab}$, which vanish *within* a sector ($\lambda_a = \lambda_b$); the inner gauge is therefore *entirely cross-sector* (verified zero within-sector gauge entries at $n_{\max} = 1, 2, 3$). Within a within-sector $(1,1)$ block the gauge does not act, so all 9 modes are non-inner-gauge at finite truncation. The continuum diffeomorphism gauge acts within the tensor field at each point; the discrete inner gauge connects different λ-sectors. These reconcile only as $n_{\max} \to \infty$, when the cross-sector directions assemble into continuum diffeomorphisms that remove 7 of the 9 within-sector modes. Thus the graviton DOF reduction is a *continuum-limit phenomenon* (Paper 38-style state-space GH), not an internal-to-the-block projection — which is the structural reason the Fierz–Pauli decomposition is the extended G6 task and not a sprint-scale one. Source: `debug/sprint_gd1_graviton_dof_memo.md`.

Which irrep carries the TT graviton (Sprint GD-4). A rep-theory diagnostic sharpens *which* sector the necessary-condition test should examine. The helicity of an $SO(4) = SU(2)_L \times SU(2)_R$ irrep (j_L, j_R) is $|j_L - j_R|$; the TT spin-2 graviton is helicity ± 2, hence lives in $|j_L - j_R| = 2$ reps (the TT tensor harmonics on S^3 are $(k{+}2, k) \oplus (k, k{+}2)$, lowest $(2,0) \oplus (0,2)$). The $(1,1)$ irrep that Proposition 1 tested has $|j_L - j_R| = 0$ — *helicity zero*, the scalar/trace-class sector. The within-sector blocks $\text{sector}_i \otimes \text{sector}_i$ do contain the helicity-2 reps $(2,0) \oplus (0,2)$ (verified present at $n = 1, 2, 3, 4$), but these were not examined. *Caveat:* in Connes–Chamseddine the physical graviton enters through the $\delta D \leftrightarrow h_{\mu\nu}$ dictionary, which can shift irrep assignments — the *local* polarization space is $(1,1)$ while the *global* TT harmonic modes are $|\Delta j| = 2$. So GD-4 does not refute the necessary-condition result; it identifies a concrete gap: the extended G6 must either examine the $|\Delta j| = 2$ δD-modes directly or establish the dictionary showing the $(1,1)$ bilinear carries the physical graviton. Together with the GD-1 finding, the extended G6 has two sharply identified must-dos: (i) test the helicity-2 sector and fix the dictionary; (ii) realize the $9 \to 2$ gauge reduction in the continuum (state-space GH) limit. Source: `debug/sprint_gd4_graviton_irrep_memo.md`.

Helicity-2 modes are positive too (Sprint GD-5). The necessary-condition half of GD-4's concern is resolved directly. The coefficient $A_\lambda = a(\lambda)(4\lambda^2/\Lambda^4 - 2/\Lambda^2)$ is a function of the sector eigenvalue λ alone (the spectral action sees only the degenerate-sector eigenvalue), hence *irrep-blind* within a block: every irrep — $(1,1)$, $(2,0)$, $(0,2)$, ... — inherits the same A_λ. Therefore the helicity-2 (TT-graviton-class) reps carry the *same positive* A_λ as the $(1,1)$ ($+0.127, +0.133, +0.066$ at $n = 1,2,3$, $\Lambda^2 = 6$). So Proposition 1's positivity is *not* confined to the scalar/trace sector; it extends to the actual TT-graviton irrep class. *Honest scope:* the irrep-blindness that makes this immediate is exactly the G6 approximation — the *sufficient* condition (a TT kinetic structure *distinct* from the trace, Fierz–Pauli) requires lifting A_λ to an irrep-resolved second variation, the extended-G6 task where the blindness must break. GD-5 closes the necessary condition for the TT sector; it does not close the TT-vs-trace distinction. Source: `debug/sprint_gd5_helicity2_memo.md`.

Sprint G6-Diag-Full driver: `debug/g6_diag_full.py`, companion first-pass `debug/g6_diag_quadratic_form.py` scoping memo `debug/g6_scoping_memo.md`, and full memo `debug/g6_diag_full_memo.md`.

8.6 Cigar spectral action and Bekenstein–Hawking entropy

The gravity arc beyond the structural-test phase (§8.2–§8.5) includes the gravity-side targets of Bekenstein–Hawking entropy ($S = A/(4G)$ from the cigar) and decompactified de Sitter vacuum (the $\beta \to \infty$ limit of §8.3). Sprint TD Track 4 (CLAUDE.md §2) already reproduced $T_H = 1/(8\pi M)$ from regularity of the cigar's τ-circle. Sprint G4 first-pass establishes the entropy via the standard Gibbons–Hawking–York thermodynamic derivation and identifies the load-bearing structural mechanism.

Cigar geometry. Euclidean Schwarzschild with metric $ds^2 = (1-2M/r)\,d\tau^2 + (1-2M/r)^{-1}dr^2 + r^2 d\Omega_2^2$ outside the horizon $r > 2M$. Regularity at $r = 2M$ forces τ-period $\beta = 8\pi M$. Horizon area $A = 4\pi r_h^2 = 16\pi M^2$.

Bekenstein–Hawking from spectral action. After Gibbons–Hawking subtraction of asymptotic flat-space reference, the Euclidean on-shell action of Schwarzschild is

$$I_E^{\mathrm{Schw}} = \frac{A}{4G}, \tag{46}$$

generated by the horizon boundary heat-kernel coefficient at the smooth conical tip. Bulk $R = 0$ (Ricci-flat) and the asymptotic boundary cancels against the flat-space subtraction. The thermodynamic entropy is

$$S_{\mathrm{thermo}} = -I_E + \beta\,\frac{\partial I_E}{\partial \beta} = \frac{A}{4G} = 4\pi M^2/G, \tag{47}$$

the Bekenstein–Hawking result. Symbolically verified in `debug/g4_cigar_BH_entropy.py` (sympy, residual exact zero).

Load-bearing mechanism. The structural piece producing $A/(4G)$ is the boundary heat-kernel coefficient at the smooth conical horizon, generating the GHY $(1/8\pi G)\oint K\sqrt{h}\,d^3x = A/(4G)$. This is the standard CC continuum derivation.

GeoVac discrete substrate. The GeoVac substrate (CH Dirac on truncated S^3 at finite $n_{\max}$) does NOT natively describe the cigar: the cigar's spatial section is $\mathbb{R} \times S^2$ (non-compact radial, S^2 instead of S^3), and the horizon is a smooth conical tip in the (τ, r) plane. Constructing the discrete

spectral triple analog on the cigar requires substantial new infrastructure: spatial S^2 truncation, non-compact radial regularization, horizon tip preservation under discretization, and verification that the boundary heat-kernel coefficient generates the correct GHY term. This is the full G4 program.

Sprint G4 first-pass: `debug/g4_cigar_BH_entropy.py` and `debug/g4_cigar_BH_entropy_memo.md`.

8.7 Decompactified $S^3 \times \mathbb{R}_\tau$ and de Sitter vacuum

Sprint G2 (§8.3) found that the joint extremum of the spectral action on $S^3_R \times S^1_\beta$ sits at $R_{\text{crit}}\Lambda = 1/\sqrt{6}$ (in R) and $\beta \to \infty$ (in β), corresponding to the zero-temperature de Sitter vacuum. Sprint G5 makes this $\beta \to \infty$ limit explicit by decompactifying the temporal direction.

Heat-kernel decompactification. On $S^3 \times \mathbb{R}_\tau$, the heat kernel factorizes per unit length of $\mathbb{R}_\tau$:

$$K_{\text{full}}(t, R) = K_{S^3}(t, R) \cdot K_{\mathbb{R}}(t), \quad K_{\mathbb{R}}(t) = \int \frac{dk}{2\pi}\, e^{-k^2 t} = \frac{1}{2\sqrt{\pi t}}. \tag{48}$$

This matches $K_{S^1}(t, \beta)/\beta$ from §8.3 in the $\beta \to \infty$ limit. Combined with the two-term exactness of K_{S^3}:

$$K_{\text{full}}(t, R) = \frac{R^3}{4}\, t^{-2} - \frac{R}{8}\, t^{-1} + O(\text{exp small}). \tag{49}$$

Two-term exactness propagates to the decompactified case.

Spectral action density. For Gaussian cutoff $f(x) = e^{-x}$:

$$s(R, \Lambda) := K_{\text{full}}(1/\Lambda^2, R) = \frac{R^3}{4}\Lambda^4 - \frac{R}{8}\Lambda^2. \tag{50}$$

De Sitter vacuum.

$$\frac{\partial s}{\partial R} = 0 \implies R_{\text{crit}}\Lambda = \frac{1}{\sqrt{6}}, \tag{51}$$

identical to G1 (§8.2). Minimum action density:

$$s_{\min} = s(R_{\text{crit}}, \Lambda) = -\frac{\Lambda}{12\sqrt{6}}. \tag{52}$$

Symbolically verified (sympy, $-\sqrt{6}\Lambda/72 = -\Lambda/(12\sqrt{6})$, residual exact zero).

Physical reading. The framework's preferred zero-temperature de Sitter vacuum has Planck-scale radius $R_{\text{crit}} \sim 1/\Lambda$ and vacuum action density $-\Lambda/(12\sqrt{6})$. This is the standard CC spectral-action-principle prediction, with the well-known cosmological-constant-scale gap. The decompactification is clean: G2's $\beta \to \infty$ minimum is manifest in G5 as the $\mathbb{R}_\tau$ vacuum.

Sprint G5: `debug/g5_decompactified_S3_x_R.py` and `debug/g5_decompactified_memo.md`.

8.8 Newton constant, cosmological constant, and background extremality

Sprint G7 consolidates the gravity arc's connection to standard Connes–Chamseddine gravity by deriving the framework's gravitational parameters in standard QFT language and verifying a structural background-extremality consistency check.

Background extremality. From the G6-Diag-Full physical $(1,1)$-irrep eigenvalues $A_\lambda = a_\lambda(4\lambda^2/\Lambda^4 - 2/\Lambda^2)$, the Gaussian-weighted integral over the spectrum vanishes exactly:

$$\int_0^\infty (4u^2 - 2)\, e^{-u^2}\, du = 4 \cdot \frac{\sqrt{\pi}}{4} - 2 \cdot \frac{\sqrt{\pi}}{2} = 0. \tag{53}$$

This confirms the standard background-extremality condition: the spectral action is stationary at the CH Dirac background, with sub-leading perturbations giving the actual graviton kinetic structure. Sympy-verified.

Newton constant and cosmological constant from G2. Matching G2's spectral action coefficients $(\beta R^3/4)\Lambda^4 - (\beta R/8)\Lambda^2$ (40) against the standard Einstein–Hilbert + cosmological constant action $S_{EH} = -(1/16\pi G_{\text{eff}}) \int (R_{\text{scalar}} - 2\Lambda_{cc})\sqrt{g}\, d^4x$ using $\text{Vol}(S^3_R \times S^1_\beta) = 2\pi^2 R^3 \beta$ and $\int R_{\text{scalar}} \sqrt{g} = 12\pi^2 R\beta$:

Proposition 2 (Newton constant + cosmological constant)**.** *The Gaussian-cutoff spectral action on $S^3_R \times S^1_\beta$ identifies the framework's effective gravitational parameters as*

$$G_{\text{eff}} = \frac{6\pi}{\Lambda^2}, \qquad \Lambda_{cc} = 6\Lambda^2. \tag{54}$$

Planck-scale identification. Setting $\Lambda \to M_{\text{Planck}}$: $G_{\text{eff}} = 6\pi/M^2_{\text{Planck}} = 6\pi G_{\text{Newton}}$ (Planck-scale Newton's constant); $\Lambda_{cc} = 6M^2_{\text{Planck}} = 6/G_{\text{eff}}$ (Planck-scale cosmological constant).

Cosmological constant scale gap. The predicted ratio $\Lambda_{cc}/M^2_{\text{Planck}} = 6$ vs. the observed cosmological dark energy $\Lambda_{cc}/M^2_{\text{Planck}} \sim 10^{-120}$ gives a ~ 120 orders of magnitude gap. This is the standard CC cosmological-constant-scale problem, inherited from continuum Connes–Chamseddine and not GeoVac-specific. The framework's discrete substrate does not modify this leading-order prediction. Standard interpretation: cosmological constant scale is Class 1 calibration data (see `memory/external_input_three_class_partition.md`).

Master Mellin engine tagging (a census and a reading). Gravity content is M2 (Seeley–DeWitt heat-kernel coefficients $\sqrt{\pi} \times \bigoplus_k \pi^{2k} \cdot \mathbb{Q}$, i.e. the pure-Tate ring once the $\sqrt{\pi}$ heat-trace normalisation is divided out; see Paper 18 §III.7). G2's normalised coefficients $1/(8\pi^2)$ and $-1/(96\pi^2)$ are M2-derived; the $(4\pi)^{-d/2}$ normalisation places them in $\pi^{-2} \cdot \mathbb{Q}$, outside the $k \geq 0$ pure-Tate ring. On the Dirac substrate, M2 terminates at two terms by (31); on scalar/tensor sectors, M2 is the full infinite series (§8.4, eq (44)). The gravity arc's content is thus the M2 sub-mechanism of the master Mellin engine, with substrate-specific termination on the Dirac bundle.

Sprint G7 driver: `debug/g7_extremality_newton.py`, memo `debug/g7_extremality_newton_memo.md`.

8.9 S^2 Dirac and the cigar's spatial spinor structure

Sprint G4-1 (first sub-sprint of the full G4 program) tests whether the framework's two-term exactness on S^3 Dirac (§8.3, Theorem 5) propagates to the S^2 Dirac that appears in the cigar's spatial section $\mathbb{R}_+ \times S^2_r$. Answer: NO.

S^2 **Dirac spectrum.** Camporesi-Higuchi for S^d Dirac: $|\lambda_n^{S^d}| = n + d/2$. For $d = 2$:

$$|\lambda_n^{S^2}| = n + 1, \quad \text{(INTEGER shift)} \tag{55}$$

$$g_n^{\text{Dirac}}|_{S^2} = 4(n+1). \tag{56}$$

In contrast, S^3 Dirac has $|\lambda_n^{S^3}| = n+3/2$ (HALF-INTEGER shift) with $g_n^{\text{Dirac}}|_{S^3} = 2(n+1)(n+2)$.

Heat trace asymptotic. The S^2 Dirac heat trace is $K(t) = 4\sum_{u\geq 1} u\, e^{-u^2 t}$. The leading asymptotic

$$K_{S^2}^{\text{Dirac}}(t) \sim \frac{2}{t} + O(1) \tag{57}$$

matches the standard CC form $(4\pi t)^{-1}\cdot\dim_S \cdot \text{Vol}(S^2)$ with $\dim_S = 2$, $V = 4\pi$. Numerically verified to rel diff 10^{-5} at $t = 10^{-4}$ (`debug/g4_1_S2_dirac.py`).

First subleading. The Seeley-DeWitt coefficient $a_1 = \dim_S \cdot (R_{\text{scalar}}/6 - E_{\text{Lich}}) \cdot V$ on unit S^2 ($R = 2$, $E = R/4 = 1/2$):

$$a_1^{S^2,\text{Dirac}} = 2(1/3 - 1/2)\cdot 4\pi = -\frac{4\pi}{3} \neq 0. \tag{58}$$

The numerical extraction $\lim_{t\to 0}[K(t) - 2/t] = a_1/(4\pi) = -1/3$ matches at rel diff $\sim 10^{-3}$ across $t = 10^{-1}$ to 10^{-3}.

Structural distinction. The two-term exactness theorem on S^3 Dirac comes from the half-integer spectrum shift, allowing Jacobi θ_2 inversion to give exactly two power-law terms. S^2 Dirac (integer shift) uses Jacobi θ_3 inversion + exponential prefactor, giving the standard CC infinite Seeley-DeWitt series with nonzero a_k at every order.

This is consistent with Sprint G3 (§8.4): the scalar Laplacian on S^3 (integer spectrum from $n = 0$) also has full SD series with $a_k = 2\pi^2/k!$ (Theorem 6). The pattern crystallizes as:

Operator	Spectrum shift	Higher SD coefficients
S^3 Dirac	half-integer $n + 3/2$	$a_k = 0$ for $k \geq 2$
S^3 scalar Δ	integer $n(n+2)$	$a_k = 2\pi^2/k!$ (G3)
S^2 Dirac	integer $n + 1$	$a_1 = -4\pi/3 \neq 0$, $a_k \neq 0$

Implication for G4 full. The cigar's spatial section $\mathbb{R}_+ \times S_r^2$ has the S^2 component inheriting standard CC continuum behavior with higher-curvature corrections at all orders. The discrete-substrate BH entropy derivation cannot exploit a clean two-term form on the spatial S^2; it must follow standard CC heat-kernel asymptotic structure. The load-bearing piece for $A/(4G)$ remains the horizon boundary a_2 coefficient (§8.6).

The framework's clean two-term exactness is *specific to the S^3 substrate sourced by the Fock projection*. It does not extend to other dimensions or the gravity arc's cigar geometry.

Sprint G4-1 driver: `debug/g4_1_S2_dirac.py` and memo `debug/g4_1_S2_dirac_memo.md`.

8.10 Conical defect / replica method for BH entropy

Sprint G4-2 (second sub-sprint of G4 full) builds on §8.9 to derive the Bekenstein–Hawking entropy via the standard conical-defect / replica method on the cigar's near-horizon geometry $D_\alpha^2 \times S_{r_h}^2$.

Setup. The cigar's near-horizon geometry is $ds^2 = d\rho^2 + \rho^2\, d\varphi^2 + r_h^2\, d\Omega_2^2$ with $\varphi \in [0, 2\pi)$ (smooth tip) or $\varphi \in [0, 2\pi\alpha)$ (conical singularity for $\alpha \neq 1$). The Bekenstein–Hawking entropy is extracted via the replica formula

$$S_{BH} = -\left.\frac{dI_E(\alpha)}{d\alpha}\right|_{\alpha=1}. \tag{59}$$

Conical heat trace contribution. The Sommerfeld/Cheeger formula for a 2D cone with apex angle $2\pi\alpha$:

$$K_{\text{cone}}(t) = K_{\text{smooth bulk}}(t) + \frac{1}{12}\left(\frac{1}{\alpha} - \alpha\right) + O(t). \tag{60}$$

The $(1/12)(1/\alpha - \alpha)$ is the conical-defect tip contribution, vanishing at smooth $\alpha = 1$.

Combined heat trace. Using G4-1's $K_{S^2_{r_h}}(t) \sim 2r_h^2/t - r_h^2/3 + O(t)$ at small t, the conical-tip contribution to the cigar's heat trace is:

$$K_{\text{tip}}(t, \alpha) = \frac{1}{12}\left(\frac{1}{\alpha} - \alpha\right)\left[\frac{2r_h^2}{t} - \frac{r_h^2}{3} + O(t)\right] \tag{61}$$

Replica entropy derivation. For Gaussian cutoff, $S = K(1/\Lambda^2)$ gives the conical-tip spectral action contribution

$$I_E^{\text{conical}}(\alpha, \Lambda, r_h) = \frac{r_h^2\Lambda^2}{6}\left(\frac{1}{\alpha} - \alpha\right) + O(1/\Lambda^2). \tag{62}$$

The α-derivative at $\alpha = 1$:

$$\left.\frac{dI_E^{\text{conical}}}{d\alpha}\right|_{\alpha=1} = \frac{r_h^2\Lambda^2}{6} \cdot (-2) = -\frac{r_h^2\Lambda^2}{3}. \tag{63}$$

Proposition 3 (Bekenstein–Hawking entropy from conical replica)**.** *The spectral action's conical-defect contribution at the cigar tip generates the Bekenstein–Hawking entropy via the replica formula:*

$$S_{BH} = -\left.\frac{dI_E^{\text{conical}}}{d\alpha}\right|_{\alpha=1} = \frac{r_h^2\Lambda^2}{3} = \frac{A\Lambda^2}{12\pi}, \tag{64}$$

where $A = 4\pi r_h^2$. Matching to $S_{BH} = A/(4G_N)$ gives $G_N = 3\pi/\Lambda^2$.

Factor-of-2 calibration with G7. The Newton constant from §8.8 (G2's spectral action matching) is $G_{\text{eff}} = 6\pi/\Lambda^2$. The factor-of-2 difference from $G_N = 3\pi/\Lambda^2$ is *not* a scalar-vs-Dirac conical-defect coefficient (the 2D Dirac and scalar cones share the coefficient $(1/12)(1/\alpha - \alpha)$; Fursaev–Miele 1996, confirmed by the G4-4c discrete extraction): it is a bookkeeping artifact forced by Wald's theorem — two-term exactness implies pure Einstein gravity, so action-G and entropy-G are the *same* constant, and the 2 lives in a 4D-bulk vs factorized-2D × 2D normalization mismatch to be closed by convention audit (Paper 51, Wald-forces-entropy-relation remark). The structural emergence of $A \cdot \Lambda^2$ from the conical tip is the load-bearing finding; the precise prefactor calibration is downstream.

Discrete-substrate analog. For G4-3 onwards (the extended program):

- Define α-deformation of the discrete substrate at the tip
- Compute the conical contribution from the discrete Dirac
- Verify the discrete substrate reproduces the $(1/\alpha - \alpha)$ form
- Apply replica method to extract S_{BH} from the discrete substrate spectral action

The conceptual framework is now in place; the full implementation program follows.
Sprint G4-2 driver: `debug/g4_2_conical_replica.py` and memo `debug/g4_2_conical_replica_memo.md`.

8.11 Cutoff-function dependence of the gravity predictions (Sprint G8)

Sprints G2 and G7 derived $G_{\text{eff}} = 6\pi/\Lambda^2$ and $\Lambda_{cc} = 6\Lambda^2$ assuming the Gaussian cutoff $f(x) = e^{-x}$. Sprint G8 generalizes these to arbitrary cutoff functions and identifies which framework predictions are cutoff-independent (structural) versus cutoff-dependent (calibration).

Mellin transform of the cutoff. For a general cutoff function f, the spectral content is captured by the Mellin transform:

$$\phi(s) := \int_0^\infty f(x)\, x^{s-1}\, dx. \tag{65}$$

The relevant moments for spectral action matching are $\phi(1)$ (Einstein-Hilbert scale) and $\phi(2)$ (cosmological-constant scale).

General formulas. From the G2 spectral action on $S^3_R \times S^1_\beta$,

$$S(R, \beta, \Lambda) = \phi(2) \cdot \frac{\beta R^3}{4}\Lambda^4 - \phi(1) \cdot \frac{\beta R}{8}\Lambda^2, \tag{66}$$

matching against Einstein-Hilbert plus cosmological constant gives:

$$G_{\text{eff}}(f, \Lambda) = \frac{6\pi}{\phi(1)\,\Lambda^2}, \quad \Lambda_{cc}(f, \Lambda) = \frac{6\,\phi(2)}{\phi(1)}\,\Lambda^2, \tag{67}$$

$$R_{\text{crit}}\,\Lambda = \sqrt{\frac{\phi(1)}{6\,\phi(2)}}. \tag{68}$$

Table 6: Gravity predictions for three cutoff functions.

Cutoff $f(x)$	$\phi(1)$	$\phi(2)$	$G_{\text{eff}}\Lambda^2$	Λ_{cc}/Λ^2
Gaussian e^{-x}	1	1	6π	6
Polynomial e^{-x^2}	$\sqrt{\pi}/2$	1/2	$12\sqrt{\pi}$	$6/\sqrt{\pi}$
Sharp $\Theta(1-x)$	1	1/2	6π	3

Three cutoff examples.

No cutoff-independent gravity prediction. Direct algebra on (67)–(68) shows that every combination of G_{eff} and Λ_{cc} depends on $\phi(1)$ and/or $\phi(2)/\phi(1)$:

- $G_{\text{eff}} \cdot \Lambda^2 = 6\pi/\phi(1)$ (depends on $\phi(1)$)
- $\Lambda_{cc}/\Lambda^2 = 6\,\phi(2)/\phi(1)$ (depends on ratio)
- $G_{\text{eff}} \cdot \Lambda_{cc} = 36\pi\,\phi(2)/(\phi(1)^2\Lambda^2)$ (depends on both)
- $R_{\text{crit}}\Lambda = \sqrt{\phi(1)/(6\,\phi(2))}$ (depends on ratio)

Class 1 calibration data. Per the framework's external-input three-class partition (see CLAUDE.md §1.7 and project memory), the cutoff function f is *Class 1 calibration data* alongside Yukawas, $Z_2/\delta m$ multi-loop renormalization data, and the Born rule. The framework determines:

- the *structural form* of $S[D]$ (spectrum, Seeley-DeWitt coefficients),
- the *proportionality* of G_{eff} and Λ_{cc} to specific ϕ-moments,
- the *functional form* of $G_{\text{eff}}(f)$, $\Lambda_{cc}(f)$, $R_{\text{crit}}(f)$.

The framework does *not* determine:

- the numerical value of G_{eff} or Λ_{cc} in absolute units,
- which cutoff function to use.

Consistency with standard CC. This is the standard situation in the Chamseddine-Connes spectral action [31, 33]: the cutoff function is external input, the framework determines the structural shape.

Structural vs numerical scope. Sprints G1, G2, G3, G4, G5, G6-Diag-Full produced *structural* findings (closed forms, theorems, mechanisms) that are cutoff-independent. Sprint G7 produced *numerical* predictions (specific values of G_{eff}, Λ_{cc}) that depend on the Gaussian cutoff choice. Sprint G8 clarifies that the framework's gravity-arc predictions are best understood as:

- structural skeleton: cutoff-independent, determined by GeoVac's spectral structure;
- numerical calibration: cutoff-dependent, proportional to ϕ-moment combinations.

Sprint G8 driver: `debug/g8_cutoff_dependence.py`; memo: `debug/g8_cutoff_dependence_memo.md`.

8.12 Discrete warped-product substrate for the cigar (Sprint G4-3 opening)

Sprints G1 through G8 produced the continuum gravity arc: closed-form spectral action on S^3_R and $S^3 \times S^1_\beta$, two-term exactness theorems, S^2 Dirac structure, Bekenstein-Hawking entropy via the conical-defect / replica method, and the cutoff-function dependence of the gravity predictions. Sprint G4-3 *opens* the discrete-substrate gravity arc: rebuild the gravity-arc results on GeoVac's discrete substrate rather than at the continuum level.

This sprint is *scoping + first-pass*: define the discrete substrate, identify the load-bearing pieces, and specify the sub-sprint sequence G4-3a through G4-6.

Discrete substrate. For the cigar's near-horizon limit $ds^2 = d\rho^2 + \rho^2 d\phi^2 + r_h^2 d\Omega_2^2$ (constant warp $r = r_h$):

$$\mathcal{G}_{\text{cigar}} = \mathbb{Z}_+(a)|_{N_\rho} \times \mathbb{Z}/N_\phi \times \text{Fock}(S^2, l_{\max}) \tag{69}$$

where:

- $\mathbb{Z}_+(a)|_{N_\rho} = N_\rho$-truncation of $\mathbb{Z}_+$ with lattice spacing a, $\rho_k = ka$;
- $\mathbb{Z}/N_\phi$ = discrete ϕ-circle, N_ϕ sites;
- $\text{Fock}(S^2, l_{\max})$ = standard S^2 Fock projection with $l \le l_{\max}$ angular cutoff.

The conical-defect parameter is $\alpha = N_\phi/N_0$ where N_0 is the reference smooth-tip count.

Constant-warp factorization. At $r(\rho) = r_h$ constant, the Laplacian factorizes:

$$\Delta_{\text{cigar}} = \Delta_{D^2}(\rho, \phi) + \frac{1}{r_h^2}\,\Delta_{S^2}.$$

The two factors act on independent sectors, so by construction:

$$K_{\text{cigar}}(t) = K_{D^2}(t) \cdot K_{S^2_{r_h}}(t).$$

This factorization holds at the operator level on the discrete substrate.

Sub-sprint sequence. G4-3 opens the extended discrete-substrate track:

- **G4-3a** (this): constant-warp substrate + scoping
- G4-3a-cleanup: Hermitian discrete polar Laplacian (engineering)
- G4-3b: variable warp $r(\rho)$ for asymptotic Schwarzschild
- G4-3c: discrete conical-defect deformation (N_ϕ sweeps)
- G4-3d: continuum-limit heat-kernel asymptotics verification
- G4-4: warped Dirac spectrum
- G4-5: discrete replica method
- G4-6: full discrete-substrate S_{BH} derivation

Estimated commitment: a substantial multi-stage program, well beyond sprint scale.

Honest scope of G4-3a. G4-3a is *scoping*: substrate definition, constant-warp factorization at operator level, and sub-sprint plan. The sprint-scale driver uses a naive non-Hermitian discrete polar Laplacian; numerical heat-trace values are quantitatively wrong but the structural framework (factorization, conical parameterization, continuum-limit targets) is intact. A Hermitian discrete radial Laplacian on $L^2(\rho\, d\rho)$ via standard symmetrization is sprint-scale engineering for G4-3a-cleanup.

Connection to G4-2 continuum derivation. G4-2 derived $S_{BH} = A\Lambda^2/(12\pi)$ at the continuum level via the conical-defect / replica method. G4-3 onwards rebuilds this on the discrete substrate. The target is to show:

1. Discrete substrate has analog conical-defect contribution $\to (1/12)(1/\alpha - \alpha)$ in continuum limit
2. Discrete Dirac on warped substrate $\to$ G4-1 S^2 Dirac at constant warp
3. Discrete replica method gives $S_{BH} \to A\Lambda^2/(12\pi)$

Connection to G8 cutoff dependence. The Λ^2 prefactor in S_{BH} depends on the cutoff function f per G8. The discrete substrate inherits this calibration: any discrete-substrate S_{BH} derivation will have prefactor $\propto \phi(2)/\phi(1)$. The cutoff dependence is structural, not a feature of G4-3 specifically.

Gravity-arc status. Continuum gravity arc: G1–G8 complete (sprint-scale closure). Discrete-substrate gravity arc: G4-3a opens; G4-3b through G4-6 run to the full S_{BH} derivation on the discrete substrate. This is the natural commitment for the next major project arc.

Sprint G4-3a driver: `debug/g4_3_warped_substrate.py`; memo: `debug/g4_3_warped_substrate_memo.md`.

9 Two-Loop Identified Constant: $S_{\min}$

Correction note (round-2 dossier, 2026-06-10). The earlier subsection heading "Extended PSLQ irreducibility" and the claim that $S_{\min}$ is "confirmed irreducible" were a basis-coverage artifact: all 15 PSLQ attempts (47-element and 100-element bases at 150–200 dps) never contained the level-2 weight-5–8 monomials ($\pi^4 \ln 2$, $\pi^6 \ln 2$, $\mathrm{Li}_5(1/2)$, $\mathrm{Li}_4(1/2)\ln 2$, $\zeta(3)\ln^2 2$, ...) that the correct room requires. The present text replaces those claims with the identified decomposition; see driver `debug/smin_dossier_round1.py` and the round-1 dossier `debug/smin_dossier_round1_memo.md` for the complete reduction chain.

The CG-weighted two-loop sunset sum is:

$$S_{\min} = \sum_{k=1}^{\infty} T(k)^2, \qquad T(k) = 2\,\zeta(2, k+\tfrac{3}{2}) - \tfrac{1}{2}\,\zeta(4, k+\tfrac{3}{2}). \tag{70}$$

This is a depth-2 multiple Hurwitz zeta value at half-integer shift $a = 3/2$, intrinsic to the Dirac spectrum on S^3.

9.1 200-digit computation

$S_{\min}$ is computed to 200 verified digits using `mpmath.nsum` with Levin u-transform acceleration. The first 80 digits agree with the Sprint 5 three-method verification (direct summation + explicit tail; Levin u-transform; Euler–Maclaurin 26-term asymptotic):

$$S_{\min} = 2.47993\,69380\,34222\,55441\,35795\,00082\,93821\,44688\ldots \tag{71}$$

9.2 Motivic identification: Hoffman double t-values

$S_{\min}$ decomposes *exactly* into Hoffman double t-values — classical level-2 alternating MZVs, periods of $\mathcal{MT}(\mathbb{Z}[1/2])$. Notation: $\lambda(s) = (1-2^{-s})\zeta(s) = \sum_{o\geq 1,\,\text{odd}} o^{-s}$; $t(b,c) = \sum_{o_2>o_1\geq 1,\,\text{odd}} o_2^{-b} o_1^{-c}$ (Hoffman [20] descending convention: first index b on the larger summation variable).

8-t-value decomposition (verified to residual 1.8×10^{-129} at 130 dps, re-verified at 260 dps):

$$S_{\min} = 64\big[t(2,1) - t(4,1) - 3\,t(2,2) + 3\,t(4,2) - t(2,3) + t(4,3) + 3\,t(2,4) - 3\,t(4,4)\big] + 32\sum_{s=3}^{8} d_s\left(\lambda(s) - 1 - 3^{-s}\right), \tag{72}$$

where the diagonal coefficients are $d_3 = 1$, $d_4 = -3$, $d_5 = -2$, $d_6 = 6$, $d_7 = 1$, $d_8 = -3$.

Stuffle reductions (exact algebraic identities from the ordering decomposition $\lambda(b)\lambda(c) = t(b,c) + t(c,b) + \lambda(b+c)$):

$$t(2,2) = \tfrac{1}{2}\big(\lambda(2)^2 - \lambda(4)\big), \tag{73}$$

$$t(4,4) = \tfrac{1}{2}\big(\lambda(4)^2 - \lambda(8)\big), \tag{74}$$

$$t(2,4) + t(4,2) = \lambda(2)\lambda(4) - \lambda(6). \tag{75}$$

(The antisymmetric combination $t(2,4) - t(4,2)$ enters $S_{\min}$ with coefficient zero, so only the reducible symmetric combination appears.)

Odd-weight depth-1 reductions (verified at $\geq 1\times 10^{-130}$; both are verbatim rows of Hoffman [20] Appendix A):

$$t(2,1) = -\tfrac{7}{16}\,\zeta(3) + \tfrac{1}{8}\,\pi^2\ln 2, \tag{76}$$

$$t(2,3) = -\tfrac{31}{64}\,\zeta(5) + \tfrac{1}{16}\,\pi^2\,\zeta(3). \tag{77}$$

Remaining odd-weight components — identified. PSLQ at 220 dps against the full conjectural level-2 bases (dim 8 at weight 5; 19 product monomials at weight 7) identifies both remaining components (residuals 2.8×10^{-221} and 3.6×10^{-222}; `debug/data/s3_pslq_stageA.json`):

$$t(4,1) = \tfrac{1}{96}\,\pi^4\ln 2 - \tfrac{1}{64}\,\pi^2\zeta(3) - \tfrac{31}{64}\,\zeta(5), \tag{78}$$

$$t(4,3) = \tfrac{1}{128}\,\pi^4\zeta(3) - \tfrac{5}{128}\,\pi^2\zeta(5) - \tfrac{127}{256}\,\zeta(7). \tag{79}$$

Both identified forms are likewise verbatim rows of Hoffman [20] Appendix A ($t(4,1)$ directly; $t(4,3)$ after converting his t-basis) — the PSLQ runs independently rediscovered the published table (grounding verified against the source PDF, 2026-06-11). Assembling the full decomposition yields the explicit closed form

$$S_{\min} = 8\pi^2\ln 2 - \tfrac{2}{3}\pi^4\ln 2 - 3\pi^2\zeta(3) + \tfrac{1}{2}\pi^4\zeta(3) - \tfrac{5}{2}\pi^2\zeta(5) + \tfrac{\pi^6}{4} - \tfrac{3}{2}\pi^4 - \tfrac{\pi^8}{96}, \tag{80}$$

verified against the Hurwitz-tail anchor at 130 dps (residual 1.7×10^{-129}; `debug/smin_final_assembly.py`). Two structural features are exact: $\zeta(7)$ cancels against the $\lambda(7)$ diagonal term, and all rational constants cancel to zero. Hence $S_{\min}$ lies in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$: the two-loop constant carries *no irreducible depth-2 content at all* — every double t-value in its decomposition reduces to depth ≤ 1.

Basis-coverage audit of the prior "irreducible" claim. The 47-element basis (all 15 PSLQ failures at 150 dps) and the 100-element basis (200 dps) both contained zero level-2 monomials in the weight-5–8 rooms. Specifically, none of $\pi^4\ln 2$, $\pi^6\ln 2$, $\mathrm{Li}_4(1/2)\ln 2$, $\mathrm{Li}_5(1/2)$, $\mathrm{Li}_6(1/2)$, $\mathrm{Li}_7(1/2)$, $\zeta(3)\ln^2 2$, … were included. Of the conjectural weight-5 level-2 basis (dimension 8), those tests contained only 2 out of 8 elements ($\zeta(5)$ and $\pi^2\zeta(3)$); at weight 7, approximately 4 out of 21. The "PSLQ-irreducible" label was a coverage artifact, not a statement about motivic depth.

9.3 Inner photon sum: closed-form channel count

The inner photon sum $I(n_1, n_2) = \sum_{q\,\text{allowed}} W(n_1, n_2, q) \cdot d_q^T/\mu_q^p$ (at $p = 1$: $d_q^T/\mu_q = 1$) counts the total SO(4) CG-weighted vertex channels. Four structural properties:

(i) $I(0, n_2) = 0$ for all n_2 (structural zero: vertex parity at $n_{\text{ext}} = 0$ forces $2n_2$ odd, impossible).

(ii) $I(n_1, n_2) = I(n_2, n_1)$ (symmetric).

(iii) $I(n, n) = 2n - 2 = 2(n - 1)$ for $n \geq 1$.

(iv) **Closed form** (proved in dossier round-1c; verified against `geovac/qed_vertex.py` for all $n_1, n_2 \leq 8$ in the frozen test `tests/test_smin_decomposition.py`; the archived dossier run swept to ≤ 12):

$$I(n_1, n_2) = 2\min(n_1, n_2) - 1 - [n_1 = n_2], \qquad n_1, n_2 \geq 1. \tag{81}$$

9.4 Exact sunset relation

The closed form (81) gives, by direct substitution $I = 2\min - 1 - \delta$ into the double sum:

$$S_{\text{sunset}} = 2\, S_{\min} - P^2 - Q, \tag{82}$$

where $S_{\text{sunset}} = \sum_{n_1, n_2 \geq 1} I(n_1, n_2)\, \phi(n_1)\, \phi(n_2)$ is the CG-exact two-loop sunset,

$$P = D(4) - \tfrac{64}{81}, \quad D(4) = \pi^2 - \tfrac{\pi^4}{12}, \tag{83}$$

$$Q = 4\,\zeta\big(4, \tfrac{5}{2}\big) - 2\,\zeta\big(6, \tfrac{5}{2}\big) + \tfrac{1}{4}\,\zeta\big(8, \tfrac{5}{2}\big)\,, \tag{84}$$

and the numerical value $S_{\text{sunset}} = 3.894516205197841044 55\ldots$ is verified at 60 dps in `tests/test_smin_decomposit`
The correction terms P^2 and Q are depth-1 elements of $\mathcal{MT}(\mathbb{Z})$ (pure-Tate, π-even); $S_{\min}$ itself reduces to depth ≤ 1 (Eq. (80)), so the two-loop sunset carries no irreducible depth-2 content.

10 Three-Loop Results

10.1 Factorized $O(N^3)$ algorithm

The three-loop iterated sunset has chain topology: three electron lines (n_1, n_2, n_3) connected by two photon lines (q_1, q_2), with vertex selection at each junction. The chain topology permits factorization into partial sums:

$$S^{(3)} = \sum_{n_2=0}^{N} \frac{g_{n_2}}{|\lambda_{n_2}|^a} \cdot L(n_2) \cdot R(n_2), \tag{85}$$

where

$$L(n_2) = \sum_{\substack{n_1, q_1 \\ \text{allowed}}} \frac{W(n_1, n_2, q_1)\; g_{n_1}\; d_{q_1}^T}{|\lambda_{n_1}|^a\; \mu_{q_1}^p}, \qquad R(n_2) = \sum_{\substack{n_3, q_2 \\ \text{allowed}}} \frac{W(n_2, n_3, q_2)\; g_{n_3}\; d_{q_2}^T}{|\lambda_{n_3}|^a\; \mu_{q_2}^p}. \tag{86}$$

$W(n_1, n_2, q)$ is NOT symmetric in (n_1, n_2), so L and R are computed independently. This reduces the five-fold sum $O(N^5)$ to $O(N^3)$ (each of L, R is $O(N^2)$ precomputed, then the outer sum is $O(N)$).

Verification: The factorized algorithm matches the direct $O(N^5)$ sum to $< 10^{-78}$ at $n_{\max} = 10, 15$ across four parameter combinations (default $a = 4, p = 1$; vertex-restricted; higher exponents $a = 6, p = 2$; with and without CG weights). The factorized algorithm at $n_{\max} = 50$ completes in < 60 seconds. Tests: 7 in `TestFactorizedMatchesDirect` and `TestFactorizedConvergence` (all passing; the frozen factorized-vs-direct gate is 10^{-25} — the $< 10^{-78}$ agreement quoted above is the measured value at the working precision, re-confirmed 2026-07-04).

10.2 Convergence analysis (corrected 2026-06-11)

At default exponents ($a = 4$, $p = 1$) with CG weights, the raw partial sums converge far too slowly for direct evaluation. An empirical power-law fit over $N \in [20, 120]$ gave

$$\Delta S^{(3)}(N) \approx 270 \cdot N^{-1.31}, \tag{87}$$

but the closure-sprint analysis showed the fitted exponent is an artifact: by the channel-count closed form (81) the inner partial sums grow as $L(n_2) = 4 \ln n_2 + O(1)$, so the increments follow $(\ln N)^2/N^2$ — a log-modulated law whose local log-slope $2 - 2/\ln N$ runs through 1.3–1.6 over the fit window and masquerades as a power law. The true remaining tail at $N = 120$ is ≈ 7 units, i.e. *zero* reliable digits from raw partial sums (cf. Table 9: the $n_{\max} = 50$ partial sum is 15.23 against a limit of 31.57).

The reliable direct evaluation is a rigorous bracket with no series acceleration anywhere: the factorized sum has positive terms, so partial sums with the closed-form full-range inner sum give a strict lower bound, and the explicit inequalities $L(m) \le 4(\ln m + 1)$, $\phi(m) \le 2/(m+1)^2$ give an integral tail bound. At $N = 4 \times 10^6$,

$$S^{(3)} \in [31.57063,\ 31.57300] \tag{88}$$

(driver `debug/s3_closure_anchor.py`). The corrected decomposition assembly (Sec. 10.4) gives the canonical value, inside the bracket:

$$S^{(3)} = 31.57256\,12075\,12022\,75476\,19295\,45069\ldots \tag{89}$$

10.3 PSLQ identification attempts (historical)

At the fast-converging exponents $a = 8$, $p = 2$, approximately 20 reliable digits are available at $n_{\max} = 50$. PSLQ with an 18-element depth-3 MZV basis (including $S_{\min}$, $\zeta(3)^2$, $\pi^2 S_{\min}$, $G \cdot S_{\min}$, and all cross-products with π^2, $\zeta(3)$, G, $\beta(4)$) returns no relation. This route is superseded by the decomposition closure below.

The unrestricted three-loop sum factorizes exactly as $D(a)^3$: at $a = 4$, $D(4)^3 = (\pi^2 - \pi^4/12)^3 \approx 5.379$, purely π^{even}.

10.4 Decomposition closure: the four trailing-1 identifications

Notation: $t_3(b_1, b_2, b_3) = \sum_{o_1 > o_2 > o_3 \ge 1 \text{ odd}} o_1^{-b_1} o_2^{-b_2} o_3^{-b_3}$, the Hoffman triple t-value in the same descending convention as $t(b, c)$. The channel-count closed form (81) expands the CG-weighted chain sum into 12 triple t-values, 26 double t-values, and λ/boundary terms (exact in Fraction arithmetic at every truncation; `tests/test_s3_decomposition.py`).

Two errors in the stage-1 numerics were diagnosed and closed on 2026-06-11 (memo `debug/sprint_s3_closure_r`) both with one root cause: Levin series acceleration silently mis-converges on summands with logarithmic factors, with *precision-dependent* error. (i) The evaluator for trailing argument $b_3 = 1$

fed the odd-harmonic factor $\widetilde{H}(j) \sim \frac{1}{2}\ln j$ into the accelerator; $t_3(2,1,1)$ was off by -4.7×10^{-4} at 130 dps and -2.9×10^{-4} at 220 dps, poisoning the earlier evaluation $S^{(3)} \approx 30.615$ through its table coefficient $+2048$ — the corrected shift $+0.9572$ is accounted for term-by-term to 1.7×10^{-15}. (ii) The same disease invalidated the factorized-anchor evaluation 30.2197 ($L(m) \sim 4\ln m$); the bracket (88) replaces it. The repaired evaluator uses Abel summation over the harmonic weight, $t_3(b_1,b_2,1) = \sum_{k\geq1}(2k-1)^{-1}A(k)$ with $A(k) = \sum_{j\geq k} a_j$ — both sums log-free. With corrected values the stuffle witnesses pass at 5×10^{-216} (R2; stage-1 residual 6.4×10^{-12}) and 8×10^{-217} (R6; was 9.8×10^{-20}); the stage-1 residuals decompose exactly into the individual evaluator errors.

PSLQ at 220/340 dps against the complete conjectural level-2 weight-homogeneous bases identifies all four previously pending components (residuals 2.6×10^{-216} to 7.3×10^{-339} in the archived dossier run; the frozen regression `tests/test_s3_decomposition.py` re-gates the identifications at 10^{-30}/45 dps):

$$t_3(2,1,1) = \tfrac{1}{2}\mathrm{Li}_4(\tfrac{1}{2}) + \tfrac{1}{48}\ln^4 2 + \tfrac{1}{24}\pi^2\ln^2 2 - \tfrac{19}{5760}\pi^4, \tag{90}$$

$$t_3(4,1,1) = \tfrac{1}{192}\pi^4\ln^2 2 - \tfrac{1}{64}\pi^2\ln 2\,\zeta(3) - \tfrac{11}{322560}\pi^6 - \tfrac{1}{2}\,t(5,1) - \tfrac{7}{128}\zeta(3)^2, \tag{91}$$

$$t_3(3,2,1) = \tfrac{3}{64}\pi^2\ln 2\,\zeta(3) - \tfrac{1}{9216}\pi^6 - \tfrac{1}{2}\,t(5,1) - \tfrac{49}{256}\zeta(3)^2, \tag{92}$$

$$\begin{aligned} t_3(3,4,1) = {} & \tfrac{1}{768}\pi^4\ln 2\,\zeta(3) + \tfrac{5}{128}\pi^2\ln 2\,\zeta(5) + \tfrac{61}{2903040}\pi^8 - \tfrac{1}{256}\pi^2\zeta(3)^2 \\ & + \tfrac{1}{2}\,t(5,3) - \tfrac{1}{2}\,t(7,1) - \tfrac{217}{512}\zeta(3)\zeta(5). \end{aligned} \tag{93}$$

Every component of the $S^{(3)}$ decomposition except the weight-10 graded piece ($W_{10} = -2048\,t_3(4,3,3) - 1024\,t_3(4,4,2) - 512\,t_2(8,2) - 1536\,t_2(4,6)$, the full four-term graded piece; level-2 basis dimension 89, not executed; parity heuristic predicts depth ≤ 2) is now an identified element of the level-2 ring, with the genuine depth-2 generators $t(5,1)$ (weight 6) and $t(5,3)$, $t(7,1)$ (weight 8) entering at coefficients $\pm\frac{1}{2}$. **Realized motivic depth at three loops: ≤ 2 on every identified component** — one rung above $S_{\min}$'s depth ≤ 1, all catalogued, nothing new minted. Falsifier: `tests/test_s3_decomposition.py` (including the rigorous-bracket regression guard for the Levin/log failure mode).

Full assembly and the weight-10 block. Assembling the complete decomposition with all identified components (symbolic assembly verified against the canonical value (89) at residual 4.1×10^{-197}) yields a stronger statement than the entry-level reductions: *every depth-2 generator cancels in the total.* $t(5,1)$, $t(5,3)$, $t(7,1)$, and $\mathrm{Li}_4(\frac{1}{2})$ all carry total coefficient zero, as does standalone $\zeta(11)$ — the $S^{(3)}$ echo of $S_{\min}$'s exact $\zeta(7)$ cancellation. The identified part of $S^{(3)}$ is a polynomial in π^2, $\ln 2$, and odd zeta values alone. The remaining weight-10 block reduces, by exact Fraction Gaussian elimination on the complete weight-10 quasi-shuffle closure (rank 157 on 255 words; identity verified numerically at 7×10^{-59}), to

$$\begin{aligned} W_{10}/256 = {} & 4\lambda(10) - \lambda(2)\lambda(8) - 3\lambda(4)\lambda(6) - 4\lambda(3)\,a(4,3) - 2\lambda(4)\,a(4,2) \\ & + a(8,2) + 4\,a(7,3) - 3\,a(6,4) - 4\,t_3(2,4,4) - 8\,t_3(3,3,4), \end{aligned} \tag{94}$$

where $a(b,c) := t(b,c) - t(c,b)$ and $\lambda(s) = (1-2^{-s})\zeta(s)$; the lower-weight $a(4,2)$, $a(4,3)$ are identified objects, leaving *five* surviving generators: the antisymmetric doubles $a(8,2)$, $a(7,3)$, $a(6,4)$ and the triples $t_3(2,4,4)$, $t_3(3,3,4)$. A structural obstruction protects this block from stuffle methods: the depth-≤ 3 relation content of the quasi-shuffle closure saturates already at the $\lambda\cdot t$ product stage, so no stuffle-product engineering at any depth can reduce it further.

The weight-10 block closed (2026-06-12). The missing input is the Charlton–Hoffman symmetry theorem [24] — level-2 regularized-double-shuffle content invisible to the stuffle-only system. Its generating-series form (their Thm. 2.21) at depth $m = 3$, $\underline{\phi} = \underline{0}$, extracted at the coefficient of $y_1^{a-1}y_2^{b-1}y_3^{c-1}$ by exact-Fraction series algebra, gives the reversal-pair sums with every product term explicit (each instance used was verified numerically at 10^{-78} or better; driver `debug/s3_w10_ident.py`, 24/24 gates):

$$t_3(4,4,2) + t_3(2,4,4) = \tfrac{5}{6144}\pi^4\zeta(3)^2 - \tfrac{5}{256}\pi^2\zeta(3)\zeta(5) - \tfrac{1}{256}\pi^2\zeta(5,3) + \tfrac{137}{99532800}\pi^{10}, \tag{95}$$

$$t_3(4,3,3) + t_3(3,3,4) = \tfrac{37}{12288}\pi^4\zeta(3)^2 - \tfrac{25}{1024}\pi^2\zeta(3)\zeta(5) - \tfrac{889}{2048}\zeta(3)\zeta(7) - \tfrac{1}{256}\pi^2\zeta(5,3) + \tfrac{251}{58060800}\pi^{10}, \tag{96}$$

where $\zeta(5,3) = \sum_{m>n\geq 1} m^{-5}n^{-3}$ is the classical weight-8 depth-2 MZV generator (conjecturally irreducible over products of single zetas), entering through the weight-8 layer of the theorem's ζ-side product terms. The pair *differences* follow from four ordinary stuffles $t(x)\,t_2(y,z)$, $(x,y,z) \in \{(2,4,4),(4,2,4),(3,3,4),(3,4,3)\}$, whose pairwise combinations eliminate the auxiliary triples $t_3(4,2,4)$, $t_3(3,4,3)$. Solving the two 2×2 systems identifies all four weight-10 triples individually:

$$t_3(2,4,4) = \tfrac{1}{4}a(8,2) - \tfrac{1}{4}a(6,4) - \tfrac{1}{6144}\pi^4\zeta(3)^2 - \tfrac{5}{512}\pi^2\zeta(3)\zeta(5) - \tfrac{1}{512}\pi^2\zeta(5,3) + \tfrac{373}{174182400}\pi^{10}, \tag{97}$$

$$t_3(3,3,4) = \tfrac{1}{4}a(7,3) - \tfrac{1}{4}a(6,4) + \tfrac{1}{12288}\pi^4\zeta(3)^2 + \tfrac{5}{1024}\pi^2\zeta(3)\zeta(5) - \tfrac{889}{4096}\zeta(3)\zeta(7) - \tfrac{1}{512}\pi^2\zeta(5,3) + \tfrac{251}{116121600}\pi^{10}, \tag{98}$$

each gated against its independently computed 200-digit value at residual $\leq 6\times 10^{-197}$ (likewise $t_3(4,4,2)$, $t_3(4,3,3)$).

Collapse theorem and the full closed form. Substituting Eqs. (97)–(98) into the full $S^{(3)}$ assembly, the three antisymmetric doubles $a(8,2)$, $a(7,3)$, $a(6,4)$ *cancel identically*: the explicit a-content of the weight-10 corollary is exactly absorbed by the a-content of the two triples (symbolic verification in the frozen falsifier). The weight-10 block therefore contributes

$$S^{(3)} - \text{Ident} \;=\; -7\pi^4\zeta(3)^2 + 70\pi^2\zeta(3)\zeta(5) + 6\pi^2\,\zeta(5,3) + \tfrac{179}{170100}\pi^{10}, \tag{99}$$

where Ident is the identified part of the symbolic assembly, and the complete closed form

$$S^{(3)} \;\in\; \mathbb{Q}\big[\pi^2,\ \ln 2,\ \zeta(3),\ \zeta(5),\ \zeta(7),\ \zeta(9),\ \zeta(5,3)\big] \tag{100}$$

is explicit, with $\zeta(5,3)$ the *only* constant beyond the depth-≤ 1 ring, at coefficient $6\pi^2$. The closed form reproduces the canonical value (89) at residual 1.2×10^{-198} (200 dps). Two structural readings: (i) the transient-generator cancellation pattern of the symbolic assembly ($t(5,1)$, $t(5,3)$, $t(7,1)$, $\mathrm{Li}_4(\frac{1}{2})$ all cancel) extends through weight 10 — the level-2 antisymmetric doubles $a(8,2)$, $a(7,3)$, $a(6,4)$ also carry total coefficient zero, and the *value* of $S^{(3)}$ is independent of these still-unreduced generators; (ii) the genuinely new depth-2 content at three loops enters on the *level-1* side ($\zeta(5,3) \in \mathcal{MT}(\mathbb{Z})$), while the level-2-specific content (the $\ln 2$ sector) stays at realized depth ≤ 1 — new depth arrives from level 1, new level stays at depth ≤ 1. Falsifier: `tests/test_s3_w10_identification.py` (symmetry-pair sums, individual triples, symbolic a-collapse, full closed form at 50 and 200 dps with $\zeta(5,3)$ computed live).

Literature grounding (verified against the sources). All weight-≤ 7 closed forms in this section and Sec. 9.2 are verbatim rows of Hoffman's Appendix A [20] — including the triples $t_3(2,1,1)$, $t_3(4,1,1)$, $t_3(3,2,1)$, after eliminating $\zeta(\bar{5},1)$ via his own $t(5,1)$ row. The PSLQ identifications above therefore independently rediscovered the published table — a two-route verification. The weight-≥ 8 reductions rest on the theorem-grade chain: Hoffman Cor. 4.1 (exact conversion to alternating MZVs) + parity/reduction theorems for alternating sums [21, 22] + the data mine's proven weight-≤ 12 reductions [23]. The explicit form Eq. (93) appears to be new (checked against the explicit families of [24], whose t-value convention reverses the argument order).

11 Depth-k Tower

Proposition 4 (Depth-k tower, corrected)**.** *At k loops on S^3 with vertex restrictions and CG weights:*

- *$k = 1$: free (unrestricted) integer-s traces of D^2 (equivalently $D(s)$ at even s) give π^{even} only (Theorem 1, proven); vertex-restricted one-loop quantities need not (the self-energy $\Sigma(1)$ is PSLQ-identified as carrying G).*
- *$k = 2$ (vertex-parity sums): the two-loop sunset vertex restriction brings G and $\beta(4)$ ($\beta(4)$ first enters here among the identified quantities; the χ_{-4} theorem).*
- *$k = 2$ (CG-weighted): $S_{\min}$ is an identified element of $\mathcal{MT}(\mathbb{Z}[1/2])$ at level 2. The depth is* bounded *by 2 (the nesting depth of the min-weight double sum) but the content decomposes into Hoffman double t-values, which are classical period-ring elements (Sec. 9.2). The prior label "irreducible" was a basis-coverage artifact.*
- *$k = 3$: $S^{(3)}$ is a fully identified element of $\mathbb{Q}[\pi^2, \ln 2, \zeta(\text{odd}), \zeta(5,3)]$ with* realized *depth ≤ 2 unconditionally (Eq. (100)): the sole depth-2 constant is the level-1 generator $\zeta(5,3)$, at coefficient $6\pi^2$; every level-2 depth-2 generator ($t(5,1)$, $t(5,3)$, $t(7,1)$, $a(8,2)$, $a(7,3)$, $a(6,4)$) appears transiently and cancels in the total (Sec. 10.4). The earlier "independence from $S_{\min}$" statement (a 20-digit PSLQ at non-physical exponents) is superseded by the identification.*
- *$k = 4$: the* realized *depth of $S^{(4)}$ is ≤ 3. Depth-4 content enters $S^{(4)}$ only through the four-chain core C_4 (the o-space relation gives C_4 coefficient $+1$; the other cores are $\leq$ depth-3), so the verdict reduces to whether C_4's depth-4 part reduces to depth ≤ 3. At weights $9, 11, 13$ the quasi-shuffle (stuffle) relation system spans the* entire *depth-4 word space (rank = number of depth-4 words: $35, 84, 165$; exact rank over two independent primes), collapsing C_4's depth-4 content to depth ≤ 3 unconditionally. At weights $5, 7$ the stuffle system is rank-deficient, but these weights lie below the threshold for any depth-≥ 3 irreducible: weight-5 values reduce to single polylog constants (depth ≤ 1) and weight-7 to depth ≤ 2 (the $w \leq 7$ GeoVac multiple-t values match Hoffman's Appendix-A reductions, verified previously). The explicit closed form of $S^{(4)}$ remains open; the depth verdict does not depend on it.*

Loop order bounds the depth of the classical level-2 multiple-t-value content; it does not mint new irreducible transcendentals automatically. Realized depth so far grows strictly more slowly than loop order ($\leq k - 1$ at $k = 2, 3, 4$).

The mechanism: each vertex coupling creates one level of Hurwitz nesting. One vertex produces a depth-2 double t-value sum ($S_{\min}$). Two vertices produce a depth-≤ 3 triple t-value sum ($S^{(3)}$). The prediction extends: k vertices produce a depth-$\leq (k+1)$ multiple t-value in $\mathcal{MT}(\mathbb{Z}[1/2])$.

Whether the surviving irreducible depth-k component is new (not reducible by parity/stuffle within the level-2 algebra) is a question for each k separately — the structural nesting is genuine, but the motivic content is classical at $k = 2$ and, on all identified components, at $k = 3$ (realized depth ≤ 2). This is the S^3 analog of the increasing motivic depth observed in flat-space multi-loop Feynman integrals [15, 19].

12 Fine-Structure Euler Sum Cancellation

The Dirac Dirichlet series $D(s)$ studied in the preceding sections sums over the *free* Camporesi–Higuchi spectrum with uniform degeneracy weights g_n. A complementary probe arises from the *exact* Dirac hydrogen spectrum, where the degeneracy-weighted energy expansion in powers of $(Z\alpha)^2$ generates Dirichlet sums with state-dependent coefficients that mix harmonic numbers of varying depth.

12.1 Setup

The exact Dirac energy for quantum numbers (n, j) is

$$E_{nj} = mc^2\left(1 + \frac{(Z\alpha)^2}{\left(n_r + \sqrt{\kappa^2 - (Z\alpha)^2}\right)^2}\right)^{-1/2} - mc^2\,, \tag{101}$$

where $\kappa = j + \frac{1}{2}$ and $n_r = n - \kappa$. Expanding in powers of $x = (Z\alpha)^2$,

$$\sum_j d(n,j)\,\frac{E_{nj}}{mc^2} = \sum_{p=1}^{\infty} c_p(n)\,x^p\,, \tag{102}$$

where $d(n, j) = 2(2j+1)$ for $j < n - \frac{1}{2}$ and $d(n, j) = 2j+1$ for $j = n - \frac{1}{2}$. The degeneracy-weighted Dirichlet sums are

$$D_p \;=\; \sum_{n=1}^{\infty} c_p(n)\,. \tag{103}$$

12.2 Closed forms for $c_p(n)$

Symbolic expansion of the exact Dirac formula and summation over j gives closed forms involving generalized harmonic numbers $H_{n-1}^{(r)} = \sum_{k=1}^{n-1} k^{-r}$:

$$c_2(n) = -\frac{5n-4}{4n^3}\,, \tag{104}$$

$$c_3(n) = \frac{19n-20}{8n^5} - \frac{3}{2}\,\frac{H_{n-1}}{n^4} - \frac{1}{2}\,\frac{H_{n-1}^{(2)}}{n^3}\,. \tag{105}$$

At order $p = 2$ and $p = 3$, the sums D_2 and D_3 collapse to pure single-zeta values:

$$D_2 = -\frac{5}{4}\,\zeta(2) + \zeta(3)\,, \qquad D_3 = \frac{19}{8}\,\zeta(4) - \frac{11}{4}\,\zeta(5)\,. \tag{106}$$

12.3 $\zeta(3)$ complementarity

The $p = 2$ coefficient defines the *fine-structure shell sum*

$$\mathcal{S}(n) \;=\; \sum_j d(n,j)\Big(\frac{n}{j+\frac{1}{2}} - \frac{3}{4}\Big) \;=\; \frac{n(5n-4)}{2}\,, \tag{107}$$

so that $c_2(n) = -\mathcal{S}(n)/(2n^4)$. This decomposes into a *conformal* piece $-3n^2/2$ (from the scalar curvature $R/8 = 3/4$ on unit S^3) and a *quantum defect* piece $2n(2n-1)$, the double hexagonal number.

In the Fock convention ($n \geq 1$), the single-chirality Dirac mode degeneracy is $g_n^{\text{Dirac}} = 2n(n+1)$.

Theorem 7 ($\zeta(3)$ complementarity)**.** *For all $n \geq 1$,*

$$\mathcal{S}(n) + g_n^{\text{Dirac}} \;=\; \frac{9n^2}{2} \;=\; |\lambda_0|^2 \times n_{\text{Schr}}^2\,, \tag{108}$$

where $|\lambda_0| = 3/2$ is the lowest Dirac eigenvalue on S^3 and $n_{\text{Schr}}^2 = 2n^2$ is the Schrödinger degeneracy (including spin). At the Dirichlet series level,

$$D_{\mathcal{S}}(s) + D_g(s) \;=\; \frac{9}{2}\,\zeta(s{-}2) \qquad \textit{for all } s\,, \tag{109}$$

where $D_{\mathcal{S}}(s) = \sum_{n\geq 1} \mathcal{S}(n)\, n^{-s}$ and $D_g(s) = \sum_{n\geq 1} g_n^{\text{Dirac}}\, n^{-s}$. The $\zeta(s{-}1)$ terms cancel identically.

Proof. $\mathcal{S}(n) = 5n^2/2 - 2n$ and $g_n^{\text{Dirac}} = 2n^2 + 2n$; their sum is $9n^2/2$. Therefore $D_{\mathcal{S}}(s) = \frac{5}{2}\,\zeta(s{-}2) - 2\,\zeta(s{-}1)$ and $D_g(s) = 2\,\zeta(s{-}2) + 2\,\zeta(s{-}1)$; summing gives $\frac{9}{2}\,\zeta(s{-}2)$. □

The cancellation of $\zeta(s{-}1)$ is the physical content: at odd s, $\zeta(s{-}1)$ is even-zeta (π^{even}) and harmless, but at even s, $\zeta(s{-}1)$ carries odd-zeta content ($\zeta(3)$, $\zeta(5)$, …). At $s = d_{\max} = 4$:

$$D_{\mathcal{S}}(4) = \frac{5}{2}\,\frac{\pi^2}{6} - 2\zeta(3) = \frac{5F}{2} - 2\zeta(3)\,, \tag{110}$$

confirming that the quantum defect (the $-2n$ piece of $\mathcal{S}$) is the physical mechanism injecting $\zeta(3)$ into the fine-structure spectrum—with sign opposite to the $+2\zeta(3)$ that appears in $D_g(4) = 2\zeta(2) + 2\zeta(3)$. Their sum $\frac{9}{2}\zeta(2) = 3\pi^2/4$ is purely π^{even}, returning to the second-order (scalar) sector.

The prefactor $9/2 = |\lambda_0|^2 \times 2 = (\text{Dirac mass gap})^2 \times (\text{spinor rank})$ links through the Lichnerowicz formula $D^2 = \Delta + R/4$ back to the S^3 geometry: the complementarity sum recovers the scalar Laplacian sector exactly, closing the chain from the packing construction through S^3 and the Dirac operator to the fine-structure Dirichlet series.

This is an analogue, in the fine-structure sector, of T9's integer-s statement (Theorem 1) — not T9 itself, which concerns the half-integer spectrum $n+3/2$, whereas the complementarity above is a sum over integer n with $g = 2n(n+1)$. Paper 18's taxonomy predicted this separation: calibration (π^{even} from the second-order curvature $R/8$) versus spinor-intrinsic (mixed zeta from the first-order quantum defect).

12.4 D_4 at weight 7: product zeta survives

At order $p = 4$,

$$c_4(n) = -\frac{5(41n-40)}{64n^7} + \frac{15}{4}\frac{H_{n-1}}{n^6} - \frac{1}{4}\frac{H^{(2)}_{n-1}}{n^5} - \frac{3}{4}\frac{H^{(3)}_{n-1}}{n^4} - \frac{1}{4}\frac{H^{(4)}_{n-1}}{n^3}\,. \tag{111}$$

The harmonic-number sums decompose into weight-7 Euler sums $S_{r,s} = \sum_{n\geq 1} H_n^{(r)}/n^s$ via the identity $H^{(r)}_{n-1} = \zeta(r) - \zeta(r,n)$ and Euler's formula $S_{1,6} = 4\zeta(7) - \zeta(2)\zeta(5) - \zeta(3)\zeta(4)$ [26].

Observation 1 (Euler sum cancellation at weight 7)**.** *At 200 dps, PSLQ identifies*

$$D_4 = -\frac{205}{64}\,\zeta(6) + \frac{71}{8}\,\zeta(7) - \frac{9}{2}\,\zeta(3)\zeta(4)\,, \tag{112}$$

with residual $< 10^{-195}$ *(archived PSLQ run; the value is pinned live in* `tests/test_paper28_gap_closure_pins.py`*, and the* D_4/D_5 *decimals were independently cross-verified by direct partial summation, 2026-07-04). The product* $\zeta(2)\zeta(5)$ *cancels* algebraically*: its coefficient vanishes exactly (verified in exact rational arithmetic from the Euler sum decomposition), while* $\zeta(3)\zeta(4)$ *survives with coefficient* $-9/2$*.*

This breaks the pure-single-zeta pattern: D_2 and D_3 involve only $\zeta(s)$ values, whereas D_4 introduces the first zeta product. The four weight-7 Euler sums, independently identified by PSLQ, are

Sum	$\zeta(7)$	$\zeta(2)\zeta(5)$	$\zeta(3)\zeta(4)$
$S_{1,6}$	4	-1	-1
$S_{2,5}$	-10	5	2
$S_{3,4}$	18	-10	0
$S_{4,3}$	-17	10	1
$S_{5,2}$	11	-4	-2

The stuffle relations $S_{r,s} + S_{s,r} = \zeta(r)\zeta(s) + \zeta(r+s)$ are satisfied to $< 10^{-200}$.

12.5 D_5 at weight 9: product survival rule

At order $p = 5$,

$$c_5(n) = \frac{7(71n-72)}{128\,n^9} - \frac{105}{16}\frac{H_{n-1}}{n^8} + \frac{45}{16}\frac{H^{(2)}_{n-1}}{n^7} + \frac{5}{4}\frac{H^{(3)}_{n-1}}{n^6} - \frac{3}{8}\frac{H^{(4)}_{n-1}}{n^5} - \frac{15}{32}\frac{H^{(5)}_{n-1}}{n^4} - \frac{5}{32}\frac{H^{(6)}_{n-1}}{n^3}\,, \tag{113}$$

verified as exact rational equality against direct sympy expansion for $n = 1, \ldots, 15$. The Dirichlet sum decomposes via weight-9 Euler sums $S_{r,s}$ for $r + s = 9$ (computed via Hurwitz zeta to 250 dps, stuffle relations verified to $< 10^{-200}$):

Sum	$\zeta(9)$	$\zeta(2)\zeta(7)$	$\zeta(3)\zeta(6)$	$\zeta(4)\zeta(5)$
$S_{2,7}$	$-35/2$	7	2	4
$S_{3,6}$	$85/2$	-21	0	-6
$S_{4,5}$	$-125/2$	35	0	5
$S_{5,4}$	$127/2$	-35	0	-4
$S_{6,3}$	$-83/2$	21	1	6
$S_{7,2}$	$37/2$	-6	-2	-4

PSLQ at 200 dps identifies

$$D_5 = \frac{497}{128}\,\zeta(8) - \frac{467}{16}\,\zeta(9) + \frac{385}{32}\,\zeta(3)\zeta(6) + \frac{75}{8}\,\zeta(4)\zeta(5)\,, \tag{114}$$

with residual $< 10^{-190}$. The product $\zeta(2)\zeta(7)$ cancels algebraically (verified from the Euler sum collection), while both $\zeta(3)\zeta(6)$ and $\zeta(4)\zeta(5)$ survive. This extends the pattern from D_4: product terms grow from 0 (at D_2, D_3) to 1 (at D_4) to 2 (at D_5).

12.6 D_6 at weight 11: product survival rule extended

At order $p = 6$ the weight-11 Euler sums $S_{r,s}$ for $r + s = 11$ are computed via Hurwitz zeta to 400 dps, with stuffle relations verified to $< 10^{-300}$. Seven nontrivial sums ($r = 1, \ldots, 8$, $r \neq 3$) contribute through the β_r coefficients of the $c_6(n)$ expansion. Individual decompositions are found by PSLQ against the weight-11 depth-1 MZV basis $\{\zeta(11), \zeta(2)\zeta(9), \zeta(3)\zeta(8), \zeta(4)\zeta(7), \zeta(5)\zeta(6)\}$:

Sum	$\zeta(11)$	$\zeta(2)\zeta(9)$	$\zeta(3)\zeta(8)$	$\zeta(4)\zeta(7)$	$\zeta(5)\zeta(6)$
$S_{1,10}$	6	-1	-1	-1	-1
$S_{2,9}$	-27	9	2	6	4
$S_{4,7}$	$-329/2$	84	0	21	4
$S_{5,6}$	$463/2$	-126	0	-21	0
$S_{6,5}$	$-461/2$	126	0	21	1
$S_{7,4}$	$331/2$	-84	0	-20	-4
$S_{8,3}$	-82	36	1	15	6

The $\zeta(3)\zeta(8)$ column has a sparse support pattern: nonzero only for $S_{1,10}$, $S_{2,9}$, and $S_{8,3}$.

Collecting through the β_r coefficients and the rational $\zeta(10)$, $\zeta(11)$ parts:

$$D_6 = -\frac{2289}{512}\,\zeta(10) + \frac{1589}{16}\,\zeta(11) - \frac{1617}{64}\,\zeta(3)\zeta(8) - \frac{1785}{64}\,\zeta(4)\zeta(7) - \frac{2065}{64}\,\zeta(5)\zeta(6)\,, \tag{115}$$

verified to 62 digits against direct Euler-sum evaluation at 400 dps. The product $\zeta(2)\zeta(9)$ cancels algebraically, and all three surviving products $\zeta(3)\zeta(8)$, $\zeta(4)\zeta(7)$, $\zeta(5)\zeta(6)$ are present, matching the prediction $\max(0, \lfloor(2 \cdot 6 - 5)/2\rfloor) = 3$.

Observation 2 (Product survival rule)**.** *At each Sommerfeld order $p \geq 2$, the Dirichlet sum D_p decomposes into weight-$(2p-1)$ zeta values. The product $\zeta(2)\zeta(2p-3)$ cancels algebraically at every order. Exactly* $\max(0, \lfloor(2p-5)/2\rfloor)$ *product terms survive, all of the form $\zeta(a)\zeta(b)$ with $a+b = 2p-1$, a odd ≥ 3, b even ≥ 4.*

Verified through D_6 ($p = 6$, weight 11). The systematic expulsion of $\zeta(2)$ products is driven by the n^2 Fock degeneracy that weights the Sommerfeld sums. This is the same n^2 that produces $F = \zeta(2) = D_{n^2}(4)$ in the Phase 4F identification (Paper 2 [9]): the Fock degeneracy protects its own Dirichlet evaluation from cross-contamination with odd-zeta products.

12.7 Transcendental structure

Substituting $\zeta(6) = \pi^6/945$ and $\zeta(4) = \pi^4/90$, equation (112) becomes

$$D_4 = -\frac{41\pi^6}{12096} + \frac{71}{8}\,\zeta(7) - \frac{\pi^4}{20}\,\zeta(3)\,. \tag{116}$$

This is a three-tier decomposition: π^6 (weight-6 even-zeta), $\zeta(7)$ (weight-7 odd-zeta), and the mixed product $\pi^4 \cdot \zeta(3)$ (weight-4 even $\times$ weight-3 odd). In Paper 18's operator-order taxonomy, π^{even} comes from 2nd-order operators and ζ(odd) from 1st-order operators. The mixed product $\pi^4 \cdot \zeta(3)$ is the first instance in the framework where these two tiers appear multiplicatively in a single observable.

The $\zeta(2)\zeta(5)$ cancellation is non-trivial: four of the five contributing Euler sums carry nonzero $\zeta(2)\zeta(5)$ coefficients ($S_{1,6}$, $S_{2,5}$, $S_{4,3}$, $S_{5,2}$), and the cancellation emerges only in the specific linear combination that the Dirac degeneracy weights and harmonic-number structure select. In contrast, $S_{3,4}$ alone is free of $\zeta(3)\zeta(4)$, but the remaining four sums contribute $\zeta(3)\zeta(4)$ terms that do not cancel.

12.8 K–Sommerfeld structural separation

The product survival rule (Observation 2) raises a natural question: does the all-orders Sommerfeld cancellation pattern explain why Paper 2's $K = \pi(B + F - \Delta)$ has pure π-content?

PSLQ at 200 dps tests whether K/π is a rational linear combination of $\{D_2, D_3, D_4, D_5, D_6, 1\}$. The result is *null*: no relation with coefficients below the PSLQ height bound at 200 dps (a null search excludes small-coefficient relations only, not membership in the $\mathbb{Q}$-span). $K \in \mathbb{Q}[\pi]$, while the D_p carry ζ(odd) and hence (conditional on the period conjecture) lie outside it.

The structural reading: K is composed of B (a rational truncated Casimir trace), $F = \zeta(2)$ (Fock Dirichlet at $s = 4$) and $\Delta^{-1} = g_3^{\text{Dirac}}$ (a Dirac mode count), so $K \in \mathbb{Q}[\pi]$ (directly, not via T9). The Sommerfeld D_p, by contrast, are sums of *physical energy coefficients* dressed with harmonic numbers $H_{n-1}^{(r)}$, which access the full Euler–Zagier MZV ring including irreducible odd-zeta at every order.

The $\zeta(2)$ cancellation mechanism is nonetheless a genuine bridge: the n^2 Fock degeneracy that generates $F = \zeta(2)$ simultaneously drives the systematic expulsion of $\zeta(2) \times \zeta$(odd) products from the Sommerfeld sums. The Fock degeneracy organizes both K (through F) and the D_p (through the cancellation pattern), but (conditional on the period conjecture) the operator/energy distinction keeps K from being a Sommerfeld functional; no small-coefficient relation was found.

13 Flat-Space Consistency

On S^3 of radius R, the Dirac eigenvalues scale as $|\lambda_n(R)| = (n + 3/2)/R$ and the Dirichlet series becomes $D(s, R) = R^s \cdot D(s, 1)$. The ratio $D(s, R)/R^s$ is R-independent *exactly*, confirming that the even/odd s-parity discriminant is structural, not a finite-R artifact.

Weyl's law is verified: the cumulative Dirac mode count $N(n_{\max}) = 2(n_{\max}+1)(n_{\max}+2)(n_{\max}+3)/3$ matches the Weyl prediction $(2/3)R^3 p_{\max}^3$ with $O(1/n_{\max})$ convergence.

$\zeta(3)$ appears in both the S^3 spectral sums and flat-space two-loop QED [13, 14]; Catalan G and Dirichlet $\beta(4)$ are S^3 vertex-parity content, absent from the flat-space a_e through three loops. Coefficient matching for the full Petermann–Sommerfield two-loop anomalous magnetic moment $a_2 = 197/144 + \pi^2/12 + 3\zeta(3)/4 - \pi^2 \ln 2/2$ [16, 17] requires matching specific diagram topologies and renormalization subtractions, which is beyond the scope of spectral sums. At one loop, the S^3 computation (Sec. 7) gives $F_2/[\alpha/(2\pi)] = 1.084$ at $n_{\text{ext}} = 1$ on unit S^3, matching the Parker–Toms leading curvature correction $1 + R/(12|\lambda|^2) = 1.080$ within 0.4%. The 5-point mode-dependent curve ($n_{\text{ext}} = 1, \ldots, 5$) follows a power law $\approx 5.91\,|\lambda|^{-1.84}$ (Table 5), consistent with the flat-space limit being $R \to \infty$ at fixed physical mass, not $|\lambda| \to \infty$ at fixed R. In the proper flat-space limit, the curvature excess vanishes, recovering the Schwinger value [10].

The sunset sum scales exactly as $R^{2s_1+2s_2+2p}$ (power = 10 at default parameters), with the ratio $S(R)/R^{\text{power}}$ R-independent to machine precision. The $D(5) = 14\zeta(3) - 31\zeta(5)/2$ identification is R-independent, confirming that the odd-zeta content is structural.

14 Comprehensive Transcendental Tables

Table 7: Even/odd decomposition of $D(s)$: PSLQ at 80+ dps.

s	$D_{\text{even}}(s)$	$D_{\text{odd}}(s)$	$D_{\text{even}} - D_{\text{odd}}$
4	$\frac{\pi^2}{2} - \frac{\pi^4}{24} - 4G + 4\beta(4)$	$\frac{\pi^2}{2} - \frac{\pi^4}{24} + 4G - 4\beta(4)$	$8(\beta(4) - G)$
5	odd-zeta + Dirichlet	odd-zeta − Dirichlet	$16(\beta(5) - \beta(3))$
6	π^{even} + Dirichlet	π^{even} − Dirichlet	$32(\beta(6) - \beta(4))$

Table 8: Verification of Theorem 3: $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$.

s	2^{s-1}	Residual |LHS − RHS|
2	2	0 (sympy)
3	4	0 (sympy)
4	8	$< 10^{-100}$
5	16	$< 10^{-100}$
6	32	$< 10^{-100}$
7	64	$< 10^{-100}$
8	128	$< 10^{-100}$
9	256	$< 10^{-100}$
10	512	$< 10^{-100}$
15	16384	$< 10^{-100}$

15 Graph-Native QED

The preceding sections computed QED on S^3 via infinite spectral sums over the Camporesi–Higuchi Dirac modes. This section reports the complementary computation: QED formulated entirely on the *finite* GeoVac graph, using no infinite sums, no regularization, and no renormalization. All quantities are finite matrix traces over the Dirac-labeled graph nodes and Fock-labeled graph edges.

15.1 Construction

The graph-native QED infrastructure consists of four layers:

1. **Fock graph** (nodes = Fock states (n, l, m) at $n \leq n_{\max}$, edges = nearest-neighbor Fock ladders) with node Laplacian $L_0 = BB^T$, edge Laplacian $L_1 = B^T B$, and exact Hodge decomposition (Paper 25 [5] vocabulary).

2. **Electron propagator** $G_e = D^{-1}$, where D is the Dirac graph operator in the (n, κ, m_j) basis with hopping parameter t. At $t = 0$, $G_e = \text{diag}(2/(2n+3))$.

Table 9: Three-loop factorized convergence at $a = 4$, $p = 1$, CG-weighted. The limit is $S^{(3)} = 31.5726\ldots$ (Eq. (89)): raw partial sums carry zero reliable digits at these truncations — the $(\ln N)^2/N^2$ increment law, not the naively fitted power law, governs the tail (Sec. 10). Rows recomputed 2026-07-04 from `geovac/qed_three_loop.py` and pinned in `tests/test_paper28_gap_closure_pins.py`; an earlier printing carried stale values (1.399/3.741/5.836/7.729/15.23) from a pre-refactor run.

$n_{\max}$	$S^{(3)}(n_{\max})$	Δ from previous	Time (s)
5	1.062	—	< 0.1
10	3.572	2.510	< 0.1
15	5.908	2.336	0.2
20	7.872	1.964	0.5
50	14.798	—	< 60
∞	$31.5726\ldots$	—	(decomposition route)

3. **Photon propagator** $G_\gamma = L_1^+$, the Moore–Penrose pseudoinverse of the edge Laplacian. This is the graph analog of the Feynman-gauge photon propagator.

4. **Vertex coupling tensor** $V[a, b, e]$, built from the Clebsch–Gordan projection matrix P that maps Dirac labels (n, κ, m_j) onto Fock nodes (n, l, m).

At $n_{\max} = 2$: 10 Dirac states, 5 Fock nodes, 3 Fock edges. At $n_{\max} = 3$: 28 Dirac states, 14 Fock nodes, 13 Fock edges.

15.2 Vacuum polarization

The graph vacuum polarization tensor is

$$\Pi_{\text{graph}}[e_1, e_2] = \sum_{a,b} V[a, b, e_1]\, G_e[a, b]\, V[b, a, e_2],$$

the complete one-loop photon self-energy on the finite graph. At $n_{\max} = 2$, $\mathrm{Tr}(\Pi) = 224/75$ (pure rational). At $n_{\max} = 3$, $\mathrm{Tr}(\Pi) = 3872/735$; the 13×13 matrix decomposes into l-sector blocks ($l = 0$: 2×2, $l = 1$: 7×7, $l = 2$: 4×4) with zero cross-block coupling. Four entries in the $l = 2$ sector contain $\sqrt{6}$ from CG coefficients; all others are rational.

15.3 Self-energy and vertex correction

The one-loop self-energy is

$$\Sigma[a, b] = \sum_{e,e'} G_\gamma[e, e']\, V_e \cdot V_{e'}^T,$$

where $V_e[a, c] = V[a, c, e]$. At $n_{\max} = 2$, $t = 0$: $\mathrm{Tr}(\Sigma) = 44/3$; eigenvalues $\{0^{(\times 5)}, 4/3, 2^{(\times 2)}, 4, 16/3\}$; positive semidefinite; Hermitian.

The one-loop vertex correction is

$$\Lambda[a, b] = \sum_{e',e''} G_\gamma[e', e'']\, V_{e'} \cdot G_e \cdot V_{e''}^T.$$

At $n_{\max} = 2$, $t = 0$: $\mathrm{Tr}(\Lambda) = 32/9$; eigenvalues $\{0^{(\times 5)}, 8/45, (28 \pm 4\sqrt{145})/45, 4/5, 4/3\}$.

The graph anomalous magnetic moment is extracted as $F_2 = \mathrm{Tr}(\Lambda)/\mathrm{Tr}(V_{\mathrm{bare}} \cdot G_e) = 5\sqrt{2}/3 \approx 2.357$ at $t = 0$.

At the matched-scale coupling $t = \kappa = -1/16$, the self-energy is *identical* (the formula does not involve G_e), but the vertex correction changes: $\mathrm{Tr}(\Lambda) = 16\sqrt{3}/8985 + 19255/5391 \approx 3.575$. Λ acquires one negative eigenvalue (-0.447), becoming indefinite. The anomalous moment shifts to

$$F_2(\kappa) = \frac{48\sqrt{3} + 96275}{3(15 + 16\sqrt{6} + 9616\sqrt{2})} \approx 2.3525,$$

a 0.2% shift from the $t = 0$ value. $F_2(\kappa)$ remains algebraic irrational in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$, the CG ring.

15.4 Neumann series structure of the electron propagator

The coupling-dependence of $F_2(t)$ admits a transparent algebraic explanation. Write the Dirac graph operator as $D = \Lambda + tA$, where $\Lambda = \mathrm{diag}(|\lambda_n|)$ is the free (diagonal) Dirac spectrum and A is the adjacency-like hopping matrix. The electron propagator admits a Born/Neumann expansion

$$G_e(t) = (\Lambda + tA)^{-1} = \Lambda^{-1} \sum_{k=0}^{\infty} \big(-t\,\Lambda^{-1} A\big)^k,$$

which converges when $\rho(t\,\Lambda^{-1} A) < 1$. The free propagator $\Lambda^{-1} = \mathrm{diag}(2/(2n+3))$ is trivially rational. Each subsequent order multiplies by the matrix $-t\,\Lambda^{-1} A$, whose entries lie in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$ (the CG ring). The expansion therefore preserves the algebraic ring at every order: no new transcendentals enter from the coupling.

By the Cayley–Hamilton theorem, the series terminates exactly at order $N - 1$, where $N = \dim(\text{Dirac space})$. Consequently, $G_e(t)$ — and hence $F_2(t)$ — is a *rational function* of t with coefficients in the CG ring. At the matched-scale coupling $t = \kappa = -1/16$, the spectral radius $\rho(\kappa\,\Lambda^{-1} A) \approx 0.2$ at $n_{\max} = 2$, so the first few Born orders dominate. The 0.2% shift $F_2(\kappa)/F_2(0) \approx 0.998$ is consistent with first-order perturbation in κ.

This structure connects to Paper 18's exchange-constant taxonomy [3]: the hopping parameter t enters F_2 as a polynomial perturbation of the free value, preserving the INTRINSIC algebraic ring at every order. The Neumann expansion makes explicit that the graph-native anomalous moment is a *finite rational expression* in κ and CG coefficients, with no transcendental content at any truncation $n_{\max}$. For larger $n_{\max}$ (where direct symbolic inversion of D becomes expensive), the rapid convergence of the Born series at $|\kappa| = 1/16$ provides an efficient perturbative alternative.

15.4.1 $F_2(t)$ is an even rational function

The Neumann expansion reveals that $F_2(t)$ is an *even* function of t: all odd Taylor coefficients vanish identically. The mechanism is the decomposition $D(t) = \Lambda + tA$, where Λ (diagonal eigenvalues $\pm|\lambda_n|$) is even in t and A (the $T_\pm$ ladder operators connecting adjacent shells) is odd. The vertex correction involves the bilinear $V \cdot G_e \cdot V^T$, which is invariant under $t \to -t$ because the adjacency contribution enters quadratically via $G_e(t) = G_e(-t)$ in the trace.

At $n_{\max} = 2$, the Taylor expansion is

$$F_2(t) = \frac{5\sqrt{2}}{3} + c_2\,t^2 + c_4\,t^4 + O(t^6), \tag{117}$$

with $c_0 = 5\sqrt{2}/3 \approx 2.357$ (algebraic), $c_1 = c_3 = c_5 = 0$ (exact), $c_2 \approx -1.163$ (algebraic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$), $c_4 \approx 1.685$ (algebraic). The even-ness means the anomalous moment depends only on $t^2 = \kappa^2$, not on the sign of the hopping coupling.

15.4.2 $F_2(\kappa)$ convergence

$n_{\max}$	N_{Dirac}	E_{Fock}	$F_2(\kappa)$	$F_2(0)$	$F_2 \cdot n_{\max}$
2	10	3	2.3525	2.3570	4.705
3	28	13	1.8731	1.8756	5.619
4	60	34	1.5892	1.5898	6.357
5	110	70	1.3958	1.3954	6.979
6	182	125	1.2532	1.2523	7.519

Table 10: Graph-native anomalous moment F_2 at the matched-scale coupling $\kappa = -1/16$. F_2 decreases monotonically; a power-law fit gives $F_2 \sim 3.507\, n_{\max}^{-0.573}$ ($R^2 = 0.99990$).

The graph-native $F_2(\kappa)$ decreases monotonically with $n_{\max}$ but remains $O(1)$, orders of magnitude above the Schwinger value $\alpha/(2\pi) \approx 1.16 \times 10^{-3}$. The suppression from graph to continuum requires both the coupling constant α (one power per loop) and the vector photon structure (SO(4) selection rules that eliminate most graph-allowed channels).

Power-law convergence (caveat: 5-point fit, asymptote inferred not derived). A least-squares fit across $n_{\max} = 2, \ldots, 6$ gives $F_2(\kappa) \sim 3.507\, n_{\max}^{-0.573}$ with $R^2 = 0.99990$. Local log-log exponents increase monotonically in magnitude: -0.562 ($n = 2{-}3$), -0.571 ($3{-}4$), -0.582 ($4{-}5$), -0.591 ($5{-}6$), consistent with a slowly steepening power law. Successive ratios $F_2(n{+}1)/F_2(n) = 0.796, 0.849, 0.878, 0.898$ approach 1 from below, confirming slow (sub-$1/n$) decay. Richardson extrapolation with adaptive exponent estimation ($p \approx 0.51$) gives the unphysical $F_\infty \approx -0.22$, which we interpret as consistent with $F_2 \to 0$ within the zero-offset power-law family.

This conclusion is fit-based, not derived: a 5-point power-law fit with high R^2 does not by itself rule out a small constant offset $F_\infty^{\text{true}} \neq 0$ that the truncated data cannot distinguish from the leading power law. Larger $n_{\max}$ data, or a derivation of the leading large-n behavior of F_2 from the spinor heat-kernel curvature expansion, would be needed to fix the asymptote on structural rather than fit grounds. The interpretation "$F_2 \to 0$ on a fixed-radius sphere" (consistent with form-factor suppression at $|\lambda| \to \infty$ on $R = 1$, see §15.4.2 discussion above) is the most natural reading of the data, but should be treated as suggestive rather than proven.

The sign of $F_2(\kappa) - F_2(0)$ flips at $n_{\max} = 5$: $\delta = -0.0045, -0.0025, -0.0006, +0.0004, +0.0009$ for $n_{\max} = 2, \ldots, 6$. This zero-crossing indicates a subtle interplay between the t-dependence and the truncation level; both sequences track within $< 0.1\%$ and converge to the same graph limit.

15.5 Broken structural zero and the pendant-edge theorem

Theorem 4 states that $\Sigma(n_{\text{ext}} = 0) = 0$ in the continuum, by vertex parity impossibility. On the finite graph, this structural zero is **broken**:

$$\Sigma^{\text{GS}}_{\text{graph}} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \neq 0.$$

The CG projection maps Dirac (n, κ, m_j) states onto Fock (n, l, m) nodes, opening vertex couplings invisible to the SO(4) channel count that enforces the continuum selection rule.

A three-layer investigation reveals the complete mechanism.

15.5.1 Layer 1: Anatomy of the ground-state self-energy

At $n_{\max} = 2$, the entire ground-state self-energy comes from a single edge pair. Let $v_0 = |1, 0, 0\rangle$ (the ground-state Fock node) and $v_1 = |2, 0, 0\rangle$ (the first radial excitation); these are connected by

the edge e_0. All Clebsch–Gordan coefficients equal 1 because $l = 0$ forces trivial angular projection. Two intermediate s-wave Dirac states ($m_j = \pm 1/2$) contribute equally. The formula reduces to

$$\Sigma[a, b] = 2 \cdot G_\gamma[e_0, e_0] \cdot 1,$$

where $G_\gamma = L_1^+$ is the photon propagator. At $n_{\max} = 2$, L_1 is a 3×3 matrix and $G_\gamma[e_0, e_0] = 1/2$, giving $\Sigma[a, b] = 1$ for all entries of the 2×2 ground-state block.

15.5.2 Layer 2: The pendant-edge theorem

The anatomy at $n_{\max} = 2$ generalizes exactly to all $n_{\max}$.

The ground-state node $v_0 = |1, 0, 0\rangle$ is a *leaf* (degree 1) of the Fock graph at every $n_{\max}$. The single edge e_0 incident on v_0 is a *pendant edge*. The s-wave ($l = 0$) radial edges form a tridiagonal sub-block of $L_1 = B^T B$ — the standard discrete Laplacian on a path graph of length $N = n_{\max} - 1$. This path-graph Laplacian T_N has an explicitly known inverse: its $(0, 0)$ entry is

$$T_N^{-1}[0, 0] \;=\; \frac{N}{N+1} \;=\; \frac{n_{\max} - 1}{n_{\max}}.$$

Because the pendant edge e_0 decouples from all non-radial edges in L_1 (s-wave edges share no endpoint with edges at $l \geq 1$), $G_\gamma[e_0, e_0] = T_N^{-1}[0, 0]$ exactly.

Proposition 5 (Pendant-edge theorem for the graph self-energy)**.** *On the GeoVac Fock graph at any finite $n_{\max} \geq 2$, the graph-native ground-state self-energy satisfies*

$$\Sigma_{\text{graph}}(\text{GS}) \;=\; \frac{2(n_{\max} - 1)}{n_{\max}}, \tag{118}$$

where the value is exact (rational) and degenerate across both $m_j = \pm 1/2$ ground-state components. In particular,

$$\lim_{n_{\max} \to \infty} \Sigma_{\text{graph}}(\text{GS}) = 2 \;\neq\; 0. \tag{119}$$

The structural zero of Theorem 4 is never recovered in the graph limit.

Proof. The ground-state Fock node $v_0 = |1, 0, 0\rangle$ has degree 1 at every $n_{\max}$ (its only neighbor is $|2, 0, 0\rangle$). The self-energy formula $\Sigma[a, b] = \sum_{e,e'} G_\gamma[e, e']\, V_e \cdot V_{e'}^T$ at the ground state receives contributions only from edges incident on v_0, of which there is exactly one (e_0). All CG coefficients at $l = 0$ equal 1, and two Dirac intermediate states ($m_j = \pm 1/2$) contribute, giving $\Sigma[a, b] = 2\, G_\gamma[e_0, e_0]$. The s-wave radial edges form a path graph of $N = n_{\max} - 1$ edges in L_1, decoupled from all $l \geq 1$ edges. The Moore–Penrose pseudoinverse of the path-graph Laplacian T_N satisfies $T_N^{-1}[0, 0] = N/(N + 1)$. Therefore $\Sigma_{\text{graph}}(\text{GS}) = 2N/(N + 1) = 2(n_{\max} - 1)/n_{\max}$. □

Verification: Eq. (118) is continuously verified by `tests/test_graph_qed_self_energy.py`: at $n_{\max} = 2$ in exact `sympy` rational arithmetic (GS block $[[1, 1], [1, 1]]$), and at $n_{\max} = 3, 4$ numerically to 10^{-10} (`TestPendantEdgeSweep`, added 2026-07-03 after the `graph_qed_photon` $n_{\max} \geq 4$ eigenvalue-sort crash fix; the exact-`sympy` photon pseudoinverse at $n_{\max} \geq 3$ exceeds any reasonable test budget, so those points are numeric). The formula itself is proven in the theorem above for all $n_{\max}$.

15.5.3 Layer 3: Selection-rule gap

The continuum possesses *two* independent protection mechanisms that enforce $\Sigma(n_{\text{ext}} = 0) = 0$:

1. **Vertex parity.** The selection rule requires $n_1 + n_2 + q$ odd (Eq. (6)). At $n_{\text{ext}} = 0$, the triangle inequality forces $q = n_{\text{int}}$, so the sum is $2\,n_{\text{int}}$ (always even). No allowed triple exists.

2. **SO(4) channel count.** The CG multiplicity $W(0, n_{\text{int}}, q) = 0$ for all q. The ground state has $j_R = 0$ in the $\text{SU}(2)_L \times \text{SU}(2)_R$ decomposition; the right-factor triangle degenerates (one component fails the $\text{SU}(2)_R$ triangle; the other passes all triangles but is killed by parity).

The finite graph has *neither* mechanism. The graph photon lives on edges—it is a 1-cochain, not a vector field. The CG projection is scalar SU(2), not the $\text{SU}(2)_L \times \text{SU}(2)_R$ double triangle of SO(4) vector harmonics. Both protection mechanisms arise from the *spin-1 vector nature* of the continuum photon, which is absent on the graph.

15.5.4 Interpretation: scalar vs. vector QED

The graph-native QED of this section is a *scalar* QED: the photon carries no spin-1 structure on the graph (it is a 1-cochain in the Hodge complex, not a vector field on the manifold). The continuum QED adds vector structure via $\text{SO}(4) = \text{SU}(2)_L \times \text{SU}(2)_R$, and Theorem 4 is a vector-QED selection rule that scalar QED does not possess.

In Paper 18's exchange-constant taxonomy [3], the photon's spin-1 vector nature is a CALIBRATION exchange constant: it is injected when projecting from the algebraic graph to the continuous manifold. The pendant-edge theorem (Proposition 5) quantifies the cost of this omission: a ground-state self-energy of 2 (in graph units) that the continuum's vector selection rules eliminate exactly. The structural zero is not a fine cancellation that the finite graph misses due to truncation—it is a *qualitative* selection rule tied to the photon's vector character, which the graph does not encode.

Independent reading: Connes–van Suijlekom operator-system gap. The same broken structural zero admits a parallel structural interpretation through the spectral-truncation framework of Connes and van Suijlekom [30]: the truncated algebra-image $\mathcal{O}_{n_{\max}} = P_{n_{\max}} C^\infty(S^3) P_{n_{\max}}$ is $*$-closed but *not* multiplicatively closed—taking its C^*-envelope $C^*(\mathcal{O}_{n_{\max}})$ collapses to a "non-informative full matrix algebra" that loses precisely the operator-system data carrying the off-diagonal Gaunt structure. The CG projection that "opens vertex couplings invisible to the SO(4) channel count" (Sec. 15.5) is exactly the failure-of-closure that distinguishes $\mathcal{O}_{n_{\max}}$ from its envelope. Read this way, the pendant-edge value $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max}$ is not an artifact of finite truncation but the operator-system content the envelope would discard.

This reading is sharpened by an independent quantitative match: the GeoVac operator system $\mathcal{O}_{n_{\max}}$ has propagation number $\text{prop}(\mathcal{O}_{n_{\max}}) = 2$ at $n_{\max} \in \{2, 3, 4\}$, matching the value Connes & van Suijlekom prove for the Toeplitz operator system $C(S^1)^{(n)}$ on the circle (their Proposition 4.2: prop = 2 independently of n); see Paper 32 [7] (propagation-number proposition). The witness pair underlying the propagation-number computation is the same boundary mechanism as the pendant-edge theorem here: a "raise-then-lower at the top shell escapes the truncation" deficit, viewed once from the QED self-energy side and once from the operator-system side. The fact that two structurally distinct constructions on the same Fock-projected S^3 graph both quantitatively realize the same boundary correction is internal evidence that the operator-system reading of GeoVac is not a formal relabelling but the structural source of both phenomena. See `debug/wh1_connes_vs_mapping_memo.md` (round-1 mapping) and `debug/wh1_round2_propagation_memo.md` (round-2 propagation computation) for the full record.

15.6 Selection rule survival census

A systematic test of 8 continuum QED selection rules on the finite Fock graph at $n_{\max} = 2$ reveals that only *one* survives:

Rule	Verdict	Tier	Key metric
Gaunt/CG sparsity	**survives**	intrinsic	density 16%→2.6%→0.6%
Angular momentum Δm_j	broken	intrinsic	20 violations
Spatial parity (E1)	broken	intrinsic	100% parity violations
Charge conjugation (C)	broken	intrinsic	8 C-violations in Σ
Furry's theorem	broken	intrinsic	tadpole $= 16\sqrt{2}/15 \neq 0$
Vertex parity ($n_1{+}n_2{+}q$ odd)	broken	structural	0% forbidden
SO(4) channel count ($W > 0$)	broken	structural	100% have $W = 0$
Ward identity	broken	structural	residual = 5.08

Table 11: Selection rule survival census at $n_{\max} = 2$. STRUCTURAL rules require vector photon structure absent from the graph; INTRINSIC rules are broken by graph kinematics.

The 7 broken rules partition cleanly into two classes:

- **3 structural** (vertex parity, SO(4) channel count, Ward identity): require the photon to be a vector harmonic on S^3. The graph's scalar 1-cochain photon has no mechanism to carry parity-flip, angular momentum transfer, or gauge derivative structure. These enter at the CALIBRATION tier when projecting to vector QED.
- **4 intrinsic** (Δm_j, spatial parity, charge conjugation, Furry): broken by graph kinematics. The CG projection allows couplings forbidden in vector QED because the scalar photon carries different angular momentum content.

Notable graph-native invariants: $\mathrm{Tr}(\Sigma)/\mathrm{Tr}(\Lambda) = 33/8$; tadpole $= 16\sqrt{2}/15$; bubble $= 32/9$; triangle $= 3584\sqrt{2}/3375$.

15.7 Native Dirac graph QED

The scalar Fock graph recovers only 1 of 8 continuum QED selection rules (Table 11). A natural question is whether building QED directly on the *Dirac* graph—with nodes labeled by $(n_{\mathrm{Fock}}, \kappa, m_j)$, the native quantum numbers of the Dirac operator on S^3—recovers additional rules by encoding spinor structure at the graph level.

15.7.1 Two adjacency rules

Two natural edge rules are tested:

- **Rule A (κ-preserving):** lifts the scalar Fock ladders $\Delta N = \pm 1$ within each κ sector. The graph decomposes into independent κ-blocks, analogous to the l-shell splitting of the scalar graph. At $n_{\max} = 2$: $V = 10$, $E = 8$, $\beta_1 = 1$.
- **Rule B (E1 dipole):** enforces $\Delta l = \pm 1$ (parity flip), $|\Delta j| \leq 1$, $|\Delta m_j| \leq 1$. Mixes κ values; the graph is a single connected component. At $n_{\max} = 2$: $V = 10$, $E = 20$, $\beta_1 = 11$.

Rule	Scalar Fock	Dirac A	Dirac B	Tier
Gaunt/CG sparsity	✓	✓	✓	always
Δm_j conservation	×	✓	✓	spinor
Spatial parity (E1)	×	×	✓	spinor
Furry's theorem	×	✓	✓	spinor
Vertex parity	×	×	×	vector
SO(4) channel count	×	×	×	vector
Ward identity	×	×	×	vector
Charge conjugation	×	×	×	vector
Total surviving	**1/8**	**3/8**	**4/8**	

Table 12: Selection rule survival on the scalar Fock graph versus the two Dirac graph rules. The 8 rules partition into three tiers: 1 always survives, 3 are spinor-recoverable, and 4 require vector photon structure.

15.7.2 Selection rule recovery

The selection rules partition cleanly into three tiers:

1. **Always survives (1):** Gaunt/CG angular sparsity, which is a property of the coupling coefficients regardless of graph topology.

2. **Spinor-recoverable (3):** Δm_j conservation (both rules—Dirac nodes carry explicit m_j), spatial parity (Rule B only—E1 edges enforce $\Delta l = \pm 1$ by construction), and Furry's theorem (both rules—the identity vertex on the Dirac graph is off-diagonal, making odd-loop traces vanish at $t = 0$). These are INTRINSIC to the spinor quantum numbers.

3. **Vector-photon-required (4):** vertex parity, SO(4) channel count, Ward identity, and charge conjugation. These require promoting the photon from a scalar 1-cochain to a vector harmonic with $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ structure. They are CALIBRATION content from the continuum embedding.

15.7.3 Self-energy structure

The ground state is **not** a pendant vertex on either Dirac graph: degree 2 (Rule A) and degree 5 (Rule B), both constant across $n_{\max} = 2, 3, 4$. Consequently, the pendant-edge theorem (Proposition 5) does not apply, and the structural zero $\Sigma(\mathrm{GS}) = 0$ is not recovered on either Dirac graph.

Exact algebraic results at $n_{\max} = 2$:

- $\mathrm{Tr}(\Sigma_A) = 17/2, \quad \mathrm{Tr}(\Sigma_B) = 103/20.$
- Rule A GS block: $\mathrm{diag}(5/8, 5/8)$ (diagonal, no m_j mixing).
- Rule B GS block: $\left(\begin{smallmatrix} 103/160 & 21/160 \\ 21/160 & 103/160 \end{smallmatrix}\right)$ (off-diagonal m_j mixing from dipole edges).
- Σ is strictly positive definite for both rules (no zero eigenvalues), in contrast to the scalar Fock graph's 5 zero eigenvalues from the photon kernel.
- Rule A eigenvalues: $\{1/2\,(\times 4),\ 3/4\,(\times 2),\ 5/4 \pm \sqrt{2}/2\,(\times 2)\}$.
- Rule B eigenvalues: $\{1/5\,(\times 2),\ 1/4\,(\times 3),\ 49/100,\ 33/40,\ 27/20,\ 267/400 \pm 3\sqrt{881}/400\}$.

15.7.4 π-free algebraic content

All Dirac graph QED quantities are π-free. Rule A lives in $\mathbb{Q}[\sqrt{2}]$—*simpler* than the scalar Fock graph's $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}]$—because the κ-preserving adjacency avoids the CG irrationals from cross-l couplings. Rule B lives in $\mathbb{Q}[\sqrt{2}, \sqrt{17}, \sqrt{41}, \sqrt{881}]$, the irrationals arising from the edge Laplacian's characteristic polynomial rather than from CG coefficients.

15.7.5 Interpretation: spinor barrier and vector barrier

The Dirac graph is an *intermediate* construction between the scalar Fock graph (1/8 rules) and continuum vector QED (8/8 rules). The spinor-recoverable rules (3 rules) are INTRINSIC to the electron's spinor quantum numbers—they are restored simply by using Dirac (n, κ, m_j) nodes instead of scalar (n, l, m) Fock nodes. The vector-photon-required rules (4 rules) are CALIBRATION content that enters when the 1-cochain photon is promoted to a vector harmonic on S^3.

In Paper 18's exchange-constant taxonomy [3], the path from graph to continuum QED therefore has two distinct barriers: (i) the *spinor barrier*, crossed by adopting Dirac node labels (graph-intrinsic, zero cost); (ii) the *vector barrier*, requiring the photon's spin-1 structure (calibration tier, continuum embedding). The three-tier partition of Table 12 makes this classification precise.

15.8 Vector QED sprint: five negative results

A systematic five-track investigation (VQ-1 through VQ-5) tested whether the 4 vector-photon-required selection rules can be recovered from the graph's intrinsic geometry without explicitly promoting the photon to a vector harmonic. All five tracks are negative for selection rule recovery.

15.8.1 Approaches tested

1. **VQ-1 ($\sigma \cdot L$ decomposition):** the spin-orbit coupling operator $\sigma \cdot L$ is zero on all graph edges because it is purely intra-shell ($L|l{=}0\rangle = 0$), producing no inter-shell coupling.

2. **VQ-2 (pure σ_μ vertex):** Pauli matrices as the vertex coupling. The graph is disconnected (no inter-shell edges); 2/8 rules survive but this is *worse* than the Dirac Rule B baseline (4/8).

3. **VQ-3 ($\sigma_\mu \otimes V_{\text{scalar}}$ vertex):** inserting σ_μ at the CG vertex while keeping the same photon propagator. The Pauli trace identity $\sum_\mu \sigma_\mu^2 = 3I$ forces $\text{Tr}(\Sigma_{\text{vec}}) = 3{\times}\text{Tr}(\Sigma_{\text{scalar}})$ and $F_{2,\text{vec}} = 3 \times F_{2,\text{scalar}}$—a trivial rescaling. Selection rules: 1/8 (same as scalar; Δm_j conservation and Furry's theorem are *broken* by the sigma dressing).

4. **VQ-4 (direction-resolved Hodge decomposition):** classifying Fock graph edges into T-channel (radial, $\Delta n = \pm 1$) and L-channel (angular, $\Delta m = \pm 1$), with per-channel sub-Laplacians and photon propagators. At $n_{\max} = 2$ the channels are spectrally anisotropic (T eigenvalues $\{2\}$ vs L eigenvalues $\{1, 3\}$) and the full propagator is block-diagonal in T/L. At $n_{\max} = 3$, cross-channel coupling emerges ($\|G_{TL}\|/\|G_{\text{full}}\| = 5.8\%$), mixing radial and angular photon modes. This gives the scalar graph photon effective "polarization" from pure topology—a positive *structural* finding, though insufficient for selection rule recovery.

5. **VQ-5 (full vector QED):** combining direction-labeled photon channels (VQ-4) with channel-specific σ_μ vertex coupling (VQ-3). All 15 σ-to-channel configurations tested (5 named mappings + exhaustive 3×3 scan). Three distinct F_2 values found (1.0×, 1.4×, 1.6× scalar), but all configurations produce 1/8 surviving selection rules.

15.8.2 Structural conclusion

The VQ sprint confirms the three-tier partition of Table 12 from a complementary direction: even when the graph's T/L edge anisotropy provides effective polarization structure and the σ_μ matrices provide spin coupling, the 4 vector-required rules (vertex parity, SO(4) channel count, Ward identity, charge conjugation) remain broken across all tested configurations. These rules require the *photon* to carry angular momentum quantum numbers (L, M_L)—promoting it from a scalar 1-cochain to a vector harmonic—which is calibration exchange constant content in Paper 18's taxonomy [3].

The positive structural finding from VQ-4 (spectral anisotropy of T/L channels, cross-channel coupling at $n_{\max} = 3$) shows that the Fock graph topology does encode directional information, but this information is insufficient to generate selection rules that depend on the photon's *spin-1* quantum numbers rather than its *propagation direction*.

15.9 Transverse photon sector from plaquettes

The preceding sections use the edge Laplacian $L_1 = B^\top B$ as the photon propagator. On a graph (simplicial 1-complex without higher-dimensional faces), this captures *only* the exact (longitudinal/gauge) sector of the Hodge decomposition. The co-exact (transverse/physical) sector requires 2-cells—faces bounded by closed edge paths.

Paper 30's SU(2) Wilson gauge construction [6] provides precisely these faces as plaquettes (minimal non-backtracking closed walks of length ≥ 6 on the Fock graph). With boundary operator d_1 mapping edges to faces, the co-exact Laplacian is $L_1^{\text{co}} = d_1 d_1^\top$ and the transverse photon propagator is its pseudoinverse:

$$G_T = (d_1 d_1^\top)^+. \tag{120}$$

15.9.1 Ground-state protection recovered

Computing the self-energy $\Sigma_T[a, b] = \sum_{e,e'} V_e G_T[e, e'] V_{e'}^\top$ with the transverse propagator:

$n_{\max}$	Σ_{scalar}(GS)	Σ_T(GS)	Plaquettes	Co-exact rank
3	4/3	**0** (exact)	4	2
4	3/2	**0** (exact)	33	8

The transverse propagator recovers the structural zero of Theorem 4 at machine precision ($< 10^{-16}$) while the scalar (longitudinal) propagator breaks it via the pendant-edge theorem (Proposition 5).

15.9.2 Mechanism: pendant-edge exclusion from plaquettes

The ground state maps to the Fock node $n = 1$, a pendant vertex (degree 1) connected to the graph by a single edge e_0. A pendant edge cannot participate in any closed walk: leaving a degree-1 vertex and returning requires immediate backtracking, which is excluded from plaquette enumeration. Therefore $d_1[e_0, f] = 0$ for all faces f, giving $(d_1 d_1^\top)[e_0, :] = 0$ and hence $G_T[e_0, :] = 0$. Since the ground-state vertex tensor $V[a, :, e]$ couples only through e_0, the self-energy vanishes:

$$\Sigma_T(\text{GS}) = 0 \qquad \text{(topological, exact at all } n_{\max}\text{)}. \tag{121}$$

This is the graph analog of the continuum vertex-parity argument: vertex parity ($n_1 + n_2 + q$ odd) with $n_2 = 0$ forces $2n_1$ odd (impossible); plaquette exclusion of pendant edges with pendant GS forces $G_T[e_0, :] = 0$ (topological). Both mechanisms protect the ground state from self-energy corrections, but via structurally different routes.

15.9.3 Refined selection rule partition

The three-tier partition of Table 12 is refined to four tiers:

1. **Always survives** (1 rule): Gaunt/CG angular sparsity.
2. **Spinor-recoverable** (3 rules): Δm_j conservation, spatial parity, Furry's theorem—from Dirac graph node labels.
3. **Plaquette-recoverable** (1 rule): ground-state structural zero—from co-exact Hodge topology of the face structure.
4. **Vector-quantum-number-required** (3 rules): SO(4) channel count, Ward identity, charge conjugation—requiring explicit (L, M_L) labeling of co-exact modes.

The boundary between "graph-intrinsic" and "calibration" has shifted: ground-state protection is recoverable from the graph's own Hodge structure (plaquettes from Paper 30), not requiring any photon angular momentum quantum numbers.

15.9.4 Mode-resolved analysis: why the final 3 rules require calibration

The co-exact eigenvalues μ_k of $d_1 d_1^\top$ approach the continuum photon eigenvalues $q(q+2)$ for low-lying modes (e.g. $\mu = 15.0 = 3 \times 5$ at $n_{\max} = 3$ matches $q = 3$ exactly; $\mu = 48.0 = 6 \times 8$ at $n_{\max} = 4$ matches $q = 6$). However, the *eigenvectors* have the wrong support. Testing the SO(4) triangle inequality $|n_1 - n_2| \le q \le n_1 + n_2 - 2$ on the mode-resolved self-energy $\Sigma_k[a,b] = \sum_{e,e'} V_e\, v_k v_k^\top G_T[e,e']\, V_{e'}^\top$:

$n_{\max}$	Mode	q_{assign}	Triangle inequality
4	0–3	$q = 3, 4$	PASS (0% violation)
4	4–5	$q = 5$	FAIL (66%)
4	6	$q = 5$	FAIL (63%)
4	7	$q = 6$	FAIL (86%)

The structural obstruction is topological: the Fock graph has only nearest-neighbor edges ($|\Delta n| \le 1$), so every co-exact eigenvector is a linear combination of $\Delta n = 0, \pm 1$ edges. A continuum photon of angular momentum q couples shells separated by up to q levels, but the graph's plaquette eigenvectors cannot span that range. The Ward identity diagnostic confirms: $\|[\Sigma_T, H_0]\|/\|\Sigma_T\| \approx 0.58$ at both $n_{\max} = 3, 4$ (zero would indicate energy conservation structure).

This identifies the Tier 1/Tier 2 boundary precisely: the graph gives the correct *count* and approximately correct *energies* of transverse modes, but not the angular momentum structure within each mode. Promoting graph 1-cochains to sections of a vector bundle indexed by (q, m_q) is the minimal calibration required for the final 3 selection rules.

15.10 Continuum bridge and projection exchange constant

The graph VP (trace) and the continuum spectral-sum VP (Sec. 3) are related at finite truncation by a *rational* projection constant $C = \Pi_{\text{cont}}/\text{Tr}(\Pi_{\text{graph}})$. At $n_{\max} = 3$:

$$C = \frac{50471424}{1779441125} \approx 0.0284.$$

C is rational. No π enters at finite truncation.

Transcendental numbers (π^{even} from Hurwitz zeta, $\text{Vol}(S^3) = 2\pi^2$) enter exclusively via two mechanisms: (i) infinite-sum regularization ($n_{\max} \to \infty$) producing π^{even} from Bernoulli numbers; (ii) volume normalization converting to dimensionful quantities. Both are Paper 18's CALIBRATION tier [3].

The graph is strictly richer than the continuum at finite truncation: at $n_{\max} = 3$, the graph captures $\sim$35$\times$ more VP content than the W-weighted continuum formula.

15.10.1 Topology-dependent projection

The VP projection constant C is specific to the vacuum polarization diagram. A natural hypothesis is that $C \times F_2 \to \alpha/(2\pi)$ as $n_{\max} \to \infty$, providing a single-constant projection from graph-native scalar QED to physical vector QED. This hypothesis is **falsified**:

$n_{\max}$	C	$F_2(\kappa)$	$C \times F_2$	$\alpha/(2\pi)$	ratio
2	0	2.3525	—	1.16×10^{-3}	—
3	2.84×10^{-2}	1.8731	5.31×10^{-2}	1.16×10^{-3}	45.7
4	4.74×10^{-2}	1.5892	7.54×10^{-2}	1.16×10^{-3}	64.9
5	—	1.3958	—	1.16×10^{-3}	—
6	—	1.2532	—	1.16×10^{-3}	—

Table 13: $C \times F_2$ grows with $n_{\max}$, diverging from $\alpha/(2\pi)$. The VP projection constant is not the vertex correction projection. C not computed at $n_{\max} = 5, 6$; the established divergence at $n_{\max} = 3, 4$ precludes convergence.

The product $C \times F_2$ *grows* with $n_{\max}$ because C increases faster than F_2 decreases. The correct graph-to-continuum projection for different diagram topologies involves different spectral-density factors: the VP projection (sensitive to photon self-energy) and the vertex projection (sensitive to electron self-energy and vertex structure) are structurally distinct. In Paper 18's taxonomy, these are independent CALIBRATION exchange constants.

15.10.2 Three-diagram projection constants

The topology-dependent projection is characterized by computing *independent* projection constants for each diagram type: $C_{\text{VP}} = \Pi_{\text{cont}}/\Pi_{\text{graph}}$, $C_{\text{SE}} = \text{Tr}(\Sigma_{\text{cont}})/\text{Tr}(\Sigma_{\text{graph}})$, and $C_{F_2} = [\alpha/(2\pi)]/F_{2,\text{graph}}$.

$n_{\max}$	C_{VP}	C_{SE}	C_{F_2}	$C_{\text{VP}}/C_{\text{SE}}$
2	0	0	4.94×10^{-4}	—
3	0.0284	0.275	6.20×10^{-4}	0.1030
4	0.0474	0.453	7.31×10^{-4}	0.1047
5	0.0599	0.582	8.32×10^{-4}	0.1029

Table 14: Cross-diagram projection constants at $n_{\max} = 2$–5. C_{VP} uses the photon-propagator trace $\text{Tr}(\Pi)$. The three constants have structurally independent power-law scaling: $C_{\text{VP}} \sim n^{1.48}$, $C_{\text{SE}} \sim n^{1.48}$, $C_{F_2} \sim n^{0.57}$.

Power-law fits give $C_{\text{VP}} \sim 0.0070 \times n^{0.93}$ ($R^2 = 0.95$), $C_{\text{SE}} \sim 0.056 \times n^{1.48}$ ($R^2 = 0.99$), and $C_{F_2} \sim 3.3 \times 10^{-4} \times n^{0.57}$ ($R^2 = 1.00$). The C_{F_2} exponent ($+0.57$) is the negative of the F_2 convergence exponent (-0.57, Table 10), so $C_{F_2} \times F_2 \to \text{const} = \alpha/(2\pi)$ by construction. However,

C_{VP} and C_{SE} grow with *different* exponents, confirming that no single focal-point constant projects all diagram types simultaneously.

VP/SE constant ratio. The ratio $C_{\mathrm{VP}}/C_{\mathrm{SE}} = 0.1035 \pm 0.0009$ (CV = 0.83%) is constant to sub-percent precision across $n_{\max} = 3, 4, 5$. The closest simple rational is $3/29 = 0.10345$. VP and SE share the same power-law exponent ($\sim n^{1.48}$), differing only by a constant prefactor $\sim 1/10$. The vertex correction C_{F_2} has fundamentally different scaling ($n^{0.57}$) because it is defined relative to the asymptotic Schwinger value. This reveals a **two-tier calibration structure**: one shared VP/SE tier (scaling as $\sim n^{1.48}$, with constant internal ratio), and one vertex-specific tier (scaling as $\sim n^{0.57}$, set by the F_2 convergence rate).

The per-mode analysis (Sec. 15.10.1) shows that the per-mode projection ratio $\rho(n_{\mathrm{ext}}, n_{\mathrm{int}})$ varies from 0 to 1.66 with no clean pattern — the graph-to-continuum projection is *topological*, not multiplicative. The root cause is that the Fock graph has nearest-neighbor connectivity ($\Delta n = \pm 1$ only), while the continuum SO(4) channel count W obeys a triangle inequality allowing non-nearest-neighbor contributions. Some modes have graph = 0, continuum $\neq 0$ (and vice versa).

The χ_{-4} Dirichlet character mechanism that produces Catalan's G and Dirichlet $\beta(4)$ in the continuum (Theorem 3) is **not accessible** from finite-graph parity decomposition: 24 PSLQ targets against $\{G, \beta(4), \pi^2, \zeta(3)\}$ all returned null. Graph F_2 even/odd ratios alternate with $n_{\max}$ (driven by degeneracy counting, not physics), while continuum even/odd ratios converge to ~ 1. The Catalan/β content lives in the continuum's Hurwitz zeta at quarter-integer shifts, which are analytic objects with no finite-graph analog.

15.11 Transcendental certificate

All graph-native QED quantities are π-free:

- VP trace: rational at every $n_{\max}$.
- Self-energy Σ: diagonal entries rational; off-diagonals contain $\sqrt{2}$, $\sqrt{3}$, $\sqrt{6}$ from CG coefficients.
- Vertex correction Λ: eigenvalues involve $\sqrt{145}$.
- Anomalous moment $F_2 = 5\sqrt{2}/3$ ($t = 0$); $(48\sqrt{3} + 96275)/[3(15 + 16\sqrt{6} + 9616\sqrt{2})]$ ($t = \kappa$): algebraic irrational at both couplings.
- Projection exchange constant: rational.

The algebraic irrationals enter through Clebsch–Gordan coefficients in the Dirac-to-Fock projection. In Paper 18's taxonomy, all graph-native QED lives in the INTRINSIC tier; transcendentals enter only when projecting onto the continuum (CALIBRATION tier).

Rational function structure. $F_2(t)$ is an even rational function of t (all odd Taylor coefficients exactly zero, from $t \to -t$ symmetry of the vertex bilinear). The degree and number field grow rapidly with $n_{\max}$: at $n_{\max} = 2$, $F_2(t)$ has degree 2/2 over $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$; at $n_{\max} = 3$, degree 16/16 over $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{10}, \sqrt{15})$, the new irrationals arising from CG coefficients involving $l = 2$ states. The rapid degree growth ($2 \to 16$) reflects the combinatorial explosion of CG coupling channels. At $n_{\max} = 3$, the Taylor coefficients c_k for $k \geq 2$ are *all negative* ($c_2 \approx -0.650$, $c_4 \approx -0.608$, $c_6 \approx -0.431$, $c_8 \approx -0.385$), in contrast to $n_{\max} = 2$ where the signs alternate. The uniform sign indicates that the poles of the $n_{\max} = 3$ rational function are farther from the origin, yielding better convergence of the Taylor series at $t = \kappa$.

15.12 Vector-photon QED: 7/8 scalar, 8/8 Dirac selection rules recovered

The preceding graph-native QED sections treat the photon as a scalar 1-cochain—a number on each edge. This produces π-free results in $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \ldots]$ but recovers only 1 of 8 continuum QED selection rules (Gaunt/CG sparsity on the scalar Fock graph) or 4 of 8 on the native Dirac graph (Rule B). The transverse photon from Wilson plaquettes (Sec. 15.9) recovers the ground-state structural zero, bringing the count to 5 of 8.

15.12.1 Pre-investigation: L_1 block-diagonalization

A natural question is whether the edge Laplacian $L_1 = B^\top B$ already contains photon angular momentum structure that can be extracted by reorganization—i.e., does L_1 block-diagonalize by (q, m_q)?

The answer is **no**. Photon quantum numbers describe the *transition* between endpoint labels (the angular momentum *difference*), not a conserved quantum number at each vertex. L_1 couples edges that share a vertex, and at that vertex, edges with different photon quantum numbers meet. The cross-sector Frobenius fraction is 20% at $n_{\max} = 3$ on the scalar Fock graph and 57% on the Dirac Rule B graph. The *only* decomposition that works is by connected components: l-sector (Fock) or κ-sector (Dirac Rule A).

Vector photon structure must therefore be added from outside. It is genuinely new structure, not a reorganization of what the graph contains.

15.12.2 Construction

The electron register uses Fock graph nodes (n, l, m) or Dirac labels (n, κ, m_j); both configurations are tested. The photon register adds modes labeled by (q, m_q) with $q = 1, \ldots, q_{\max}$ and $m_q = -q, \ldots, q$. The photon propagator is $G_\gamma(q) = 1/[q(q+2)]$, the vector Laplacian eigenvalues on S^3.

The vertex coupling is

$$V(a, b, q, m_q) = \begin{pmatrix} l_a & q & l_b \\ -m_a & m_q & m_b \end{pmatrix} \times \delta_{l_a + l_b + q \text{ odd}}, \tag{122}$$

where the Wigner $3j$ symbol enforces the triangle inequality and m-conservation, and the parity constraint $l_a + l_b + q$ odd enforces E1 vector coupling. Radial matrix elements are set to unity ($R = 1$), isolating angular structure.

The self-energy is

$$\Sigma(a, c) = \sum_{b, q, m_q} V(a, b, q, m_q)\, G_e(b)\, G_\gamma(q)\, V(c, b, q, m_q), \tag{123}$$

where $G_e(b)$ is the electron propagator.

15.12.3 Selection rule census

15.12.4 Ground-state structural zero: angular momentum mechanism

$\Sigma(\text{GS}, \text{GS}) = 0$ exactly (machine precision) at all tested $n_{\max}$. The mechanism: for the ground state with $l = 0$, the triangle inequality forces $q = l_{\text{int}}$, and the parity rule requires $l_{\text{int}} + l_{\text{int}} + q = 2l_{\text{int}} + q$ odd. With $q = l_{\text{int}}$, this gives $3l_{\text{int}}$ odd — but more precisely, with $l_{\text{ext}} = 0$: $l_{\text{int}} + q$ odd, and the triangle with $l_{\text{ext}} = 0$ gives $q = l_{\text{int}}$, so $2l_{\text{int}}$ must be odd. Impossible for any integer.

This is the rule broken by the pendant-edge theorem on the scalar graph ($\Sigma_{\text{scalar}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max} \to 2$). The vector photon restores it through angular momentum conservation alone.

Table 15: Selection rule recovery with vector photon quantum numbers. Scalar electrons recover 7 of 8 rules; Dirac electrons recover **8 of 8** at $n_{\max} = 2$.

#	Rule	Scalar + vec	Dirac + vec	Mechanism
1	Gaunt/CG sparsity	PASS	PASS	$3j$ symbol
2	GS structural zero	PASS	PASS	$2l_{\text{int}}$ odd
3	SO(4) channel count	PASS	PASS	triangle ineq.
4	Δm_j conservation	PASS	PASS	$3j$ m-rule
5	Spatial parity	PASS	PASS	E1 parity
6	Furry's theorem	FAIL	PASS	Dirac spinor phase
7	Ward identity	PASS[a]	PASS[a]	Σ block-diag.
8	Charge conjugation	PASS	PASS	Σ Hermiticity

[a] Ward identity

passes at $n_{\max} = 2$; truncation artifacts at $n_{\max} \geq 3$ (ratio 0.61 Dirac, 3.1 scalar).

15.12.5 Calibration exchange constant: $1/(4\pi)$

The $l = 1$ self-energy at $n_{\max} = 2$ is **exactly** $1/(4\pi)$ to machine precision. This is the Weyl exchange constant for S^2 (the density-of-states constant for the 2-sphere), entering through the normalization of the vector spherical harmonics: $\sqrt{(2l+1)(2q+1)(2l'+1)/(4\pi)}$ in the vertex coupling.

The entire scalar graph QED is π-free ($\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \ldots]$). The vector photon introduces **exactly one transcendental per loop**: π, through the S^2 solid angle normalization. In Paper 18's taxonomy [3], this is precisely where calibration content enters: the $1 \to 7$ transition from graph topology to angular momentum conservation costs exactly one transcendental (π, via $1/(4\pi)$ per loop).

15.12.6 Furry's theorem: scalar obstruction, Dirac resolution

For **scalar electrons**, Furry's theorem fails: the diagonal vertex coupling $V(a, a, q, m_q)$ is nonzero when q is odd (parity $2l + q$ odd is satisfied), and the $\left(\begin{smallmatrix} j & q & j \\ -m & 0 & m \end{smallmatrix}\right)$ does not vanish for generic m.

For **Dirac electrons**, the diagonal coupling vanishes identically by the *Dirac spinor phase constraint*. The physical mechanism: the Dirac spinor $\psi = (f \cdot \Omega_\kappa,\ ig \cdot \Omega_{-\kappa})$ has an i-factor on the small component, and the Dirac α-matrix $\alpha = \left(\begin{smallmatrix} 0 & \sigma \\ \sigma & 0 \end{smallmatrix}\right)$ is off-diagonal in large/small component space. For $a = b$, the vertex decomposes as a Hermitian pair whose small-component i phase reduces it to an imaginary part, $V_{LS} + V_{SL} \propto fg\,\mathrm{Im}\,\langle\Omega_\kappa|\sigma \cdot \varepsilon|\Omega_{-\kappa}\rangle$, and m_l conservation forces that imaginary part to vanish — the mechanism is m_l conservation, not Hermiticity alone (Paper 33's mechanism remark; sympy-exact in `tests/test_paper33_furry_derivation.py`). The cross terms cancel exactly, making $V(a, a, q, m_q) = 0$ for all q and m_q.

This is *not* Furry's theorem in the field-theoretic sense. In continuum QED, Furry works because for every electron loop there is a positron loop that exactly cancels — charge conjugation symmetry of the *field*, not the *state labels*. The Dirac spinor phase constraint is a kinematic identity of the single-particle vertex: it kills the tadpole at the vertex level, before any loop sum is performed. The two mechanisms are independent, and the Dirac constraint is the stronger one.

15.12.7 The 1–7–8 partition

The 8 continuum QED selection rules partition cleanly into three tiers. The partition depends on whether the electron basis is scalar or Dirac:

Tier	Rules	Source	Scalar / Dirac
Graph topology	1	Gaunt/CG sparsity (always survives)	both
Angular momentum	6	CG + E1 parity + triangle ineq.	both
8th rule	1	Furry / Dirac spinor phase	FAIL / PASS
		Total:	7/8 / 8/8

The $1 \to 7$ transition costs exactly one transcendental: π, entering as $1/(4\pi)$ per loop from the S^2 solid angle normalization. For **scalar electrons**, the $7 \to 8$ transition requires the full field-theoretic structure of charge conjugation — a categorically different ingredient that is absent from the graph/vector construction. For **Dirac electrons**, the $7 \to 8$ transition is closed by the Dirac spinor phase constraint at zero additional cost: the off-diagonal α-matrix combined with the i-factor on the small component reduces $V(a, a, q, m_q)$ to an imaginary part that m_l conservation forces to zero identically, killing the tadpole at the vertex level. No second-quantized field theory is needed.

16 Discussion

The five theorems, two propositions, and two observations of this paper, together with the anomalous magnetic moment computation (Sec. 7), the graph-native QED infrastructure (Sec. 15), and the vector-photon QED construction (Sec. 15.12), complete the first comprehensive treatment of perturbative QED on S^3 in both the continuum spectral-sum and finite-graph formulations. The vector-photon construction recovers 7 of 8 continuum selection rules for scalar electrons and **8 of 8 for Dirac electrons**, with a single transcendental cost ($1/(4\pi)$ per loop). For Dirac electrons, the 1–7–8 partition closes entirely at the kinematic level: graph topology $\to$ angular momentum conservation $\to$ Dirac spinor phase constraint. The three-axis taxonomy extends Paper 18's exchange-constant classification [3] from free propagators to interaction vertices: *diagram topology* is an independent source of transcendental content beyond operator order and bundle type.

The self-energy structural zero (Theorem 4) and the anomalous magnetic moment extraction are the first physical-process QED results in the GeoVac framework, complementing the vacuum diagrams of the prior QED modules. The self-energy demonstrates that the vertex parity selection rule has direct consequences for scattering processes on S^3. The pendant-edge theorem (Proposition 5) sharpens the broken-structural-zero finding: the graph ground-state self-energy $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max}$ is an exact rational expression that converges to 2, not 0, as $n_{\max} \to \infty$. This is not a finite-size artifact that heals in the continuum limit; it reflects a qualitative difference between graph QED (scalar, 1-cochain photon) and continuum QED (vector, $\text{SO}(4) = \text{SU}(2)_L \times \text{SU}(2)_R$ photon). The photon's spin-1 vector nature is thereby classified as a CALIBRATION exchange constant in Paper 18's taxonomy [3]. The selection rule survival census (Table 11) makes this classification precise: of 8 continuum QED selection rules, only Gaunt/CG sparsity survives on the scalar Fock graph; the 7 broken rules partition cleanly into 3 STRUCTURAL (requiring vector photon structure) and 4 INTRINSIC (graph kinematics). The native Dirac graph QED (Sec. 15.7) refines this classification: building QED on the (n, κ, m_j) Dirac graph with E1 dipole adjacency (Rule B) recovers 4 of 8 rules, identifying a three-tier partition—1 always survives (Gaunt/CG), 3 are spinor-recoverable (Δm_j, spatial parity, Furry), and 4 require vector photon structure (vertex parity, SO(4), Ward, charge conjugation). The vector-photon QED construction (Sec. 15.12) then recovers 7 of 8 for scalar electrons and **8 of 8 for Dirac electrons** by adding explicit photon angular momentum labels (q, m_q) with Clebsch–Gordan vertex coupling and E1 parity enforcement. The path from graph to continuum QED has three distinct barriers: the spinor barrier (crossed by adopting Dirac labels, zero cost), the angular momentum barrier

(crossed by promoting the photon from scalar 1-cochain to vector harmonic, calibration cost $1/(4\pi)$ per loop), and — for scalar electrons only — the field-theory barrier (requiring second-quantized charge conjugation for Furry's theorem). For Dirac electrons, no field-theory barrier exists: the Dirac spinor phase constraint ($V(a, a, q, m_q) = 0$ identically from the off-diagonal α-matrix, the i-factor on the small component, and m_l conservation) closes the $7 \to 8$ gap at the single-particle vertex level, without invoking particle–antiparticle cancellation. The vector QED sprint (Sec. 15.8) provides five independent negative results confirming the three-tier selection rule partition from a complementary direction: neither Pauli spin matrices at the vertex (VQ-1–3), nor direction-resolved edge decomposition (VQ-4), nor their combination (VQ-5) recovers any of the 4 vector-required rules. The positive structural finding—spectral anisotropy of the T/L edge channels and cross-channel coupling at $n_{\max} = 3$—shows that the graph topology encodes directional information, but directional propagation is not equivalent to spin-1 quantum numbers.

The even-ness of $F_2(t)$ (Sec. 15.4.1) — all odd Taylor coefficients vanish — reflects the $t \to -t$ symmetry of the vertex bilinear and confirms that the anomalous moment depends only on κ^2, not on the sign of the hopping coupling.

The topology-dependent projection result (Tables 13 and 14) falsifies the hypothesis that a single projection constant C converts graph-native QED to physical QED: $C \times F_2$ grows with $n_{\max}$ ($45.7\times$ and $64.9\times$ above $\alpha/(2\pi)$ at $n_{\max} = 3, 4$), not converges. The three-diagram analysis (Table 14) sharpens this: C_{VP}, C_{SE}, and C_{F_2} have structurally independent power-law scalings ($n^{1.48}$, $n^{1.48}$, $n^{0.57}$ respectively), but VP and SE share the *same* exponent with a constant prefactor ratio $C_{\mathrm{VP}}/C_{\mathrm{SE}} = 0.1035 \pm 0.0009$ (CV = 0.83%, closest rational 3/29). This reveals a two-tier calibration structure: one shared tier for VP and SE (vector photon structure and non-nearest-neighbor connectivity calibrate both identically; the $\sim 10\times$ prefactor difference is a pure topology-of-diagram effect), and one vertex-specific tier for F_2 (requiring the full vertex correction projection). The extended F_2 convergence data (Table 10) confirms this picture: $F_2(\kappa)$ decreases as a slow power law $\sim n_{\max}^{-0.57}$, with Richardson extrapolation giving an unphysical negative limit, confirming $F_2 \to 0$ rather than converging to $\alpha/(2\pi)$. The graph-native anomalous moment vanishes in the infinite-graph limit; the physical Schwinger value requires both the coupling constant α and the vector photon selection rules, neither of which is present on the finite graph.

The anomalous magnetic moment ($F_2/[\alpha/(2\pi)] = 1.084$ at $n_{\mathrm{ext}} = 1$) provides the first quantitative curvature correction to a measurable QED observable in the framework, matching the Parker–Toms prediction within 0.4%.

The curvature coefficient identification $c_2 = (2 - B\Delta - F\Delta - F/B)/5$ (Sec. 7.7) is the first concrete numerical bridge between Papers 2 and 28 (a numerical coincidence, not a derivation — §7.7): the three spectral objects $B = 42$, $F = \pi^2/6$, $\Delta = 1/40$ that compose the α construction $K = \pi(B + F - \Delta)$ also appear as the geometric invariants controlling the second curvature correction to the anomalous magnetic moment on S^3. Since α cancels in the ratio $F_2/[\alpha/(2\pi)]$, this is a statement about the spectral geometry of the Dirac operator on S^3, not about QED per se. The high-precision $n_{\mathrm{int}} = 50$ extraction (Sec. 7.7) gives $c_3 = -5.946(3) \times 10^{-7}$ at $200.9\,\sigma$, confirming the expansion does not terminate at second order; the ratio $|c_3 x^3|/|c_2 x^2| \approx 5.4 \times 10^{-6}$ indicates rapid but non-terminating convergence. Its closed-form identification remains open (a π^{even}-polynomial form is an ansatz, not a T9 consequence).

The 5-point mode-dependent curve (Table 5) reveals that the anomalous moment on a compact space is strongly mode-dependent: enhanced for the lowest mode and progressively suppressed for higher modes, with a power-law $|\lambda|^{-1.84\pm0.06}$. The dominant internal mode ($n_{\mathrm{int}} = 1$) approaches the Parker–Toms λ^{-2} scaling asymptotically, but higher internal modes contribute increasingly at larger n_{ext}, pulling the effective exponent above -2. This non-perturbative mode dependence is a structural consequence of the discrete spectral geometry on S^3.

The χ_{-4} identity (Theorem 3) is a concrete spectral-zeta to Dirichlet-L bridge. Combined with the RH-sprint finding that the spectral zeros of $D(s)$, $D_{\text{even}}(s)$, and $D_{\text{odd}}(s)$ exhibit GUE-like spacing statistics (CV ≈ 0.35–0.40, KS rejects Poisson; Paper 29 [4]), the Dirac spectral geometry on S^3 realizes both axes of the Weil dictionary — though GUE spacing without a single critical line means the analogy to the Riemann hypothesis remains structural, not literal.

The depth-k tower has a natural interpretation in the framework of Zagier's motivic weight conjecture [18]: each loop order increases the motivic weight by the vertex coupling, and the algebra of multiple zeta values provides the natural receptacle for the resulting transcendentals. The half-integer shift $a = 3/2$ from the Dirac spectrum on S^3 produces Hurwitz (rather than Riemann) multiple zeta values. (An earlier version read the failure of 100-element PSLQ bases as evidence that the S^3 spectral geometry accesses an MZV sector not reached by flat-space Feynman integrals; that failure was a basis-coverage artifact, and $S_{\min}$ is resolved in closed form in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$.)

The fine-structure Euler sum cancellation (Observations 1 and 2) connects the free-propagator Dirichlet series studied in Secs. 3–5 to the *interacting* Dirac hydrogen spectrum. The product survival rule — $\zeta(2) \times \zeta(\text{odd})$ cancelled at every order, with exactly $\max(0, \lfloor (2p-5)/2 \rfloor)$ products surviving — is verified through D_6 and driven by the same n^2 Fock degeneracy that produces $F = \zeta(2)$ in Paper 2 [9]. Despite this structural link, PSLQ finds no small-coefficient relation placing K/π in the $\mathbb{Q}$-span of $\{D_2, \ldots, D_6\}$: K lies in $\mathbb{Q}[\pi]$, while the Sommerfeld sums carry odd-zeta content from the Euler–Zagier MZV algebra at every order. The operator/energy distinction — spectral zeta versus physical energy coefficients — is the structural separation between K and the D_p.

The three-loop factorized algorithm (Sec. 10) reduces the computational complexity from $O(N^5)$ to $O(N^3)$, but raw partial sums at physical exponents carry zero reliable digits at any practical $n_{\max}$ (the $(\ln N)^2/N^2$ increment law; the fitted $N^{-1.31}$ was its local masquerade). The decomposition route supersedes direct summation entirely: the closure sprint (2026-06-11) gives $S^{(3)} = 31.5726\ldots$ inside the rigorous bracket $[31.57063, 31.57300]$, and the W10-identification sprint (2026-06-12) closes the full form: $S^{(3)} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(\text{odd}), \zeta(5,3)]$, realized depth ≤ 2, with $\zeta(5,3)$ the only depth-2 constant (Eq. (100)). The earlier "$S^{(3)}$ independent of $S_{\min}$" 20-digit test at non-physical exponents is superseded.

16.1 Open questions

1. **Resolved (2026-06-11):** $S_{\min}$ is identified in closed form, Eq. (80). The 100-element PSLQ null was a basis-coverage artifact, not irreducibility.

2. Does the depth-k tower stabilize (i.e., do all constants beyond some k_0 reduce to products of lower-depth constants)? **Updated (2026-06-12, W10-identification sprint):** at $k = 2$ everything reduces to depth ≤ 1; at $k = 3$ the full closed form is explicit with realized depth ≤ 2, the sole depth-2 constant being the level-1 generator $\zeta(5,3)$ (Eq. (100)); all level-2 depth-2 generators cancel in the total. Realized depth $\leq k-1$ at $k = 2, 3$. **Updated (2026-06-13, $S^{(4)}$ stage-2):** the depth bound extends to $k = 4$ — realized depth ≤ 3 for $S^{(4)}$ (Prop. 4, $k = 4$ bullet), proven numerics-free at weights $9, 11, 13$ (stuffle spans the full depth-4 space, two-prime exact rank) and settled at weights $5, 7$ by the low-weight depth filtration. So realized depth $\leq k - 1$ holds at $k = 2, 3, 4$. The explicit closed form of $S^{(4)}$ (the depth-≤ 3 survivors) and the $k \geq 5$ continuation remain open.

3. Is there a Picard–Fuchs interpretation of $S_{\min}$, analogous to Apéry's recurrence for $\zeta(3)$ [27]?

4. Can the self-energy structural zero at $n_{\text{ext}} = 0$ be lifted by going to higher loop order, or does

vertex parity enforce $\Sigma^{(k)}(0) = 0$ at every order? **Partial resolution:** The structural zero is broken on the finite graph (Sec. 15.5): the pendant-edge theorem (Proposition 5) proves $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max} \to 2$ exactly, never healing. The root cause is that the graph photon is a scalar 1-cochain lacking the spin-1 vector selection rules (vertex parity + SO(4) channel count) that enforce the continuum zero. **Further resolution:** Adding explicit vector photon quantum numbers (q, m_q) (Sec. 15.12) recovers the GS structural zero exactly through angular momentum conservation alone ($2l_{\text{int}}$ odd is impossible), confirming that the broken zero was due to the scalar photon, not a fundamental graph obstruction. The continuum question at higher loop order remains open.

5. The three-loop CG-weighted sum at physical exponents has convergence rate $\alpha \approx -1.31$. Is there a resummation technique (e.g., Padé, conformal mapping) that can extract more digits from the partial sums?

6. The 5-point mode-dependent curve $F_2/[\alpha/(2\pi)] \approx 5.91\,|\lambda|^{-1.84\pm0.06}$ has an effective exponent 2.8σ from -2. The dominant internal mode's local slope converges toward -2, but higher internal modes ($n_{\text{int}} \geq 2$) contribute increasingly at larger n_{ext} and pull the effective exponent above -2. Is the true asymptotic exponent exactly -2 (Parker–Toms), or a distinct value? This could be resolved by computing $n_{\text{ext}} = 6$–10 or by analytical large-λ asymptotics of the CG vertex sums.

7. The tree-level magnetic amplitude satisfies $V_{\text{mag}} \to (4/3)/|\lambda|$ asymptotically (Richardson-extrapolated). Can this 4/3 prefactor be derived from the CG coefficient structure in closed form?

8. The exact algebraic expressions for V_{mag} and $B(n_{\text{int}} = 1)$ at each n_{ext} are sums of square roots with rational coefficients, exhibiting a telescoping recurrence: the negative sqrt argument at step n equals the positive sqrt argument at step $n - 1$. Can this recurrence be proven from the CG coupling rules?

9. Can the flat-space limit $R \to \infty$ be verified explicitly by computing F_2 on spheres of increasing radius with n scaling as $n \sim mR$ to keep the physical mass fixed?

10. The curvature coefficient identification $c_2 = (2 - B\Delta - F\Delta - F/B)/5$ matches the spectral sum to 8 digits (Sec. 7.7). **RESOLVED (partially):** Extending to $n_{\text{int}} = 50$ confirms c_2 and gives $c_3 = -5.946(3) \times 10^{-7}$ at $200.9\,\sigma$. Deriving c_2 analytically from the Schwinger–DeWitt heat kernel remains open.

11. **RESOLVED.** $c_3 = -5.946(3) \times 10^{-7}$ from $n_{\text{int}} = 50$, nonzero at $200.9\,\sigma$. The curvature expansion does not terminate at second order; the ratio $|c_3 x^3|/|c_2 x^2| \approx 5.4 \times 10^{-6}$ shows rapid but non-terminating convergence. The closed-form identification of c_3 (a π^{even}-polynomial ansatz) remains open.

12. The multi-n_{ext} extraction of curvature coefficients requires Wigner–Eckart decomposition into multipole form factors for higher-j sectors. At $n_{\text{ext}} \geq 2$, the $j = 1/2$ component is a $2/[(n + 1)(n + 2)]$ fraction of the total degeneracy. Can the j-dependent form factors be separated cleanly to give independent c_2 extractions at each n_{ext}?

13. The VP/SE projection ratio $C_{\text{VP}}/C_{\text{SE}} = 0.1035 \pm 0.0009$ (Table 14) is constant to sub-percent across $n_{\max} = 3, 4, 5$, with closest simple rational $3/29 = 0.10345$. Is this a rational number? If so, what structural invariant fixes it? The constancy implies that VP and SE probe the same calibration content with a universal diagram-topology prefactor.

14. **RESOLVED.** The Euler sum cancellation extends to D_5 (weight 9): $\zeta(2)\zeta(7)$ cancels while $\zeta(3)\zeta(6)$ and $\zeta(4)\zeta(5)$ survive (Observation 2). The product survival rule $\max(0, \lfloor(2p-5)/2\rfloor)$ is verified through $p = 5$. The S_N-symmetry origin remains open, but the mechanism is identified: the n^2 Fock degeneracy weight systematically expels $\zeta(2)$ products via Euler sum reflection identities.

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